



r/IGCSE Economics

Complete Syllabus Guide Resource — Cambridge 0455

2027-2029 Syllabus Revised

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“**Economics** is the social science that describes the factors that determine the production, distribution and consumption of goods and services.”
(Source: Wikipedia)

The Nature of the Economic Problem

Resources: are the inputs required for the production of goods and services.

Scarcity: a lack of something (in this context, resources).

The fundamental economic problem is that there is a scarcity of resources to satisfy all human wants and needs. There are **finite resources** and **unlimited wants**. This is applicable to consumers, producers, workers and the government, in how they manage their resources.

Economic goods are those which are scarce in supply and so can only be produced with an economic cost and/or consumed with a price. In other words, an economic good is a good with an opportunity cost. All the goods we buy are economic goods, from bottled water to clothes.

Free goods, on the other hand, are those which are abundant in supply, usually referring to natural sources such as air and sunlight.

The Factors of Production

Resources are also called ‘factors of production’ (especially in Business). They are:

- **Land:** all **natural resources** in an economy. This includes the surface of the earth, lakes, rivers, forests, mineral deposits, climate etc.
 - The **reward** for land is the **rent** it receives.
 - Since, the amount of land in existence stays the same, its **supply is said to be fixed**. But in relation to a country or business, when it takes over or expands to a new area, you can say that the supply of land has increased, but the supply is not depended on its price, i.e. rent.
 - The **quality** of land depends upon the soil type, fertility, weather and so on.
 - Since land can’t be moved around, it is **geographically immobile** but since it can be used for a variety of economic activities it is **occupationally mobile**.
- **Labour:** all the **human resources** available in an economy. That is, the mental and physical efforts and skills of workers/labourers.
 - The **reward** for work is **wages/salaries**.
 - The **supply** of labour depends upon the number of workers available (which is in turn influenced by population size, no. of years of schooling, retirement age, age structure of the population, attitude towards women working etc.) and the number of hours they work (which is influenced by number of hours to work in a single day/week, number of holidays, length of sick leaves, maternity/paternity leaves, whether the job is part-time or full-time etc.).
 - The **quality** of labour will depend upon the skills, education and qualification of labour.
 - Labour **mobility** can depend up on various factors. Labour can achieve high occupational mobility (ability to change jobs) if they have the right skills and qualifications. It can achieve geographical mobility (ability to move to a place for a

Basic Economic Problem

The basic economic problem deals with how to best allocate resources, with unlimited human wants, and finite (scarce) resources.

Economic goods are scarce goods produced from limited factors of production, and thus pose an opportunity cost with their production. **Free goods** are goods present in unlimited quantities and don't require any limited factors of production to be made, and thus pose no opportunity cost.

The problem shows up at every level of the economy:

- Consumers have finite money, so it's either a phone or a laptop
- Workers have limited time, skills and energy and must make a decision about which industry and job to work — and also must choose between leisure time and working overtime
- Producers can either make consumer goods or capital goods
- Governments must choose either education or healthcare to use its taxpayer funds

The Factors of Production

The four factors of production are **Land** (reward: rent), **Capital** (reward: interest), **Enterprise** (reward: profit), and **Labor** (reward: salaries/wages).

- **Land** refers to the natural resources used for the production of goods and services.
- **Capital** refers to the human-made produced products utilized in the production process for the production of goods and services.
- **Labor** refers to the human effort, both physical and mental, used to produce goods and services.
- **Enterprise** refers to the willingness and ability to take risks and organize the other 3 aforementioned factors of production in order to produce goods and services, usually in the pursuit of profit.

Mobility of factors refers to the ease with which a factor of production can shift between jobs, industries, and locations.

1. Labor

Influences on the geographical mobility of labor:

- **Wage differences** — higher wages attract workers
- **Cultural Restrictions** — In some cultures, straying out of the “homeland” makes you impure.
- **Apprehension**- Due to precarious geopolitical conditions
- **Living Standards**- Living standards of the other place could be low.
- **Cost of living** — high living costs reduce mobility
- **Housing availability/Cost of Shifting** — lack of housing limits movement
- **Family and social ties** — reduce willingness to move
- **Transport links** — better transport increases mobility
- **Immigration laws** — restrictions reduce mobility

Influences on the occupational mobility of labor:

- **Level of education and skills** — having higher education and skills decreases retraining time and increases mobility
- **Training time and retraining opportunities** — more training increases mobility
- **Qualifications requirements** — strict qualifications (like needing a doctorate for performing medicine) reduce mobility
- **Age** — younger workers are usually more mobile, because:

- They usually have fewer family commitments (e.g. no children or mortgages), so it is easier to move location
- They are more willing to take risks and try new jobs or industries
- They find it easier to retrain and learn new skills
- They are often more adaptable to new technology
- They have lower opportunity costs of changing jobs early in their careers
- **Technological change** — can reduce mobility if skills become outdated (What if IT jobs become redundant because of the advent of AI?)

2. Capital

Influences on the mobility of capital:

- **Nature of capital** — fixed capital (machines, buildings) is less mobile; working capital (money, raw materials) is more mobile
- **Cost of moving machinery** — high costs reduce mobility
- **Demand for the product** — higher demand increases mobility
- **Government regulations and taxes** — can restrict movement
- **Time factor** — capital often takes time to transfer or replace

3. Land

- Land is geographically immobile (obviously since you can't shift the location of a piece of land)
- Depends on government regulation (what all can you use land for [occupational mobility])
- Its fertility will indicate if its mobile for agriculture
- Generally, is very mobile occupationally as you can use it for many things (above 2 are the only main points)

4. Enterprise

Influences on the mobility of enterprise:

- **Availability of finance** — access to loans increases mobility
- **Level of risk and uncertainty (of economy)** — high risk reduces mobility
- **Government support and regulations** — incentives (like lower taxes and lower regulation) increase mobility
- **Skills and experience of entrepreneurs** — more skills increase mobility
- **Market conditions** — profitable markets attract enterprise

Factors affecting quantity and quality of factors of production

Another cause of change in quantity of labor can be the change in level of qualifications needed, or even increased wages to incentivize dormant workers to come back to work.

Factor	Causes of change in Quantity	Causes of change in Quality
Labor	• Population growth or decline • Migration (immigration increases, Emigration decreases) • Retirement age changes	• Education level • Training and retraining • Work experience • Health and nutrition • Technological familiarity

	- Female participation in the workforce - Health crises or pandemics	
Land	- Land reclamation (increases usable land) - Natural disasters (reduce usable land) - Urbanization (reduces agricultural land) - Environmental degradation Note: Total land supply is fixed; only usable land changes	- Soil fertility - Use of fertilizers and irrigation - Climate and weather - Pollution and overuse - Conservation and sustainable practices
Capital	- Investment by firms - Savings and availability of finance - Government spending - Depreciation and wear and tear (decreases) - Economic growth or recession	Technological progress Maintenance of machinery Replacement of old equipment; Efficiency and automation Infrastructure development
Enterprise	- Number of entrepreneurs - Profit opportunities - Access to finance - Government incentives or taxes - Risk and economic stability. - High barriers to entry (with high corporate taxes and regulations.)	Education and experience of entrepreneurs Management skills Innovation and creativity Risk-assessment ability Access to information and technology

Opportunity Cost

Opportunity cost is the value of the next best opportunity foregone. Opportunity cost is generally expressed in terms of *benefit*.

How Opportunity Cost Works	
Scarce resources available Limited money / time / land / capital	
▼	
Decision: Option A or Option B? You can only choose one	
▼	
You choose Option A	
▼	
Opportunity Cost = the benefit of Option B The benefit you gave up by not choosing B	

1. Firms (Producers)

Why useful:

- **Helps decide which product to produce** when resources are limited
- **Encourages efficient allocation of capital and labor** (as opportunity cost helps in producing maximum benefit)
- **Highlights trade-offs between short-term profit and long-term investment** (Option A: Pay shareholders \$1 million → short-term profit now, Option B: Buy new machinery → long-term efficiency and higher future profits)
- **Can improve profitability** by comparing opportunity costs of different projects (helps decide which one is more profitable)

Why not useful:

- **Some benefits are difficult to measure** (brand value, customer loyalty)
- Opportunity cost ignores external factors like market fluctuations or changing government regulations

2. Governments

Why useful:

- Helps priorities spending between competing needs (e.g., healthcare vs education)
- Encourages efficient use of scarce tax revenue
- Guides long-term planning and policy-making

Why not useful:

- Many external benefits are hard to measure in monetary terms (safety, equality, environment)
- Political pressures may override economic efficiency
- Emergency decisions may ignore opportunity cost
- Social and ethical considerations sometimes outweigh purely economic calculations

3. Consumers

Why useful:

- Helps make rational choices with limited income
- Shows the true cost of spending decisions
- Highlights trade-offs between present and future consumption

Why not useful:

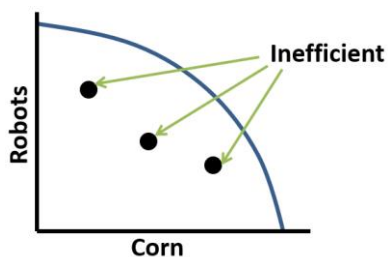
- Consumers may lack information about alternatives
- Some benefits (happiness, satisfaction) are subjective and hard to quantify
- Emotional or social factors may override rational calculations
- Not always practical for every small decision

PPC Graphs

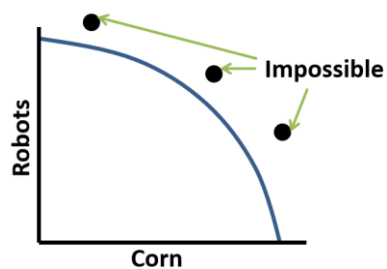
A Production Possibility Curve (PPC) shows the maximum possible output of two goods or services that an economy can produce, given its available resources and technology, and the trade-offs between them. *(For marks mention: maximum possible output, tradeoff, at the same level of technology, at the same level of quantity and quality of resources)*

Point on diagram	Meaning
On the curve	Productively efficient — all resources are fully and efficiently employed
Inside the curve (below)	Productively inefficient — some resources are unemployed or wasted (e.g. during a recession)
Outside the curve (beyond)	Currently unattainable — impossible with current resources and technology

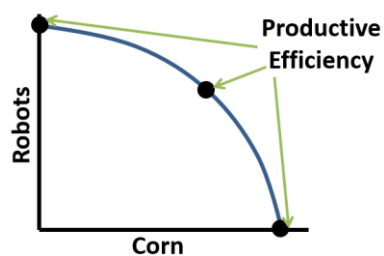
Inside the curve



Outside the curve

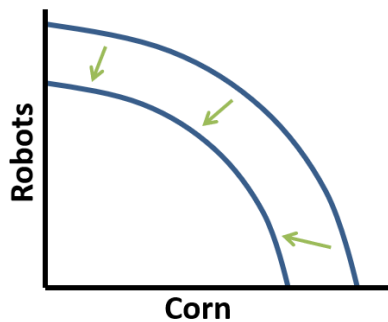


On the curve (perfectly efficient)

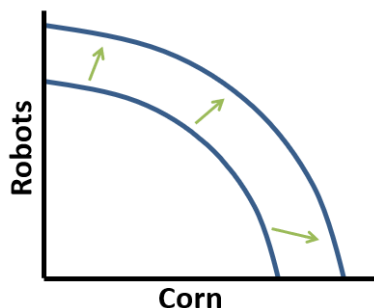


Points to note for PPC graphs:

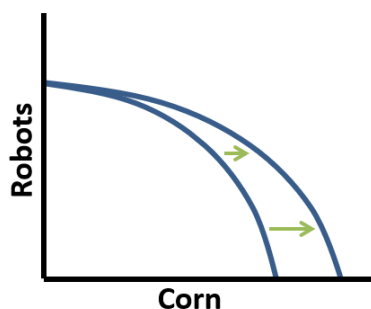
- **If there is an inward shift:** The quantity and quality of factors of production are decreasing / degrading level of technology is there in the economy
(Here suppose the technology to make **both** corn and robots worsened)/(Or the quality/quantity of the factors of production used for producing **both** corn and robots worsened/decreased)



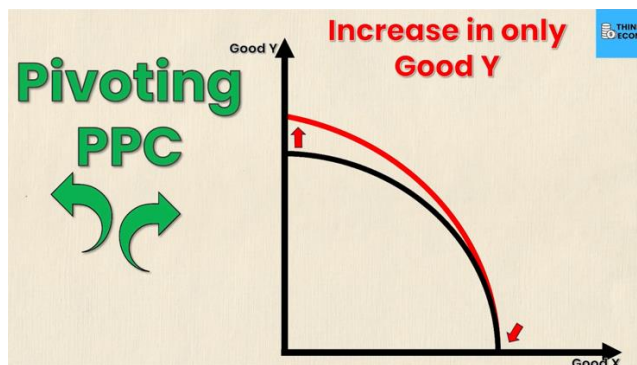
- **If there is an outward shift:** The quantity and quality of factors of production are increasing / better level of technology is there in the economy
(Here suppose the technology to make **both** corn and robots ameliorated (got better))/(Or the quality/quantity of the factors of production to make **both** corn and robots ameliorated/increased)



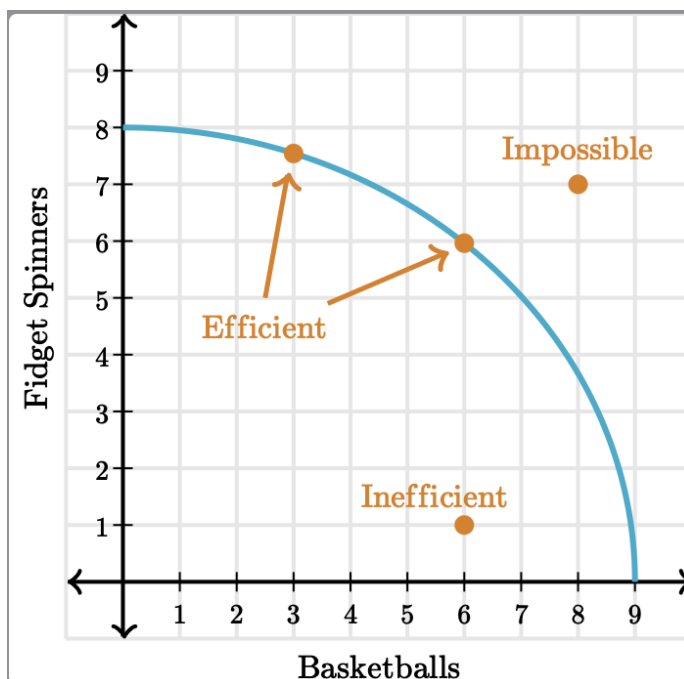
- **If shift is only towards the x axis,** that means the resources and technology that can **ONLY** be used for goods in the x axis are increasing
(**Only** the technology/Factors of production used for corns ameliorated/increased (which is on the x axis), the shift is towards the x axis)



- **If shift is only towards the y axis**, that means the resources and technology that can ONLY be used for goods in the y axis are increasing
(**Only** the technology/Factors of production used for robots ameliorated/increased (which is on the y axis), the shift is towards the y axis)



Also note, the increasing trade-offs lead to increasing opportunity costs, as the mobility of factors of production decreases (you use the more adept and efficient resources first).



If you want to learn this topic more diagrammatically, be sure to follow this section of the course on **Khan Academy** [Here](#).

Still confused? Read a more elaborate blog [Here](#).

The Allocation of Resources

Microeconomics studies the economic behavior of individual consumers, households, firms and specific markets or industries, rather than the economy as a whole.

Macroeconomics concerns itself with the economic behavior of the entire country's economy. The decision makers are the government and the central bank.

Allocation of resources is done with 3 main questions in mind:

- What is to be produced?
- How is it to be produced? (Using labor, or capital? In private sector or public sector [for public goods, for example])
- For whom is it to be produced? (Welfare or those capable of buying it?)

Price mechanism is the way in which price is adjusted due to changes in supply and demand, helping in the allocation of resources. The market ideally is supposed to work at equilibrium, and perpetually tries to adjust prices to achieve equilibrium from disequilibrium.

How the Price Mechanism Allocates Resources
Signal A price rise signals scarcity/high demand; a price fall signals surplus/low demand
▼
Incentive High prices incentivize producers to supply more and consumers to buy less
▼
Reallocation Resources flow toward high-price (profitable) markets and away from low-price ones
▼
Equilibrium Quantity demanded = quantity supplied — market clears

Demand

Demand is the quantity of a good or service that consumers are willing and able to buy at a given price over a given period of time.

Contractions/Extensions ALONG the demand line happen **SOLELY** when price changes.

Non-price factors include the following for demand (aka which cause SHIFT (increase and decrease) of demand line):

- Income
- Tastes, Fashion, Preference
- Advertising
- Government Policies (Taxation on a good)
- Economy (The condition of the economy)
- Price of related goods (Substitutes and Complements)
- Seasonal Factors (Sweater's demand more during winters)
- Expectations of future prices (If expected there will be higher prices in the future, demand increases now et converse)
- Population Size and Demographics (More old people means more demand for old-age related drugs)

Market Demand = Sum of Individual Demands. (Know this in graph setting, where you must add the shapes of the graph!)

Supply

Supply is the quantity of goods or services that suppliers are willing and able to supply at a given price over a given period of time.

Same contraction and extension along the line rules as applied in demand.

Non price factors include:

- **Cost of production** (If they increase, supply decreases)
- **Technology** (Increased technology leads to more supply)
- **Number of firms** (The higher the number of firms, the bigger the supply)
- **Government Policies** (If corporate taxes increase, supply decreases)
- **Increased Availability of Factors of Production** (increased availability shifts supply to right)
- Natural Factors like Weather/Climate
- **Price of related goods** (If price of substitute-in-production increases, the original good's supply decreases as producers switch)
- **Expectations of Future prices** (if prices may increase, suppliers reduce supply to hoard stock; if they may decrease, suppliers increase stock to quickly sell whatever they can)
- **Productivity of Factors of Production** (If they increase, so does supply)

*Same rule of adding up individual supplies to form total/market supply (Graph form too!)

Price Determination

Price changes happen due to shifts in demand or supply. Consequences (which you should put in your analysis questions) are:

- Changes in quantity demanded
- Changes in quantity supplied
- Changes in sales (equilibrium quantity) — talk about previous equilibrium and new equilibrium
- Possible shortages or surpluses
- Effects on consumers and producers (higher prices, lower prices, etc.)

What Happens at Disequilibrium
Price ABOVE Equilibrium $Q_s > Q_d \rightarrow$ Surplus / Excess Supply
▼
Firms cannot sell all stock $\rightarrow$ price starts to fall
▼
Price falls $\rightarrow$ Q_d rises, Q_s falls
▼
New Equilibrium: $Q_d = Q_s$ Market clears

What Happens with a Shortage
Price BELOW Equilibrium $Q_d > Q_s \rightarrow \text{Shortage / Excess Demand}$
▼
Consumers compete for limited supply $\rightarrow$ price gets bid up
▼
Price rises $\rightarrow Q_d$ falls, Q_s rises
▼
New Equilibrium: $Q_d = Q_s$ Market clears

Price Elasticity of Demand

Price Elasticity of Demand (PED) measures the percentage change in quantity demanded resulting from a percentage change in price.

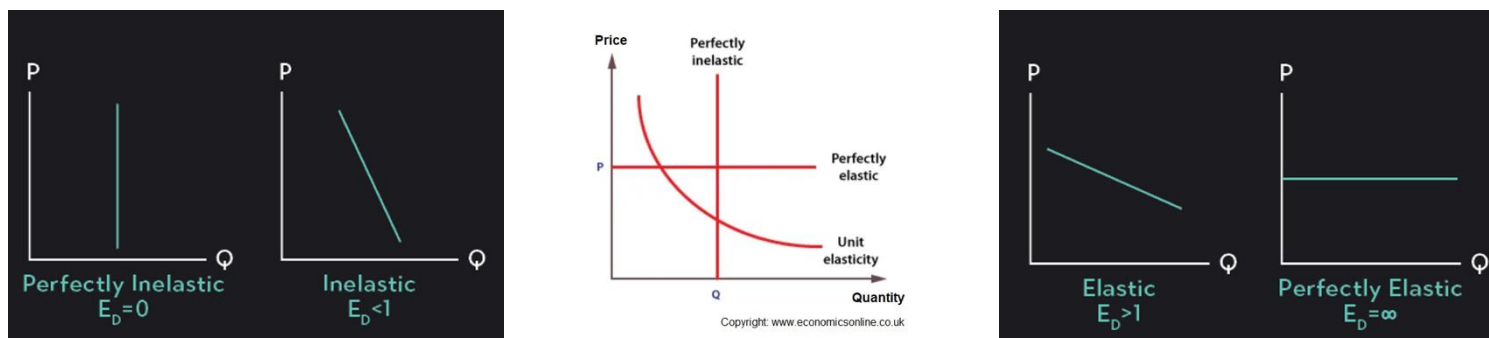
PED	$\frac{\% \text{ Change in Quantity Demanded}}{\% \text{ Change in Price}}$
	=
	$\frac{\frac{\text{New Quantity Demanded} - \text{Original Quantity Demanded}}{\text{Original Quantity Demanded}}}{\frac{\text{New Price} - \text{Original Price}}{\text{Original Price}}}$

Factors affecting PED:

- Addictiveness
- Proportion of Income
- Availability of Substitutes
- **Durability of Product** (If it is durable, the purchase is prolonged if prices increase, making it more elastic)
- **Definition of Product** (Milk is more elastic than Amul milk)
- Necessity or Luxury
- **Time Period** (Inelastic in the short term — it's harder to adjust consumption; long term, consumers find other alternatives)
- **Brand Loyalty/Cost of switching** [Apple to Android] (inelastic if high)

PED Value	Effect on Revenue when Price Rises
Inelastic ($0 > PED > -1$)	Revenue INCREASES
Elastic ($PED < -1$)	Revenue DECREASES
Unit elastic ($PED = -1$)	Revenue stays the same

Here is a more diagrammatic representation of the above theory:



The line here signifies the demand line.

(You may notice the unusual curve pattern of unit elasticity, and to understand why that happens, you will need some mathematical understanding of graphs, so be sure to explore this if you're inquisitive)

Price discrimination refers to charging different customers different prices for the same product due to varying PEDs.

Significance of PED for different groups:

1. Producers / Firms

Why it matters:

- Helps set pricing strategies
- Helps forecast revenue changes

Examples:

- **Elastic demand ($PED > 1$)** → price rise ↓ total revenue, price fall ↑ total revenue. E.g., a soft drink company raising price may lose many customers
- **Inelastic demand ($PED < 1$)** → price rise ↑ total revenue, price fall ↓ total revenue. E.g., a life-saving drug company can raise price without losing much sales

Implication: Firms can increase profit by knowing whether their product is elastic or inelastic.

2. Consumers

Why it matters:

- Helps consumers make informed choices
- Shows how sensitive they are to price changes

Example:

- If the price of petrol rises and PED is inelastic → consumers have limited ability to reduce consumption
- If PED is elastic → consumers can switch to public transport or alternatives

Implication: Consumers can plan spending and decide whether to substitute or continue buying.

3. Government

Why it matters:

- Helps design effective tax policy — taxing inelastic goods (cigarettes, alcohol, petrol) raises more revenue
- If the aim is to reduce consumption of a demerit good, elastic demand is preferable — consumers respond more to price changes
- Must consider PED when setting indirect taxes, otherwise the policy may raise little revenue or change little behavior

Price Elasticity of Supply

Price Elasticity of Supply (PES) measures the percentage change in quantity supplied resulting from a percentage change in price. It shows how responsive producers are to price changes.

PES	$\frac{\% \text{ Change in Quantity Supplied}}{\% \text{ Change in Price}}$
	=
	$\frac{\frac{\text{New Quantity Supplied} - \text{Original Quantity Supplied}}{\text{Original Quantity Supplied}}}{\frac{\text{New Price} - \text{Original Price}}{\text{Original Price}}}$

PES is always positive (direct relationship — higher price → more supplied). Values:

- **PES = 0** → Perfectly inelastic (supply cannot change at all — e.g. original art, land in the short run)
- **0 < PES < 1** → Inelastic supply (% Qs change < % price change)
- **PES = 1** → Unit elastic
- **PES > 1** → Elastic supply (% Qs change > % price change)

Factors affecting PES

- **Time period / Production lead time** — In the short run, supply is more inelastic as firms can't quickly scale up. In the long run, firms can adjust capacity, making supply more elastic.
- **Availability of stocks / Inventories** — If a firm holds large stocks, it can respond to price rises quickly → more elastic
- **Spare production capacity** — Firms operating below full capacity can increase output easily → more elastic; firms at full capacity cannot → more inelastic
- **Factor substitutability** — If inputs are widely available and easy to substitute, supply is more elastic
- **Perishability of product** — Perishable goods (fresh food) must be sold immediately regardless of price → inelastic. Non-perishables can be stored and timed.
- **Length of production period** — Long production cycles (wine, aircraft) make supply highly inelastic in the short run

Significance of PES

Why it matters for firms:

- Inelastic PES: firms can't quickly respond to price rises → may miss profit opportunities. Guides investment in spare capacity to improve flexibility over time.
- Elastic PES: firms can rapidly expand output when prices rise, gaining from market opportunities.

Why it matters for government:

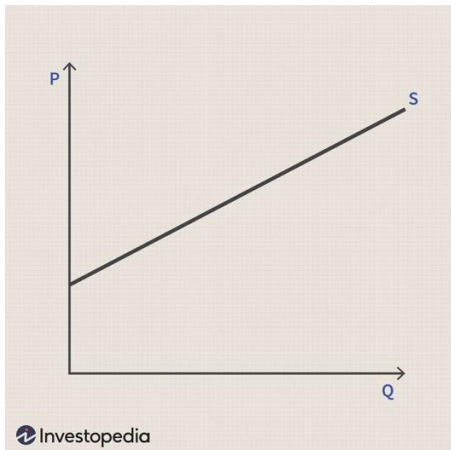
- Tax incidence: with inelastic supply, more of the tax burden falls on producers (they can't reduce supply easily). With elastic supply, producers can cut output and pass more of the burden to consumers.

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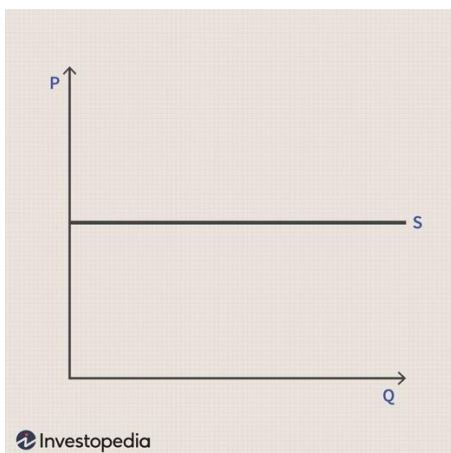
- Agricultural policy: farm supply is often inelastic in the short run → price supports and subsidies may be needed to stabilize incomes.
- Housing: supply of housing is very inelastic (planning, construction time) → demand increases lead to sharp price rises.

Here is a more diagrammatic representation of the above theory:

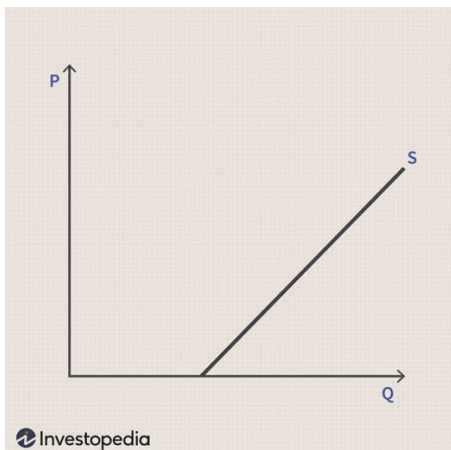
Elastic Supply:



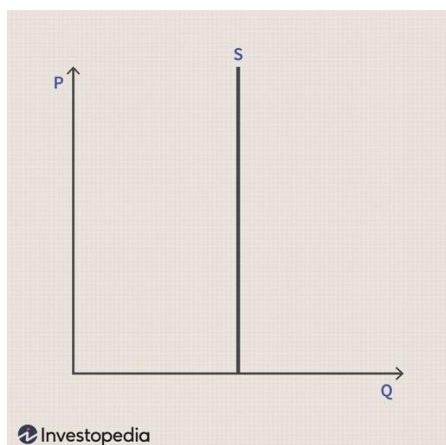
Perfectly Elastic Supply:



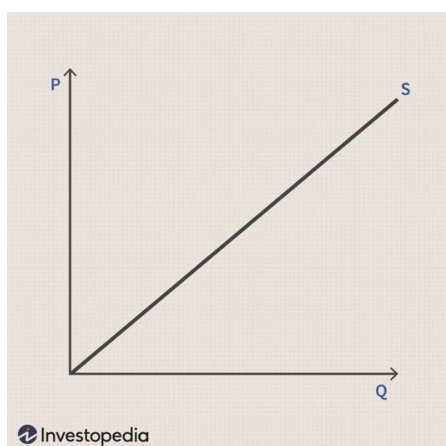
Inelastic Supply:



Perfectly Inelastic Supply:



Unit Elastic:



Note: The unit elastic graph of supply should always pass through the origin (0,0).

Tip: Don't think the PES graph is unitary JUST because its gradient=1, it **must** pass through 0,0. That's the only criteria.

Eyeballing the nature of the graph

For PES graphs, you need to see the slope of the graph to determine its nature. If a change in the y value (of price) gives a relatively big change in the x value (horizontal value), it's elastic, because it is **very responsive**. If the same doesn't happen, it is relatively **non-responsive** and is inelastic.

Market Economic System

A **market economic system** (also called a free market or capitalist economy) is one where the allocation of resources is determined primarily by the price mechanism — the interaction of supply and demand — with minimal government intervention.

Advantages	Disadvantages
Price mechanism allocates resources efficiently toward what consumers want (consumer sovereignty)	Public goods won't be provided — non-excludable, non-rival nature creates the free rider problem

Competition drives down prices and improves quality and innovation	Merit goods are under-consumed as people underestimate their benefits (e.g. education, vaccines)
Profit motive incentivizes entrepreneurship and efficiency	Demerit goods are over-consumed as people ignore external costs (e.g. cigarettes, alcohol)
Flexible and responsive to changing consumer preferences	Monopoly power can arise — firms exploit consumers with high prices and low quality
Resources tend to flow to where they're most valued	Income inequality: wealth concentrates in few hands, leaving some unable to meet basic needs
Examples: USA, Singapore, Hong Kong — strong growth and living standards	Negative externalities (pollution, congestion) ignored by profit-seeking firms

Market Failure

Market failure occurs when the free market fails to allocate resources efficiently, resulting in a misallocation — too much or too little of a good is produced from society's perspective.

Key Terms

Term	Definition
Public Good	Non-excludable (can't stop non-payers using it) AND non-rival (one person's use doesn't reduce availability for others). The free rider problem means the market won't provide it. Examples: national defense, street lighting.
Merit Good	A good whose benefits are underestimated by consumers → under-consumed in the free market. Has positive externalities. Examples: education, healthcare, vaccinations.
Demerit Good	A good whose costs (to user and society) are underestimated → over-consumed in the free market. Has negative externalities. Examples: cigarettes, alcohol.
Private Cost	Cost borne directly by the producer or consumer. E.g. a firm's production costs.
External Cost	Cost imposed on third parties not involved in the transaction. E.g. pollution from a factory affecting local residents.
Social Cost	Total cost to society = Private Cost + External Cost
Private Benefit	Benefit enjoyed directly by the producer or consumer. E.g. profit for a firm; satisfaction for a consumer.
External Benefit	Benefit enjoyed by third parties. E.g. employers benefit from an educated workforce without paying for the education.
Social Benefit	Total benefit to society = Private Benefit + External Benefit

Causes of Market Failure

Public Goods: Because public goods are non-excludable and non-rival, individuals can benefit without paying — the free rider problem. Since no private profit can be made, the market will not provide them. The government must step in and fund through taxation.

Merit Goods: Consumers underestimate private AND social benefits, so the free market under-produces and under-consumes them relative to what is socially optimal. Example: people may not vaccinate because they ignore the herd immunity benefit to others.

Demerit Goods: Consumers underestimate private costs AND ignore external costs to others. The free market over-produces these goods. Example: smokers don't fully account for their own long-term health costs or the passive smoking costs imposed on others.

Negative Externalities: External costs mean social cost > private cost. The free market produces more than is socially optimal — e.g. a factory producing steel also produces air pollution borne by nearby residents, who have no say in the transaction.

Positive Externalities: External benefits mean social benefit > private benefit. The market produces less than is socially optimal — e.g. education benefits employers who hire the graduates without paying for their training.

Monopoly Power: A dominant firm can restrict output and raise prices above competitive levels, leading to allocative inefficiency. Consumers face higher prices, less choice and lower quality.

Factor Immobility: When factors can't easily move between industries or regions, resources stay stuck in declining sectors and can't flow to where they're more productive. Results in structural unemployment and inefficient allocation.

Mixed Economic System

A mixed economic system combines elements of both the free market and government intervention, aiming to gain the benefits of markets while correcting their failures.

Government Intervention to Address Market Failure

Maximum Price (Price Ceiling): Set below the equilibrium price to keep essential goods affordable (e.g. rent controls, food price controls). Effect: $Q_d > Q_s \rightarrow$ a shortage is created. Problem: black markets may emerge and quality may fall.

Minimum Price (Price Floor): Set above the equilibrium price to protect producers or discourage consumption (e.g. minimum wage, minimum alcohol pricing). Effect: $Q_s > Q_d \rightarrow$ a surplus is created. Example: agricultural price floors lead to food stockpiles.

Indirect Taxation: A tax on production or sale of goods shifts supply left. Used to discourage consumption of demerit goods and raise revenue. The tax burden is split between consumers and producers depending on relative PED and PES — whoever is more inelastic bears more of it.

Subsidies: Government payment to producers shifts supply right. Used to encourage production and consumption of merit goods or support domestic industries. Consumers benefit from lower prices; producers receive a guaranteed return.

Regulation: Laws and rules that control firm behavior — e.g. pollution limits, safety standards, banning certain products.

Privatization: Transferring ownership from public to private sector to introduce competition, profit motive and efficiency.

Nationalization: Transferring ownership from private to public sector to ensure universal access to essential services (e.g. NHS, railways).

Direct Provision: Government directly produces and provides goods/services — used for public goods and merit goods (education, defense, public healthcare).

Effectiveness of Government Intervention

Government intervention can work well but has real limitations. Information failure is a big one — the government may not know the correct tax or subsidy level, and setting it wrong can make things worse. Unintended consequences are common: rent controls create housing shortages; agricultural price floors create food surpluses. There are also political pressures — policies are often driven by short-term electoral interests rather than economic efficiency. And all intervention has an opportunity cost: subsidies and direct provision require tax revenue. Black markets can emerge around price controls, and most policies have time lags before they take effect.

Money

Characteristics:

- Divisible
- Durable
- Acceptable
- Scarce
- Uniform
- Portable

Functions:

- **Medium of Exchange** — Used to buy and sell goods and services and avoids barter problems (double coincidence of wants)
- **Standard of Deferred Payment** — Used to pay back debt
- **Store of Value** — Can be saved for future use, it retains its value over time
- **Unit of Account** — Used for measuring the value of each good. Prices, debts, profits are expressed in money.

Central Bank

Functions:

- Acts as lender of last resort to commercial banks
- Controls the monetary policy of the country
- Regulates commercial banks (repo rates, cash reserve ratios)
- Is the banker for the government
- Issues notes and coins (currency)
- Controls the country's foreign reserve and gold reserve

Importance for:

Government:

- Central bank manages the government's accounts and debts
- Controls inflation and maintains economic stability with monetary policy

Firms:

- Sets interest rates which affect their borrowing
- Controls inflation, retaining their confidence

Consumers:

- Sets interest rates which affect their borrowing
- Controls inflation, protecting their purchasing power

Functions of Commercial Banks

Primary Functions:

- Accepting Deposits
- Giving Loans
- Credit Creation (giving out loans is credit creation, as even though they give out debited money, the debit is still there, and the loan is still there, both have independent capability to spend them, thus money is created)

Secondary Functions:

- Additional financial services like investment advice, help in ITR forms, etc.
- Offering credit card facilities for the convenience of customers (private individuals and commercial clients).
- Providing safety deposit boxes for customers to safeguard highly valued possessions, including jewellery and important documents such as wills.
- Providing money transfer facilities, such as transferring money to an overseas bank account or to pay various bills, such as telephone, electricity, gas and water bills.

Influences on spending, saving, borrowing:

- Income
- Interest rates (affecting their ability to borrow)
- Consumer confidence
- Wealth (assets can be given as collateral)
- Availability of credit (how easily can you get loans)
- Inflation — expected inflation encourages spending now
- Age — Old people spend more on healthcare, young people spend more on education, middle-aged people save and borrow more for their childrens' education

Workers

Factors affecting an Individual's choice of occupation:

Wage factors (ANYTHING that affects your income over time):

- Level of wage/salary
- Overtime pay/bonus
- Promotion prospects (more money, thus in wage factors)
- Job security (security means you work for longer, thus gain more wage)
- Fringe benefits (saves you money + adds monetary value)

Non-Wage Factors:

1. Working conditions
2. Working hours
3. Status and prestige of the job
4. Training and career development
5. Location (may be far from home)
6. Risk of the job (different from working conditions) / stress levels
7. Discrimination or social norms

Key Terms:

Current expenditure is money spent on goods and services consumed within the current year. Unlike capital expenditure, it is often recurrent, such as the spending on food, clothing, entertainment and haircuts.

Capital expenditure is money spent on fixed assets (items owned by an individual or firm which last more than 12 months), such as computers, cars, furniture, buildings and equipment.

Wage Determination

Why do Skilled Workers earn more than Unskilled Workers?

The demand for skilled workers is likely to be higher (1) and inelastic (1) due to their higher productivity / education (1) expectation of a higher rate of return / sales (1) better quality goods and services (1) resulting in greater profits (1). The supply of skilled workers is likely to be lower / more inelastic (1) due to the shortage of people with skills / qualifications (1). Skilled workers may have more bargaining power (1) harder to replace / may cause more disruption by taking industrial action (1). Skilled workers may be more occupationally and geographically mobile (1) and work in the tertiary sector (1).

Points to note:

- Their demand, its elasticity, their supply, its elasticity (and reason)
- Bargaining power (reason)
- Sector they work in
- Their mobility
- Their advantages (higher rate of return, better quality, more productivity, greater profits)

Why do Unskilled Workers sometimes earn more than Skilled Workers?

Skilled workers may have less experience (1) and may be in less promoted positions (1). Skilled workers may be in declining industries (1). Skilled workers may place more importance on non-wage factors when deciding what jobs to do (1). Unskilled workers with longer working hours and poor working conditions may be higher paid (1). Skilled workers in poorer countries may have lower wages than unskilled workers in richer countries [cost of living may be different] (1). Unskilled jobs may be more dangerous e.g. miners (1) and firms will have to pay more wages to attract workers (1). Skilled workers may face discrimination (1).

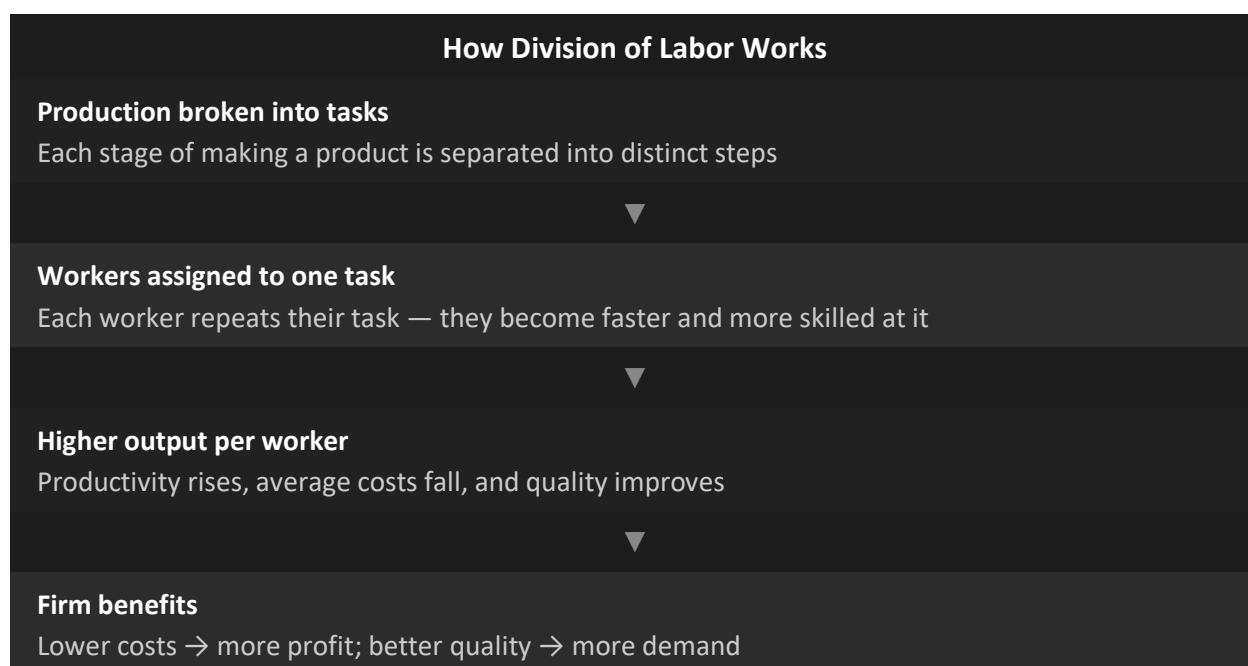
Reasons for Differences in earnings:

1. The sector they work in
2. Their Demand, Supply
3. Education
4. Discrimination
5. Occupation/Geographical Mobility
6. Declining Industries
7. Richer, Poorer Countries [Difference in Cost of Living]

Specialization and Division of Labor

Division of labor is the breakdown of the production process into separate tasks, with each worker specializing in a specific task.

Specialization occurs when an individual, firm, or country focuses on producing a particular good or a limited range of goods in which it is most efficient. (A doctor specializing in cardiology or a country specializing in harvesting coffee beans)



Advantages of Specialization

For Workers

- Higher pay — specialized jobs tend to have high demand and low supply
- Workers can do what they love
- May need lower training time (for general specialization)
- Improved skills with repetition

For Firms

- Lower training costs (don't need to teach every worker how to operate every tool)
- Higher productivity and efficiency (lower average costs) → more profit
- Better quality as workers become more skilled in what they make

For the Economy

- More efficiency leads to more productive capacity, leading to higher GDP
- Lower costs for products leads to more accessibility, less poverty
- More international competitiveness (reduced current account deficit)
- More efficient use of scarce resources — other countries don't waste their resources producing something they are not efficient in

Disadvantages of Specialization

For Workers

- Lower occupational mobility
- May have longer training time (for specialization in medicine)
- Work may become monotonous, reducing worker morale and efficiency, lower job satisfaction
- Higher risk of structural unemployment

For Firms

- Overspecialization — if one worker is absent or the industry of a specialized product fails, the entire production cycle could get hampered
- Lower worker morale leads to inefficiency
- Higher costs of labor as they become specialized

For the Economy

- Overspecialization — product industry may suffer from reduced demand
- Lower variety of goods (as they produce only one or some products)
- May be very susceptible to global shocks
- Overdependence on other countries (e.g. depending on Russia solely for oil)
- High risks of structural unemployment
- Other industries may not be very well developed as they are not specialized — inequality between sectors or regions, as some sectors may be less developed and some regions may not have the advantage of specialization

Trade Unions

Trade unions are organizations formed by workers that negotiate for better wages, working conditions and protect worker rights, using collective bargaining.

Role of Trade Unions:

Engaging in collective bargaining on wages, working hours, working conditions, protecting employment (job security) and influencing government policies.

Factors influencing the strength of trade unions:

- Number of workers
- Government policies
- Level of unemployment — if unemployment is low, trade union power is more; if unemployment is high, trade union power is low as workers are easily dispensable
- Skill level of workers
- The sector they work in

Advantages and Disadvantages of Trade Unions

	For Workers	For Firms	For Government
Advantages	More wage; better working conditions; more job security; more legal support; stronger voice in negotiations; provide services like training; can influence government policies to favor workers	Lesser cost of negotiation (don't have to meet every employee); lesser disputes as discontent can be resolved peacefully; more labor supply as better conditions attract applicants; provides channel for communication; may promote training, increasing labor productivity	Improved working conditions and labor standards; reduces exploitation by companies; reduced inequality as workers keep getting wage hikes
Disadvantages	Industrial actions may lead to loss of income; membership fees; may lose their jobs as higher wages increase cost of production, leading to lower demand for labor; may reduce	Higher cost of production, lower profits due to higher wages and reduced productivity; industrial action like strikes can disrupt production and lose revenue; may reduce flexibility as unions insist	Can make industries uncompetitive → fewer exports → increased current account deficit; inflationary pressure as workers get more wages; political conflict with unions; strikes can disrupt production; can

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	competitiveness, leading to lower investment	workers only undertake tasks in their job descriptions	increase public sector spending if they protest for wage hikes
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Firms

Firms can either be classified as firms in the private/public sector or in the primary/secondary/tertiary sector.

Benefits and Drawbacks of Small Firms

Benefits of Small Firms	Disadvantages of Small Firms
Can operate in a niche market, facing no competition from bigger firms	Can't benefit from economies of scale like bigger firms, leading to higher cost of production and reduced competitiveness
Can benefit from economies of scale by partnering with other local smaller firms	Small firms can be driven out of business by larger firms, and may not have enough capital to survive recessions
Can be a local monopoly	Generally, tend to close down as they are sole traders
Can provide a personalized service — inelastic demand as customer loyalty is there	Difficulty in raising capital as they have no collateral — cannot invest in R&D and equipment
Generally, there isn't any trade union to oppose smaller firms	Government subsidies may only be short term
Government helps smaller firms with tax concessions and subsidies	Large firms may take over to increase market power and take advantage of economies of scale
Small firms can have smaller transport costs as most customers are local	
More responsive to customer demand as smaller firms are more flexible and don't have to consult many people — efficient communication	
Can benefit from external economies of scale	

Internal Growth (Organic Growth)

Internal growth occurs when a firm expands using its own resources by increasing output, sales, or capacity.

How it happens:

- Increasing production
- Opening new branches or stores
- Introducing new products
- Investing in new machinery
- Hiring more workers

External Growth

External growth occurs when a firm grows by joining with or taking over another firm.

Forms:

- **Merger** — two firms agree to join
- **Takeover (acquisition)** — one firm buy another

Types of Mergers

Horizontal Mergers

Two firms at the same stage of production in the same industry merge (e.g. two car manufacturers).

Advantages	Disadvantages
More market share	Diseconomies of scale
Higher price-setting ability	Culture clashes between the two firms — coordination and communication problems, leading to inefficiencies and conflicts
Economies of scale	Mergers tend to create duplication of resources (like staff), which can lead to laying off workers, creating anxiety among remaining workers — inefficiency and lower staff morale
Operating with fewer employees (no need to maintain 2 financial departments)	

Backwards Vertical Mergers

A firm merge with a supplier at an earlier stage of production (e.g. a car manufacturer merges with a steel producer).

Advantages	Disadvantages
Secure source of raw materials	Higher transport costs as the firm now has to pay for transportation
More control over the quality of raw materials	Firm may not have experience in processing or producing at the primary stage, leading to unproductivity
Can restrict supply for competitors	Costs of running operations in the primary sector increase total costs as more land, labor and capital are required
Lower costs as firm does not have to pay for raw materials at a marked-up price	

Conglomerate Mergers

A firm merges with a firm in a completely different industry (e.g. a food company merging with a tech company).

Advantages	Disadvantages
Diversification of risk — if one product in one industry is losing profit, profit from other industries might offset it	Lack of experience in new market
Redundant Factors of Production (no need for 2 financial teams), lowering costs	Culture clashes
	Diseconomies of scale
	Management difficulties
	Ridding of redundant labor can lead to lower staff morale

Economies of Scale

Economies of scale occur when a firm's long-run average costs fall as it increases its output. The opposite — diseconomies of scale — occurs when average costs rise as the firm grows too large.

Internal Economies of Scale

Type	Explanation
Financial	Can easily get loans as they have more collateral — often at lower interest rates
Bulk Buying	Large firms can buy raw materials in bulk, reducing average cost per unit
Marketing	Can spread marketing costs over a larger output — cost per unit falls
Technical	Can employ large-scale capital equipment that smaller firms can't afford
Managerial	Can employ specialized managers for each department, improving efficiency
Risk-spreading	Diversification into different markets reduces overall risk

External Economies of Scale

- **Infrastructure** — government may build better infrastructure in the area, lowering average costs for all firms
- **Ancillary industries** — supporting industries develop around the firm (e.g. specialist suppliers, maintenance firms)
- **Specialized labor** — firms can employ trained workers from other local industries without having to train them

Firms can spread **fixed**/variable costs over a greater amount of output — this is the core reason why average costs fall.

Internal Diseconomies of Scale

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- Coordination and communication problems / ineffective management
- Lower labor morale / alienation in a big firm
- Poor industrial relationships — causing strikes and decreasing productivity

External Diseconomies of Scale

- Pressure on infrastructure, causing congestion
- More demand for scarce resources, increasing cost of production due to higher raw material costs

Competitive vs Monopolistic Markets

Competitive Markets

Advantages	Disadvantages
Lower prices due to strong competition	Normal profits only, which may reduce incentive to invest
Greater efficiency as firms minimize costs to survive	Wasteful duplication of advertising and branding
Consumer sovereignty — firms respond to consumer wants	Resource depletion due to overproduction
Wide choice and variety of products	Small firms may fail, causing unemployment
Better quality and innovation to attract customers	Lack of economies of scale compared to large firms
Higher economic welfare and living standards	Excessive competition can cause instability in prices and output
More international competitiveness → higher exports and GDP	
Prevents exploitation of consumers (no price-setting power)	

Monopolistic Markets

Advantages	Disadvantages
Economies of scale → lower average costs; more internationally competitive	Higher prices for consumers
Lower prices in natural monopolies (e.g. utilities — one set of pipes is cheaper than four)	Exploitation of consumers (price-setting power) → market failure
Greater investment in R&D due to high profits	Complacency leading to inefficiency
Avoids duplication of infrastructure	Diseconomies of scale
Potentially efficient if regulated by the government	Poor quality and limited choice
	Barriers to entry reduce competition

	Income inequality due to excess profits
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Firms and Production

The demand for all factors of production is **DERIVED** demand — derived from the demand of the final product. If no one wants the product, there is no demand for the labor or capital needed to make it.

Factors influencing the demand for Factors of Production:

- Demand for the product
- Relative price of other Factors of Production / increasing cost of Factors of Production = lesser demand
- Availability of Factors of Production — if a factor is **readily available**, firms can hire more of it; if a factor is **scarce**, demand may be limited even if firms want more
- Productivity of Factors of Production — more productive factors are in higher demand as they generate more output per unit cost

Costs of Production

The **cost of production** refers to the total expenses incurred by a company to manufacture and deliver a product or service.

$$\text{Total Cost (TC)} = \text{Fixed Cost (FC)} + \text{Variable Cost (VC)}$$

$$\text{Average Cost (AC)} = \text{TC} \div \text{Output}$$

$$\text{AFC or AVC} = (\text{FC or VC}) \div \text{Output}$$

Fixed Costs	Variable Costs
Definition: Costs that do NOT change with the level of output in the short run — they exist even when output is zero	Definition: Costs that change DIRECTLY with the level of output
Remain the same whether you produce 1 unit or 10,000 units	Increase when output increases; decrease when output decreases
Examples: Rent of factory; salaries of permanent staff; insurance; interest on loans	Examples: Raw materials; wages of hourly or piece-rate workers; electricity used in production; fuel

Revenue and Profit

$$\text{Revenue} = \text{Price of Output} \times \text{Output}$$

$$\text{Profit} = \text{Revenue} - \text{Cost of Production}$$

Revenue is the total income generated from the sale of the output. Revenue is NOT profit — you need to subtract costs.

Objectives of a Firm

Objective	What it means
Profit Maximization	Producing where marginal revenue = marginal cost — maximizing the difference between revenue and total costs
Growth	Expanding market share, output, revenue — often through mergers or new markets
Survival	Getting enough revenue to not shut down — typically the aim during recessions
Social Welfare	Working for the betterment of living standards of society — common in social enterprises and NGOs
Profit Satisficing	Gaining enough profit to keep shareholders satisfied, then focusing on other objectives like staff welfare or sustainability

Unit 4 — The Role of the Government

Local Role	National Role	International Role
Providing local public goods	Regulation of markets	Trade blocs
Providing local merit goods (education, healthcare)	Correcting market failure	International organizations like WHO
Collecting local taxes	Managing fiscal policy	International aid
	Providing national public goods (national defense)	Global peace efforts and international relations

Macroeconomic Aims of the Government

- **Price stability** — keeping inflation low and steady
- **Economic growth** — increasing real GDP over time
- **Stable balance of payments** — keeping exports and imports in balance
- **Lower unemployment** — better standards of living, less poverty

Other plausible goals:

- **Environmental protection** — less pollution, higher living standards
- **Redistribution of income** — reduce relative poverty and inequality

Possible conflicts between aims: full employment (causes demand-pull inflation) versus stable prices; economic growth (more exports demanded as disposable income rises) versus balance of payments stability; and full employment (once again, more exports) versus balance of payments stability.

Government Budget

The **government budget** is an annual plan showing the government's expected income (revenue) and expenditure (spending) for a financial year.

Budget Surplus	Budget Deficit	Balanced Budget
Revenue > Spending	Revenue < Spending	Revenue = Spending
Government can save or pay off debt	Government must borrow (increases national debt)	Neutral fiscal stance

Why does the government spend?

To create jobs, provide merit goods and public goods, better infrastructure, fund welfare, and stabilize the economy.

Taxation

Reasons for levying taxation:

- Government revenue

- To discourage consumption of an item
- Correct market failure (putting tax on cigars)
- Reduce inequality (progressive taxation)
- Economic stabilization (higher tax to reduce consumption, leading to lower inflationary pressure)

Type of Tax	How it works	Example
Regressive	Takes a higher % of income from lower earners	£8 tax on a good — a person earning £200 pays proportionally far more than someone earning £100,000
Progressive	Takes a higher % of income from higher earners	10% for income under £12,000; 20% for income under £50,000; 45% for income above £150,000
Proportional	Takes the same % from everyone	25% of everyone's income
Indirect Tax	Tax on goods/services — can be passed on to consumers through higher prices	VAT, excise duty on petrol or cigarettes
Direct Tax	Tax paid directly by individuals or firms — cannot be passed on	Income tax, corporation tax

Qualities of a good tax (Adam Smith's Canons):

- **Equity / Fairness** — proportional to ability to pay
- **Convenience** — convenient to pay so lesser chance of tax evasion (e.g. pay online)
- **Economical** — cost of collecting should not exceed the revenue itself
- **Certainty** — taxpayers should know how much, where, when, and to whom they pay
- **Flexible** — can be changed according to economic conditions
- Does not discourage enterprise or work too much

Fiscal Policy

Fiscal policy is the use of government spending and taxation to influence the level of economic activity in an economy. It uses two tools: (1) Government Spending and (2) Taxation.

Macroeconomic Aim	How Fiscal Policy Achieves It
Price Stability	Increased taxation reduces disposable income → lower purchasing power → decreased aggregate demand → lower inflationary pressure. Decreased government spending also reduces aggregate demand.
Economic Growth	Decreased taxation and increased government spending raise aggregate demand → more output → higher GDP.
Balance of Payments Stability	Reduced disposable income also leads to lower demand for imports, improving the current account.

Low Unemployment	Decreased taxation (incentive to work) and increased government spending on projects that create jobs.
Redistribution of Income	Progressive taxation (higher taxes for rich) and increased government spending on welfare and construction jobs leads to a more equitable society.

Monetary Policy

Monetary policy is the use of interest rates, money supply, and exchange-rate measures by the central bank to influence economic activity in an economy.

Money supply is the total amount of money available in an economy at a given time — it includes cash in circulation and money held in bank accounts.

Monetary policy is a bit vaster per se. Central banks, yes, do change interest rates. But they also use **quantitative easing** (injecting more money into the economy) in some measures (very rare). Additionally, they may make more money (which they do, as sometimes money is often lost through bad loans). They don't just inject the money by saying "Oh! You're poor, here, take some of mah money!" No. They sell government bonds to take money from people, buy back government bonds with interest to inject money. They set **cash reserve ratios**, **repo rates** (the rate at which commercial banks take loans from the central bank). Moreover, they even use **foreign exchange rates** (when they peg values in the fixed exchange rate system). Depreciating the currency makes exports cheaper for other countries → more demand. Appreciating does the opposite.

How Interest Rate Changes Work

Central bank raises interest rates

Borrowing becomes more expensive; saving becomes more attractive



Consumers borrow less, save more

Disposable income falls → consumption falls



Firms invest less

Cost of borrowing for investment rises → fewer projects undertaken



Aggregate demand falls

Less spending in the economy → downward pressure on prices (reduces inflation)



Lower interest rates — reverse

Cheaper borrowing → more spending and investment → demand rises → economic growth stimulated

Supply-Side Policy

Supply-side policy refers to government policies aimed at increasing the productive capacity of an economy by improving the quantity and quality of its factors of production.

- **Education and Training** — training and educating workers increases quality of labor
- **Labor Market Reforms** — flexible contracts; lower hiring/firing costs (this makes labor more flexible and not rigidly stuck in unproductive positions, they can be easily placed in more productive jobs, leading to more productivity and lesser allocative inefficiency; firms will also be more willing to expand since they can hire and fire more easily with lesser costs); lesser power of trade unions; minimum wages
- **Lower Direct Taxes** — increased incentive to expand
- **Deregulation** — if regulation decreases, what was once a monopolistic market could turn more competitive, leading to more efficiency and productivity
- **Improving incentives to work** — lower welfare benefits, lower income tax
- **Improving incentive to invest** — lower corporate tax, subsidies
- **Privatization** — reducing inefficiency of public corporations, increasing productivity and efficiency

Supply-Side Policy Effect	Macroeconomic Aim Achieved
Education and training → more skilled workers	Economic growth — more productive capacity; Full employment — reduces structural unemployment
Labor market reforms → more flexible workforce	Economic growth — resources move to most productive uses
Lower taxes → more incentive to work and invest	Economic growth; Low unemployment — more hiring
Deregulation → more competition	Price stability — lower costs → lower prices (without demand-pull inflation)
Increased productive capacity	Balance of payments — more competitive exports; Stable prices — supply grows alongside demand

The main criticism of supply-side policies is **time lag** — it might take decades for a nation to enjoy the benefits of an improved education system or better infrastructure.

Economic Growth

Economic growth refers to an increase in a country's real Gross Domestic Product (real GDP) over time. We use REAL GDP (adjusted for inflation).

At equilibrium: Aggregate Demand = Real GDP (Spending = Output)

$$AD = GDP = C + I + G + (X - M)$$

Recession is a period when an economy experiences negative economic growth for at least two consecutive quarters.

Causes of Economic Growth

- Increased demand → more output
- Increased quantity of Factors of Production
- Increased productivity of Factors of Production
- Supply-side policies
- Increased technology
- Increased investment

Consequences of Economic Growth

Benefits	Costs
More output → higher GDP → lower prices → more affordability → less poverty	Resource depletion
Lower unemployment	Negative externalities like increased pollution
Decreased balance of payments deficit	Economic growth can lead to inflation, reducing price stability and FDI
More tax revenue → more government spending on welfare	Income inequality — income gets distributed unevenly
More FDI and investment	Labor exploitation can happen from deregulation
More power in trade blocs and international organizations	
Higher standards of living	

Policies to promote economic growth:

Expansionary fiscal policy (lower taxes, more government spending) + expansionary monetary policy (lower interest rates) + supply-side policies.

Employment and Unemployment

Employment is when people of working age (not children, old people, or undocumented immigrants even if they have economic activity) have a paying job, either full-time or part-time.

Unemployment occurs when people of working age are without a job, are available for work, and are actively seeking work.

Full employment occurs when all the people of working age who are actively looking and available for work have a job.

$$\text{Unemployment Rate} = (\text{Number of People Unemployed} \div \text{Workforce Strength}) \times 100$$

How unemployment is measured:

- **Claimant count** — number of people claiming unemployment benefits
- **Labor Force Survey (LFS)** — a household survey asking about employment status

Types of Unemployment

Type	Definition	Cause	Solution
Frictional	Workers temporarily unemployed while moving between jobs	Transitioning between jobs — always exists in any economy	Job centers; better information matching workers to vacancies
Structural	Workers' skills don't match jobs available in the economy	Decline of an industry; technological change; shift in consumer demand — this is LONG TERM unemployment	Retraining schemes; supply-side policies; regional development
Cyclical	People lose jobs because the economy is performing badly, usually during a recession	Low aggregate demand → firms cut output → fewer workers needed	Expansionary fiscal and monetary policy to boost demand

Changing Patterns of Employment

Concerned with the changing patterns of WHERE labor works (primary, secondary, tertiary; public, private; in the country or abroad; formal or informal economy; part-time; self-employed) and WHO works (older workers, women, immigrants, disabled people).

- Changing technology can reduce demand for labor in an industry, making workers shift (e.g. from secondary to tertiary)
- Reduced demand for an industry will make labor shift to other industries → structural unemployment

- Changing social norms (e.g. stigmas against women working) can incentivize more women to join the workforce
- Nationalization / privatization can make people shift to public / private sector
- Government regulations like the national minimum wage can reduce formal labor demand → shift to informal economy
- Lenient immigration procedures lead to more immigrants in the workforce
- Better infrastructure for disabled people and older workers incentivizes them to join
- Increasing the retirement age leads to more older workers in the workforce
- Better wage conditions abroad can lead to emigration
- Changes in education mean more people in the tertiary / secondary sector
- Economic development means more people in tertiary / secondary sector
- More desire for flexibility leads to more part-time jobs
- More business opportunities leads to more self-employment

Effects of Unemployment

On Individuals	On Firms	On the Economy
Loss of income → lower consumption → lower standards of living	Cheaper labor → cheaper cost of production → more profits (in a manner it's good for firms)	Lower standards of living; reduced HDI
Lower self-esteem and morale	Reduced aggregate demand in the economy → lower profits	Inefficient utilization of resources → lower productive capacity
Loss of skills if unemployed for long → harder to get a job (employers have a bad perception about long hiatuses from work)		Economy becomes less internationally competitive; decrease in GDP and probable recession
Fall into poverty; increased income inequality		More spending on welfare → opportunity cost; lower tax revenue
		Increased crime and social unrest

Inflation and Deflation

Inflation is the **sustained** rise in the general price level of a basket of goods over time. (If prices rise then stop rising, it ISN'T inflation.)

Deflation is the sustained decrease in the general price level of a basket of goods over time.

Both are measured by the **Consumer Price Index (CPI)**. The CPI first measures price changes (if it was \$100 before and now \$125, the price index is 125 with a base of 100), then applies a **weightage** to reflect how much each good makes up in a typical household's spending.

Causes of Inflation

Type	Cause	Example
Demand-Pull Inflation	Aggregate demand rises faster than the economy's productive capacity — too much money chasing too few goods	Economy booms; government cuts taxes → consumer spending surges
Cost-Push Inflation	Rising costs of production force firms to raise prices	Oil price spike → higher transport costs → higher prices across the economy

Consequences of Inflation

- Lower purchasing power — money buys less
- Decreased consumer confidence
- Demand increases before inflation kicks in (anticipation effect — people buy now before prices rise further)
- Less saving, more borrowing
- Beneficial for borrowers (debt is worth less in real terms) — bad for lenders and savers
- Less investment by firms as predicting future revenue becomes difficult
- Economy becomes less internationally competitive (exports become more expensive)
- Workers with lower bargaining power (blue-collar) suffer most — wages may not keep up with prices
- **Wage-price spiral** — workers demand higher wages to keep up with prices → firms raise prices further → a self-reinforcing cycle
- **Menu costs** — costs to constantly update prices in systems, menus, etc.
- **Shoe leather costs** — people must shift money to banks offering higher interest rates; firms spend more time managing money rather than producing goods; people keep less cash in hand (but this is contradictory to the spending-quickly behavior — that only applies to a few products; you can't buy a year's worth of groceries in a day, so you must keep your money where interest rates are higher!)

Causes of Deflation

Type	Nature	Explanation
Demand-side (malign)	Bad deflation	Caused by falling aggregate demand — consumers and firms spend less, leading to falling prices, lower profits, unemployment, and a deflationary spiral
Supply-side (benign)	Good deflation	Caused by improved productivity / lower production costs — prices fall because goods are cheaper to make, not because demand collapsed

Consequences of Deflation

Demand-side (Malign) Deflation

- Lower demand for products → lower investment → lower profits → unemployment → even lower demand → **deflationary spiral** (the cycle of lower demand → unemployment → further lower demand)
- Lower economic growth and decreasing GDP → potential recession
- Higher pressure of debts — the real value of debt rises
- Lenders and savers benefit; borrowers suffer
- Business and consumer confidence falls
- Lower tax revenue and more spending on welfare → increasing budget deficits
- **Menu costs** (NOTE: not a main point — businesses often just use discounts rather than changing prices repeatedly; also, no shoe leather costs here)

One benefit of deflation:

- Products become more internationally competitive as they are cheaper for foreign buyers

Economic Development

Indicators of Living Standards

There are two main ways to measure living standards: **Real GDP per capita** and the **Human Development Index (HDI)**.

Indicator	What it measures	Limitation
Real GDP per capita	Average income per person (real GDP ÷ population)	Doesn't account for inequality — average can be distorted by very rich or very poor; doesn't capture health or education
HDI (0 to 1)	Composite of income, education, and life expectancy	Doesn't capture inequality, political freedom, environmental quality, or happiness

HDI is made up of:

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- Mean years of education (past education of adults now) and expected years of education (future education prospects of children — just because their parents are uneducated doesn't mean children will be too)
- Life expectancy at birth
- Gross National Income (GNI = GDP + income from abroad) per capita

Differences in Living Standards Between Countries

- Political freedom and less corruption
- Healthcare quality
- Education
- Productivity (some countries have more technology and capital)
- Resource endowment (e.g. mineral resources)
- Infrastructure
- International trade — open economies doing international trade grow faster

Differences in Living Standards Within Countries

- Education and skills may be different and of better quality in certain parts of the country
- Some parts may have tertiary sector domination while others have primary sector domination (e.g. Mumbai vs Punjab)
- Inheritance and family background — wealthy families pass on advantages such as education and assets
- Ownership of wealth — people who own land, businesses, or shares earn income from assets; those without assets get less, increasing income inequality
- Infrastructure — some places may have better transport links, utilities, etc.
- Discrimination — gender, disability or ethnic discrimination can limit job opportunities

Poverty

Absolute poverty refers to the condition in which an individual does not have enough income to purchase basic human needs like food, water, shelter, clothing, etc.

Relative poverty refers to the condition in which an individual earns significantly lower than the average income in a country, and thus cannot afford what is considered the normal standard of living in that country.

Causes of Poverty:

- Education — people with low education earn less
- Illness — ailing people tend to not get jobs and fall into poverty
- Low wages — can lead to poverty even for employed people
- Low economic growth — more unemployment
- Unequal distribution of wealth (causes relative poverty)
- Age — older people often don't get many jobs

Policies to alleviate poverty and redistribute income:

- Progressive taxation — tax the rich more
- Welfare benefits — direct income support for low earners

- National minimum wage — ensures a floor on earnings
- Investment in education and healthcare — improves earning potential
- Government subsidies on essential goods — reduces cost of living
- Supply-side policies — improve productive capacity and employment

Population

Population size in a country change through three components:

$$\text{Population Change} = (\text{Births} - \text{Deaths}) + \text{Net Migration}$$

$$\text{Net Migration} = \text{Immigration} - \text{Emigration}$$

Reasons for Different Rates of Population Growth Between Countries

Factor	How it affects population growth
Education	Contraception knowledge; more women in labor force → lower birth rates
Health / Healthcare	Better healthcare → lower death rates → population grows faster
Income levels	Higher income attracts immigrants; lower income may push people to emigrate
Standards of living	Better standards of living attract more immigration
Wars, famine, political instability	Drive emigration and increase death rates, reducing population

Optimum Population

Optimum population is the population that allows a country to achieve the highest possible GDP per person, given its available resources and level of technology.

Overpopulation (above optimum)	Underpopulation (below optimum)
More strain on scarce resources	Resources not being utilized efficiently
Congestion and pressure on infrastructure	Reduced GDP as labor Factor of Production decreases → slower economic growth
If dependency ratio rises → more government funds allocated to welfare	Smaller tax revenue → less public spending
Can lead to demand-pull inflation	Less pressure on resources is an advantage

Dependency ratio = $(\text{Non-working population} \div \text{Working population}) \times 100$. A higher dependency ratio means more strain on government welfare spending.

Differences in Economic Development Between Countries

Factor	Explanation
Factor Endowment	Countries with more natural resources, skilled labor, and capital have higher productive capacity
Productivity and Technology	More advanced technology → more output from same Factor of Production → higher GDP
Healthcare	Healthier workforce → more productive → faster growth
Education	Better educated workers → higher productivity and innovation
Sector Dominance	Countries dominated by tertiary sector tend to have higher GDP than those dominated by primary sector
FDI and Investment	Foreign direct investment brings capital, technology, and jobs
Infrastructure	Better roads, ports, and utilities lower costs and attract investment
Growing Population	More labor supply (if productive) increases output and GDP

International Specialization

Why Do Countries Specialize?

There are two main reasons why countries specialize in producing certain goods:

Basis	What it means	Example
Comparative Advantage	A country should specialize in the good where it has a lower opportunity cost than other countries — even if another country is better at producing everything, trade is still mutually beneficial	Country A is more efficient at producing electronics than clothing, so it specializes in electronics and imports clothing from Country B which has a comparative advantage in textiles
Absolute Advantage	A country can produce more of a good using the same amount of resources — due to better technology, infrastructure, more skilled labor, or lower wage costs (note: lower wage costs here means workers produce more output per labor hour — same Factors of Production, more output)	Country A produces 10 t-shirts per worker per day; Country B produces only 5. Country A has an absolute advantage in t-shirts

You may wonder why lower wage costs matter if they're still using the same workers. Money isn't a factor of production, so utilizing less of it doesn't factor in anything in the traditional sense. When they say cheaper

production methods, they don't mean cheaper factors of production. They mean the production process costs less because workers are more efficient — in the same labor hours (1 day of salary), they create 10 t-shirts in Country A instead of 5 in Country B. Same Factors of Production, more output. That's absolute advantage.

Advantages of International Specialization

- More efficient allocation of scarce resources — less waste in the economy
- Higher productivity of workers as they focus on a narrow task — the economy's productive capacity grows as it focuses on one industry, leading to lower average costs
- Better quality of goods and services
- Encourages international trade as they become internationally competitive → decreased current account deficit
- Economies of scale as the industry grows

Disadvantages of International Specialization

- One industry becomes more developed than others → structural unemployment in other sectors
- Overspecialization
- Overdependence on other countries for products — risky during times of war and political instability
- Some areas of the country may be able to specialize, others may not — leading to them not benefiting from economies of scale, facing fierce competition from firms abroad, and small firms shutting down
- Specialization is risky as international prices may be unstable — reducing investment as firms can't plan revenue with unstable costs
- Diseconomies of scale might set in
- Trade restrictions like embargos and tariffs may be imposed on them
- They may face fluctuating demand and rise in competitors

Advantages and Disadvantages of National Specialization

	Advantages	Disadvantages
For Consumers	Lower prices due to higher efficiency and economies of scale; better quality goods due to skilled production	Dependence on imports — risk of shortages if trade is disrupted; prices may rise if world prices increase; lower choice of domestic goods
For Firms	Higher productivity and efficiency; lower average costs; access to larger international markets; higher profits and competitiveness	Over-dependence on a single product or industry; vulnerable to changes in global demand or prices; risk of closures and unemployment if industry declines
For the Economy	More efficient allocation of resources; higher GDP and economic growth; encourages international trade; improves living standards	Greater exposure to external shocks (global recessions, trade restrictions); risk of balance of payments problems if export prices fall; regional inequality; less economic risk diversification

Globalization, Free Trade, and Protectionism

Globalization is the growing economic interdependence and interconnectedness of countries through trade, investment, and movement of labor and information.

Causes and consequences of changes in globalization

Causes of changes	Consequences of Changes
Changes in trade restrictions: When governments reduce tariffs, quotas, and other barriers (through agreements like the WTO or regional trade blocs), it becomes cheaper and easier for goods and services to cross borders, so the volume of international trade rises and countries become more economically integrated. Conversely, rising protectionism — tariffs, sanctions, trade wars — reverses this, shrinking global economic interdependence.	International trade: Greater globalisation increases the volume and variety of goods and services traded across borders. Countries specialise more deeply in their comparative advantages, leading to lower prices for consumers. However, domestic industries exposed to cheaper foreign competition may decline, and countries can become heavily dependent on trading partners.
Changes in transport costs: Containerisation, larger cargo ships, and more efficient logistics have dramatically cut the cost of moving goods across the world, making it viable to source components from one continent and sell the finished product on another. If fuel costs rise sharply or new environmental regulations increase shipping costs, globalization slows.	Competition: Globalisation exposes domestic firms to international competition, which pressures them to become more efficient, innovate, and reduce costs — benefiting consumers through lower prices and better quality. The downside is that smaller or less efficient domestic firms may be unable to compete and shut down, causing unemployment.
Changes in communication costs: The internet, cheap international calls, and digital platforms have made it almost costless to coordinate business across borders — a firm in the UK can manage a supply chain in Vietnam in real time. This has especially accelerated trade in services (software, finance, customer support). Higher communication costs or internet restrictions work in the opposite direction.	The environment: Globalisation has mixed environmental effects. More trade means more shipping and air freight, increasing carbon emissions and pollution. MNCs sometimes relocate to countries with weaker environmental regulations (a "pollution haven"), worsening local environmental damage. On the other hand, globalisation can spread cleaner technologies and international environmental agreements become easier to coordinate.
Movement of multinational companies (MNCs): When MNCs expand into new countries — building factories, opening offices, acquiring local firms — they bring capital, technology, and trade flows with them, deepening global integration. MNC expansion is both a cause and a consequence of globalisation; as it becomes easier to operate internationally, more firms do so, which further integrates economies.	Migration: Greater economic integration encourages the movement of workers across borders. Workers move from lower-wage to higher-wage countries, which can ease labour shortages in receiving countries and send remittances back to home countries. However, receiving countries may face pressure on public services, housing, and wages for low-skilled workers, while sending countries risk a "brain drain" — losing their most skilled workers.

	Income distribution: Globalisation tends to raise average incomes by growing GDP, but its benefits are unevenly distributed. Skilled workers and capital owners in both developed and developing countries often gain the most, while unskilled workers in developed countries may see wages squeezed by competition from cheaper labour abroad. Within developing countries, those regions connected to global supply chains benefit more than rural or isolated areas, widening internal inequality.
	Economic development: For developing countries, globalisation can accelerate development by attracting FDI, creating jobs, transferring technology and skills, and expanding export markets. This raises GDP, living standards, and HDI. However, over-dependence on a single export industry makes economies vulnerable to global price shocks, and profit repatriation by MNCs means not all gains stay in the host country. Some argue that developing countries remain trapped in low-value primary sector exports while developed countries capture the higher-value stages of production.

Effects of Globalization on MNCs

	Benefits	Costs
Host Country (where MNC operates)	Employment creation — new jobs for local workers; brings capital, skills, and advanced technology; infrastructure development (roads, ports, factories); boosts foreign exchange earnings from exports	Profit repatriation — profits often go back to the home country, not the local economy; local firms may struggle to compete; environmental concerns — some MNCs exploit natural resources or pollute
Home Country (where MNC is based)	Home country owners earn profits and returns from foreign operations; easier access to raw materials and markets; jobs in headquarters and related industries	Job outsourcing — jobs may move abroad, causing unemployment at home; economic dependence — risk if too reliant on foreign operations

Benefits of Free Trade

For Consumers	For Producers	For the Economy
Lower prices due to increased competition and lower production costs	Larger markets and increased sales	Higher economic growth and GDP
Greater choice of goods and services	Economies of scale → lower average costs	Encourages Specialization and more efficiency
Higher quality products as firms compete	Increased profits and efficiency	Increased employment in export industries
Access to goods not produced domestically	Access to cheaper raw materials and components	Improved international relations and cooperation
More responsiveness to consumer demand		Higher living standards

Note: free trade **encourages** Specialization, not the other way around.

Protectionism

Protectionism is when a government uses policies to restrict imports and protect domestic industries from foreign competition.

Type of Protection	What it is	Who gets the money
Tariff	A tax on imported goods, raising their price	Goes to the government
Embargo	A complete ban on imports from a specific country	N/A
Subsidy	Government payment to domestic firms to lower their costs and improve competitiveness	Goes to domestic businesses
Import Quota	A quantitative limit on how much of a good can be imported	N/A — limits quantity, not price

Reasons for protectionism:

- Saving infant industries (new industries that need time to develop), strategic industries (food, water, communications, national defense, energy), and declining industries
- Reducing current account deficit
- To increase demand for domestic goods and protect jobs
- Increased government revenue (from tariffs)
- Avoidance of dumping — when foreign firms sell goods below cost price to undercut domestic firms

Consequences of protectionism:

- Reduced choices and lower living standards for consumers
- Higher prices as supply falls — less affordability and thus more poverty
- Retaliatory measures by other governments — trade wars
- Makes domestic firms more inefficient as they slack off with less competition → slower economic growth
- May increase prices further as cost of production rises due to higher prices of imported resources
- MNCs may relocate to countries with fewer restrictions
- Loss of revenue for trading partners and unemployment in those countries — may even increase their current account deficit if no retaliatory measures are taken

Foreign Exchange Rates

A **foreign exchange rate** is the price of one currency in terms of another currency.

If £1 = \$1.25 → the exchange rate of GBP against USD is 1.25

Appreciation = currency becomes worth more | **Depreciation** = currency becomes worth less

Causes of changes in the foreign exchange rate:

- **Speculation** — traders buying/selling currencies based on expected future movements
- **Entry/departure of MNCs** — MNCs converting currency when investing or withdrawing
- **Changes in demand for imports and exports** — more exports → more demand for domestic currency → appreciation
- **Interest rates** — higher interest rates attract foreign investment → more demand for domestic currency → appreciation

Depreciation and **appreciation** happen in **floating** exchange rate economies. **Revaluation** and **devaluation** happen in **fixed** exchange rate economies.

Fixed vs Floating Exchange Rate Systems

	Fixed Exchange Rate	Floating Exchange Rate
Definition	Currency value is set and maintained by the government or central bank — controlled by managing supply and demand of the currency	Currency value is determined entirely by market forces of demand and supply
Advantage 1	Reduces speculation — currency remains pegged at a certain value and doesn't fluctuate with short-term speculation	Imbalances in the balance of payments get fixed automatically — no need to deplete foreign reserves
Advantage 2	Certainty for businesses as currency remains stable — encourages more investment	No need to hold large amounts of foreign reserves — they can be invested instead
Advantage 3	Constrains irresponsible government policy — can't decrease interest rates excessively or create money recklessly as this would deplete foreign reserves	Smaller impact of speculation as there's no fixed peg to defend — less foreign reserve expenditure
Advantage 4	Can maintain competitiveness by devaluing the currency	Monetary policy independence — can adjust interest rates freely to meet domestic goals
Disadvantage 1	Holding large foreign reserves has a big opportunity cost — those reserves could be invested	Uncertainty for businesses — volatile exchange rates make planning difficult
Disadvantage 2	Constraints on monetary policy — can't raise interest rates freely as this attracts FDI, destabilizing the exchange rate and hindering economic growth policies	Speculation can lead to volatile and unpredictable exchange rates
Disadvantage 3	Speculation still exists on whether the government will revalue or devalue — may cost the government a lot in foreign reserves	Less FDI and investment due to currency uncertainty
Disadvantage 4	Balance of payments crises can make it very hard to maintain the peg — no automatic adjustment mechanism exists	Imported inflation if the currency depreciates significantly

Effects of Exchange Rate Changes

	Depreciation / Devaluation (currency weakens)	Appreciation / Revaluation (currency strengthens)
Exports	Cheaper for foreign buyers → export demand rises → current account improves	More expensive for foreign buyers → export demand falls → current account worsens
Imports	More expensive for domestic buyers → import demand falls → reduces deficit	Cheaper for domestic buyers → import demand rises → deficit may worsen
Inflation	Cost-push inflation — imported raw materials and goods cost more	Lower import prices → reduces inflation
Economic growth	Boost from higher net exports → more demand → higher GDP	May reduce GDP if exports fall significantly
Living standards	Imported goods become pricier — consumers feel a squeeze	Consumers benefit from cheaper imports and foreign travel

Current Account of the Balance of Payments

The **current account** measures a country's net trade and income flows with the rest of the world.

Component	What it includes
Balance of Trade in Goods (Visible)	Exports and imports of physical goods — cars, food, oil, machinery
Balance of Trade in Services (Invisible)	Exports and imports of services — banking, tourism, insurance, education
Primary Income	Net flow of income earned by residents from foreign investments, and income paid to foreign investors (e.g. dividends, profits, interest on loans)
Secondary Income	Transfers like foreign aid, remittances (money sent home by workers abroad), donations

$$\text{Current Account Balance} = \text{Trade in Goods} + \text{Trade in Services} + \text{Primary Income} + \text{Secondary Income}$$

$$\text{Surplus} = \text{Receipts} > \text{Payments} \quad | \quad \text{Deficit} = \text{Payments} > \text{Receipts}$$

Reasons for a Current Account Deficit

- Decrease in demand for exports (e.g. trading partners in recession)
- Lower international competitiveness — domestic goods too expensive or poor quality
- Strong currency (appreciation) — makes exports more expensive and imports cheaper
- Profit repatriation by MNCs → primary income outflow

Reasons for a Current Account Surplus

- Weak domestic currency — makes exports cheaper and imports expensive
- Trade policies like subsidies and tariffs reduce imports, increasing surplus
- High export demand from strong international competitiveness
- High income from foreign investments (primary income inflow)
- Lower inflation than trading partners — products more competitively priced

Consequences of Surplus and Deficit

Consequence	Current Account Deficit	Current Account Surplus
GDP	Reduces GDP as net exports (X-M) is negative — less domestic demand	Increases GDP as net exports contribute positively
Employment	Decreases as less domestic demand → firms produce less → fewer jobs	Increases as more demand for domestically produced exports
Inflation	Lower demand → less demand-pull inflation → deflationary pressure	More demand → demand-pull inflation may occur; but a later currency appreciation may reduce it
Exchange Rate	Decreases (depreciates) as less demand for domestic currency	Increases (appreciates) as more demand for domestic currency

Important Clarifications

FDI vs Portfolio Investment: FDI is investment in building factories and buying controlling stakes in companies in other countries. Investment from other countries in a host country's banks purely for profit is NOT FDI — that is **portfolio investment**.

Why capital reward = interest: You deposit money → bank lends it to firms → firms invest in machines → they generate productive output → the return from that activity flows back as interest on your deposit. Your interest is indirectly a return from productive activity — that's kind of how capital reward = interest.

job) depending on transport facilities and costs, housing facilities and costs, family and personal priorities, regional or national laws and regulations on travel and work etc.

- **Capital:** all the **man-made resources** available in an economy. All man-made goods (which help to produce other goods – capital goods) from a simple spade to a complex car assembly plant are included in this. Capital is usually denoted in monetary terms as the total value of all the capital goods needed in production.
 - The **reward** for capital is the **interest** it receives.
 - The supply of capital depends upon the demand for goods and services, how well businesses are doing, and savings in the economy (since capital for investment is financed by loans from banks which are sourced from savings).
 - The **quality** of capital depends on how many good quality products can be produced using the given capital. For example, the capital is said to be of much more quality in a car manufacturing plant that uses mechanisation and technology to produce cars rather than one in which manual labour does the work.
 - Capital **mobility** can depend upon the nature and use of the capital. For example, an office building is geographically immobile but occupationally mobile. On the other hand, a pen is geographically and occupationally mobile.
 - **Enterprise:** the **ability to take risks and run a business venture** or a firm is called enterprise. A person who has enterprise is called an entrepreneur. In short, they are the people who start a business. Entrepreneurs organize all the other factors of production and take the risks and decisions necessary to make a firm run successfully.
 - The **reward** to enterprise is the **profit** generated from the business.
 - The **supply** of enterprise is dependent on entrepreneurial skills (risk-taking, innovation, effective communication etc.), education, corporate taxes (if taxes on profits are too high, nobody will want to start a business), regulations in doing business and so on.
 - The quality of enterprise will depend on how well it is able to satisfy and expand demand in the economy in cost-effective and innovative ways.
 - Enterprise is usually highly mobile, both geographically and occupationally.
- All the above factors of productions are scarce because the time people have to spend working, the different skills they have, the land on which firms operate, the natural resources they use etc. are all in limited in supply; which brings us to the topic of opportunity cost.

Opportunity Cost

The scarcity of resources means that there are not sufficient goods and services to satisfy all our needs and wants; we are forced to choose some over the others. Choice is necessary because these resources have alternative uses- they can be used to produce many things. But since there are only a finite number of resources, we have to choose.

When we choose something over the other, the choice that was given up is called the opportunity cost. **Opportunity cost**, by definition, **is the next best alternative that is sacrificed/forgone in order to satisfy the other.**

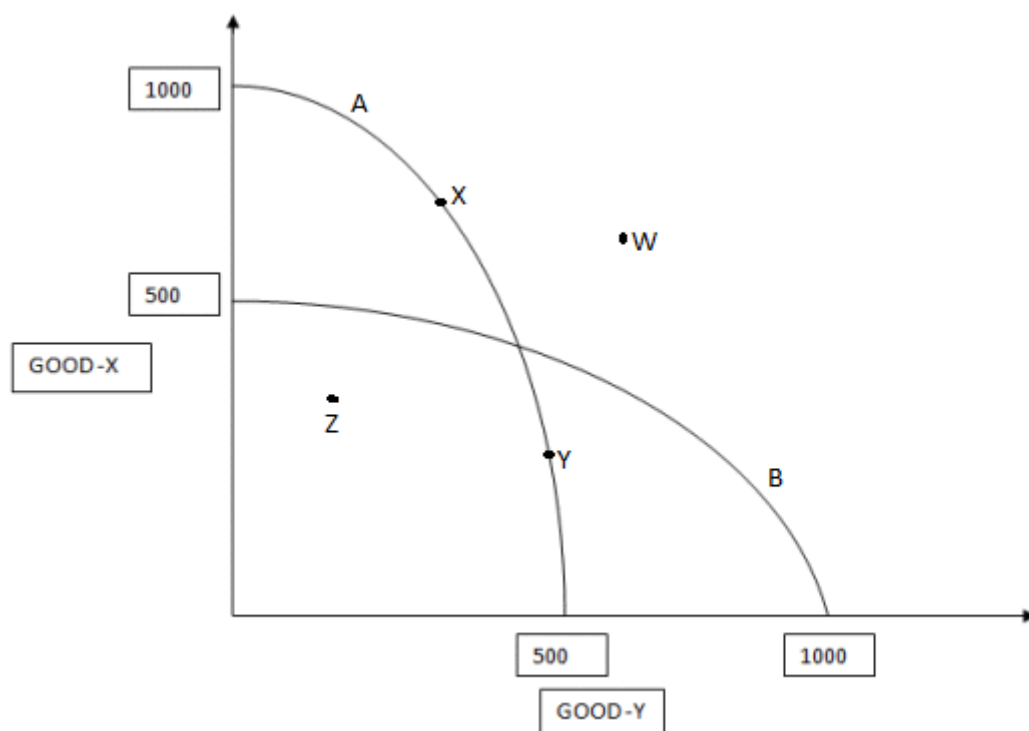
Example 1: the government has a certain amount of money and it has two options: to build a school or a hospital, with that money. The govt. decides to build the hospital.

The school, then, becomes the opportunity cost as it was given up. In a wider perspective, the opportunity cost is the education the children could have received, as it is the actual cost to the economy of giving up the school.

Example 2: you have to decide whether to stay up and study or go to bed and not study. If you chose to go to bed, the knowledge and preparation you could have gained by choosing to stay up and study is the opportunity cost.

Production Possibility Curve Diagrams (PPC)

Because resources are scarce and have alternative uses, a decision to devote more resources to producing one product means fewer resources are available to produce other goods. A Production Possibility Curve diagram shows this, that is, **the maximum combination of two goods that can be produced by an economy with all the available resources.**



The PPC diagram above shows the production capacities of two goods- X and Y- against each other. When 500 units of good X are produced, 1000 units of good Y can be produced. But when the units of good X increases to 1000, only 500 units good Y can be produced.

Let's look at the PPC named A. At point X and Y it can produce certain combinations of good X and good Y. These are points on the curve- they are attainable, given the resources. The economy can move between points on a PPC simply by reallocating resources between the two goods.

If the economy were producing at point Z, which is inside/below the PPC, the economy is said to be **inefficient**, because it is producing less than what it can. Point W, outside/above the PPC, is unattainable because it is beyond the scope of the economy's existing resources. In order to produce at point W, the economy would need to see a shift in the PPC towards the right.

For an **outward shift** to occur, an economy would need to:

- discover or develop new raw materials. Example: discover new oil fields
- employ new technology and production methods to increase productivity
- increase labour force by encouraging birth and immigration, increasing retirement age etc.

An outward shift in PPC, that is higher production possibility, will lead to economic growth.

In the same way, an **inward shift** can occur in the PPC due to:

- natural disasters, that erode infrastructure and kill the population
- very low investment in new technologies will cause productivity to fall over time
- running out of resources, especially non-renewable ones like oil or water

An inward shift in the PPC will lead to the economy shrinking.

How is opportunity cost linked to PPC?

Individuals, businessmen and the government can calculate the **opportunity cost** from PPC diagrams. In the above example, if the firm decided to increase production of good Y from 500 to 750, it can calculate the opportunity cost of the decision to be 250 units of good X (as production of good X falls from 1000 to 750). They are able to compare the opportunity cost for different decisions.

Economy: an area where people and firms produce, trade and consume goods and services. This can vary in size- from your local town to your country, or the globe itself.

Microeconomics and Macroeconomics

Microeconomics is the study of individual markets. For example: studying the effect of a price change on the demand for a good. Microeconomic decision makers are producers and consumers (who directly operate in markets)

Macroeconomics is the study of an entire economy, as a whole. Examples include studying the total size of the economy or the unemployment rate, among other things. Macroeconomic decisions are made by the government of the particular economy – a town, state or country)

The Role of Markets in Allocating Resources

Resource allocation: the way in which economies decide what goods and services to provide, how to produce them and who to produce them for.

These questions- **what to produce, how to produce, and for whom to produce** – are termed '**the basic economic questions**'. In short, resource allocation is the way in which economies solve the three basic economics questions.

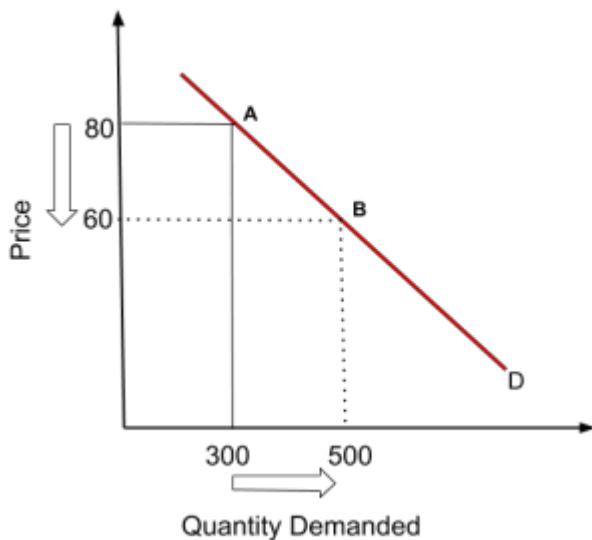
Market is any set of arrangement that brings together all the producers and consumers of a good or service, so they may engage in exchange. Example: a market for soft drinks.

Goods and services are bought and sold in a market at an equilibrium price where demand and supply are equal. This is called the price mechanism. It helps answer the three basic economic questions. Producers will produce the good that consumers demand the most, it will be produced in a way that is cost-efficient, and will be produced for those who are willing and able to buy the product. More on these topics below:

Demand

Demand is the want and willingness of consumers to buy a good or services at a given price. Effective demand is where the willingness to buy is backed by the ability to pay. For example, when you want a laptop but you don't have the money, it is called demand. When you do have the money to buy it, it is called effective demand. The effective demand for a particular good or service is called **quantity demanded**. (**Individual demand** is the demand from one consumer, while **market demand** for a product is the total (aggregate) demand for the product, or the sum of all individual demands of consumers).

The law of demand states that an increase in price leads to a decrease in demand, and a decrease in price leads to an increase in demand (it's an inverse relationship between price and demand. However it's worth noting that an increase in demand leads to an increase in price and a decrease in demand leads to a decrease in price. The law of demand is established with respect to changes in price, not demand, hence the difference).



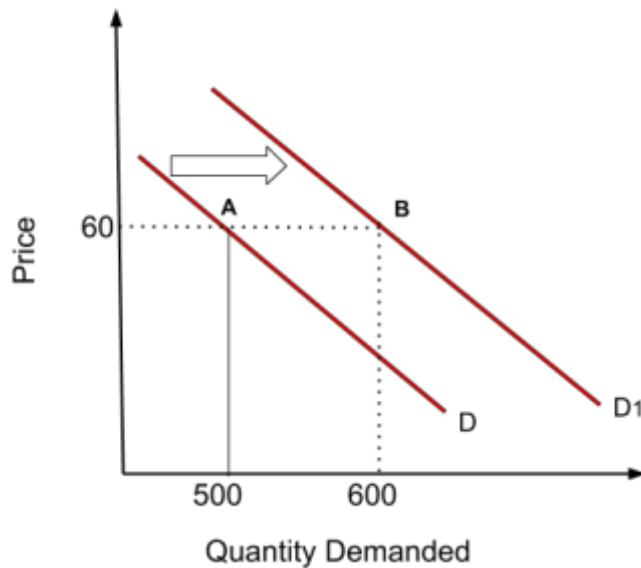
Coca-Cola.

Here, a decrease in price from 80 to 60 has increased its demand from 300 to 500.

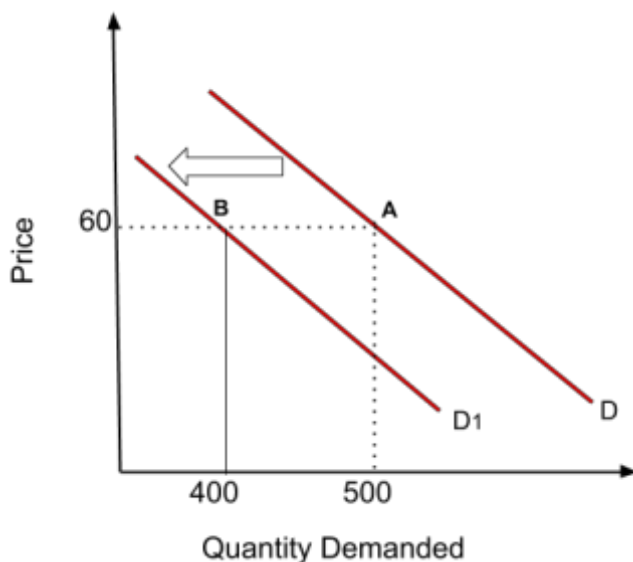
The increase in demand due to changes in price (without changes in other factors) **is called an extension in demand**. Here the extension in demand is from A to B.

In the above example, an increase in price from 60 to 80, will decreased the demand from 500 to 300. **The decrease in demand due to the changes in price** (without changes in other factors) **is called a contraction in demand**. Here the contraction in demand will be from B to A.

This is an example of a demand curve for



In this example, there is a rise in the demand of Coca-Cola from 500 to 600, without any change in price. **A rise in the demand for a product due to the changes in other factors (excluding price) causes the demand curve to shift to the right (from A to B).**



In this example, there is a fall in demand of Coca-Cola from 500 to 400, without any change in price.

A fall in demand for a product due to the changes in other factors (excluding price) causes the demand curve to shift to the left (from A to B).

Factors that cause shifts in a demand curve:

- **Consumer incomes:** a rise in consumers' incomes increases demand, causing a shift to right. Similarly, a fall in incomes will shift the demand curve to the left.
- **Taxes on incomes:** a rise in tax on incomes means less demand, causing a shift to the left; and vice versa.
- **Price of substitutes:** **Substitutes** are goods that can be used instead of a particular product. Example: tea and coffee are substitutes (they are used for similar purposes). A rise in the price of a substitute causes a rise in the demand for the product, causing the demand curve to shift to the right; and vice versa.
- **Price of complements:** **Complements** are goods that are used along with another product. For example, printers and ink cartridges are complements. A rise in the price of a complementary

good will reduce the demand for the particular product, causing the demand curve to shift to the left; and vice versa.

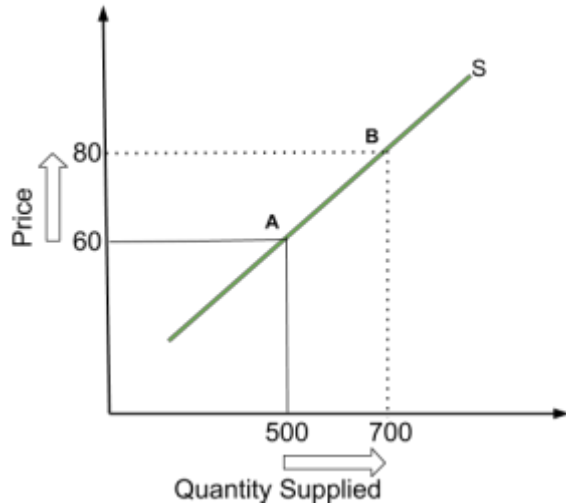
- **Changes in consumer tastes and fashion:** for example, the demand for DVDs have fallen since the advent of streaming services like Netflix, which has caused the demand curve for DVDs to shift to the left.
- **Degree of Advertising:** when a good is very effectively advertised (Coke and Pepsi are good examples), its demand rises, causing a shift to the right. Lower advertising shifts the demand curve to the left.
- **Change in population:** A rise in the population will raise demand, and vice versa.
- **Other factors,** such as weather, natural disasters, laws, interest rates etc. can also shift the demand curve.

Supply

Supply is the want and willingness of producers to supply a good or services at a given price. The amount of goods or services producers are willing to make and supply is called quantity supplied.

(Market supply refers to the amount of goods and services all producers supplying that particular product are willing to supply or the sum of individual supplies of all producers).

The law of supply states that **an increase in price leads to a increase in supply, and a decrease in price leads to an decrease in supply** (there is a positive relationship between price and supply. However it's also worth noting that, an increase in supply leads to a decrease in price and a decrease in supply leads to an increase in price. The law of supply is established with respect to changes in price, not supply, hence the difference).



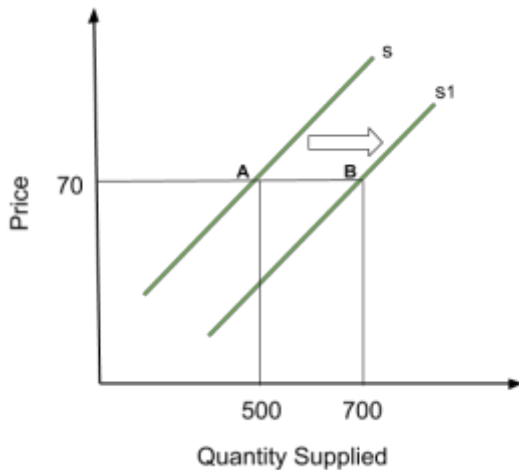
product.

Here, an increase in price from 60 to 80, has increased its supply from 500 to 700.

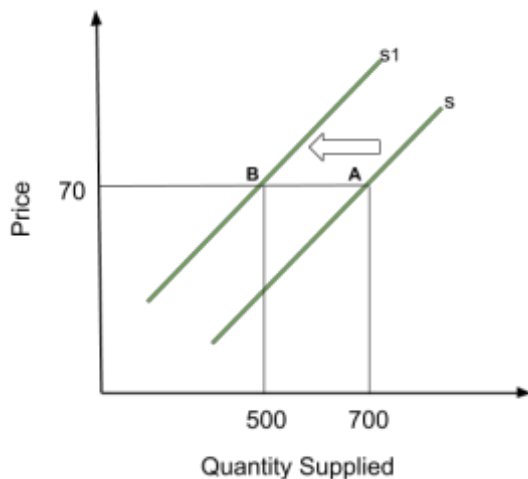
The increase in supply due to changes in price (without changes in other factors) is called an **extension in supply**.

A decrease in price from 80 to 60, will decreased the supply from 700 to 500. **The decrease in supply due to changes in price** (without the changes in other factors) is called a **contraction in supply**.

This is an example of a supply curve for a



In this example, there is a rise in the supply of a product from 500 to 700, without any change in price. **A rise in the supply for a product due to the changes in other factors (excluding price) causes a shift to the right.**

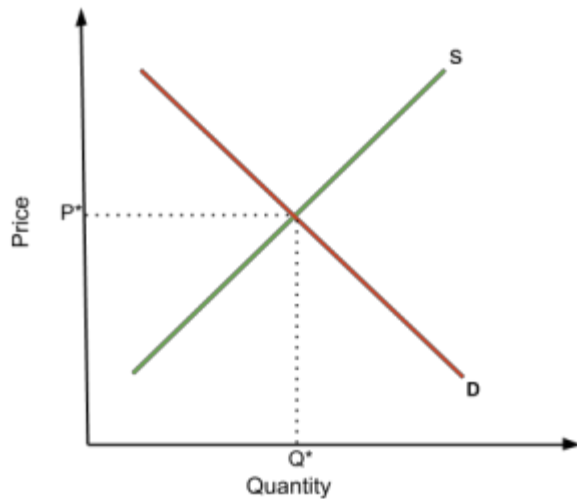


A fall in supply from 500 to 300, without any changes in price is also shown. **A fall in the supply for a product due to the changes in other factors (excluding price) causes a shift to the left.**

Factors that cause shifts in supply curve:

- **Changes in cost of production:** when the cost of factors to produce the good falls, producers can produce and supply more products cheaply, causing a shift in the supply curve to the right. A subsidy*, which lowers the cost of production also shifts the supply curve right. When cost of production rises, supply falls, causing the supply curve to shift to the left.
- **Changes in the quantity of resources available:** when the amount of resources available rises, the supply rises; and vice versa.
- **Technological changes:** an introduction of new technology will increase the ability to produce more products, causing a shift to the right in the supply curve.
- **The profitability of other products:** if a certain product is seen to be more profitable than the one currently being produced, producers might shift to producing the more profitable product, reducing supply of the initial product (causing a shift to the left).
- Other factors: weather, natural disasters, wars etc. can shift the supply curve left.

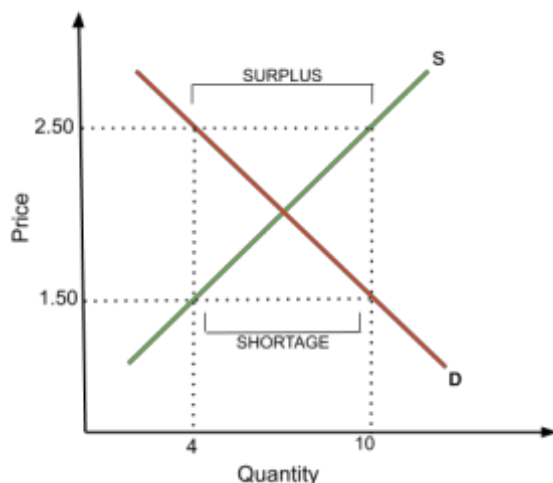
Market Price



The market **equilibrium price** is the price at which the demand and supply curves in a given market meet. In this diagram, P^* is the equilibrium price.

Disequilibrium price is the price at which market demand and supply curves do not meet, which in this diagram, is any price other than P^* .

Price Changes



In this diagram, two disequilibrium prices are marked- 2.50 and 1.50.

At price 2.50, the demand is 4 while the supply is 10. There is excess supply relative to the demand. **When the price is above the equilibrium price, a surplus is experienced.** (Surplus means 'excess').

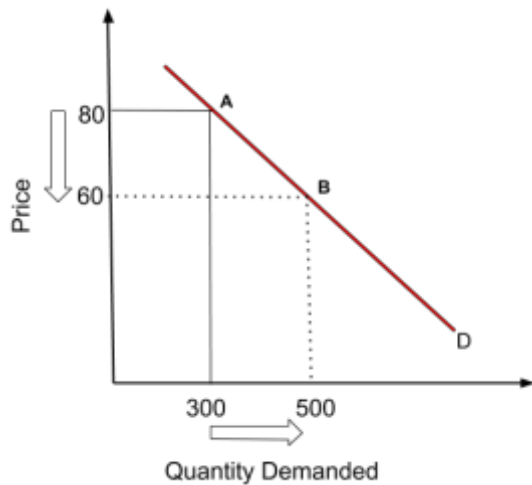
At price 1.50, the demand is 10 while the supply is only 4. There is excess demand relative to supply. **When the price is below the equilibrium price, a shortage is experienced.**

(This shortage and surplus is said in terms of the supply being short or excess respectively).

Price Elasticity of Demand (PED)

The PED of a product refers to **the responsiveness of the quantity demanded to changes in its price.**

PED (of a product) = % change in quantity demanded / % change in price

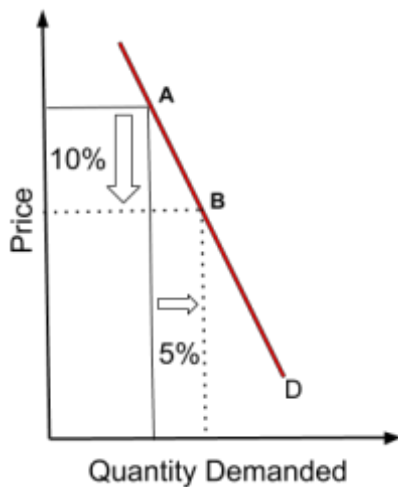


For example, calculate the price elasticity of demand of Coca-Cola from this diagram.

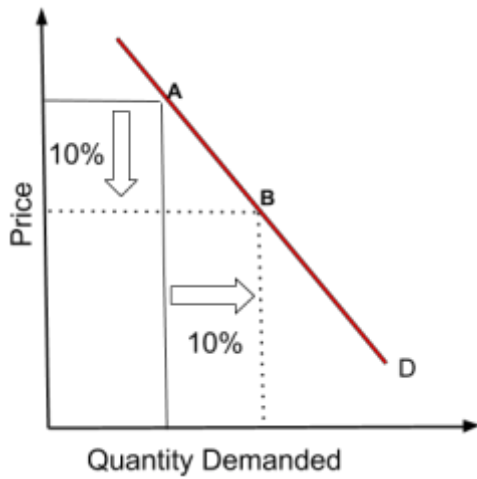
$$PED = [(500-300/300)*100] / [(80-60/80)*100]$$

$$= 66.67 / 25 = 2.67$$

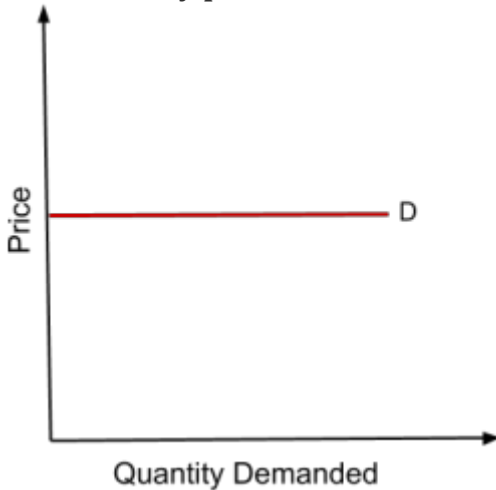
In this example, the PED is 2.67, that is, the % change in quantity demanded was higher than the % change in the price. This means, **a change in price makes a higher change in quantity demanded**. These products have a **price elastic demand**. Their values are always above 1.



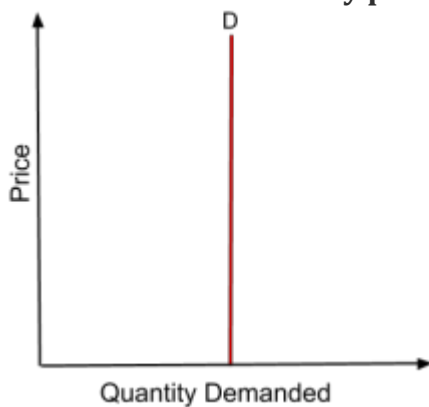
When the % change in quantity demanded is lesser than the % change in price, it is said to have a price **inelastic demand**. Their values are always below 1. **A change in price makes a smaller change in demand.**



When the % change in demand and price are equal, that is value is 1, it is called **unitary price elastic demand**.



When the quantity demanded changes without any changes in price itself, it is said to have an **infinitely price elastic demand**. Their values are infinite.



When the price changes have no effect on demand whatsoever, it is said to have a **perfect price inelastic demand**. Their elasticity is 0.
What affects PED?

- **No. of substitutes:** if a product has many substitute products it will have an elastic demand. For example, Coca-Cola has many substitutes such as Pepsi and Mountain Dew. Thus a change in price will have a greater effect on its demand (If price rises, consumers will quickly move to the substitutes and if price lowers, more consumers will buy Coca-Cola).
- **Time period:** demand for a product is more likely to be elastic in the long run. For example, if the price rises, consumers will search for cheaper substitutes. The longer they have, the more likely they are to find one.
- **Proportion of income spend on commodity:** goods such as rice, water (necessities) will have an inelastic demand as a change in price won't have any significant effect on its demand, as it will only take up a very small proportion of their income. Luxury goods such as cars on the other hand, will have a high price elastic demand as it takes up a huge proportion of consumers' incomes.

Relationship between PED and revenue and how it is helpful to producers:

Producers can calculate the PED of their product and take a suitable action to make the product more profitable.

Revenue is the amount of money a producer/firm generates from sales, i.e., the total number of units sold multiplied by the price per unit. So, as the price or the quantity sold changes, those changes have a direct effect on revenue.

If the product is found to **have an elastic demand, the producer can lower prices to increase revenue**. The law of demand states that a price fall increases the demand. And since it is an elastic product (change in demand is higher than change in price), the demand of the product will increase highly. The producers get more revenue. If the product is found to **have an inelastic demand, the producer can raise prices to increase revenue**. Since quantity demanded wouldn't fall much as it is inelastic, the high prices will make way for higher revenue and thus higher profits.

Price Elasticity of Supply (PES)

The PES of a product refers to **the responsiveness of its quantity supplied to changes in its price**.

PES of a product = %change in quantity supplied / %change in price

Similar to PED, PES too can be categorized into price elastic supply, price inelastic supply, perfectly price inelastic supply, infinitely price elastic supply and unitary price elastic supply. (See if you can figure out what each supply elasticity means using the demand elasticities above as reference, and draw the diagrams as well!)

What affects PES?

- **Time of production:** If the product can be quickly produced, it will have a price elastic supply as the product can be quickly supplied at any price. For example, juice at a restaurant. But products which take a longer time to produce, such as cars, will have a price inelastic supply as it will take a longer time for supply to adjust to price.
- **Availability of resources:** More resource (land, labour, capital) will make way for an elastic supply. If there are not enough resources, producers will find it difficult to adjust to the price changes, and supply will become price inelastic.

Market Economic System

In a market economic system or free market economic system, **all resources are allocated by the market – private producers and consumers**; that is, there is no or very little government intervention in resource allocation. (There are virtually no economies in the world that follow this system – there is a government control everywhere, although Hong Kong and Singapore do come close – check out the [Index of Economic Freedom](#) to see the ranking of economies on the basis of how market-friendly they are).

Features:

- All resources are owned and allocated by private individuals.
- Government refrains from regulating markets. It instead tries to create very business-friendly environments and any intervention is mostly limited to protecting private property. The demand and supply fixes the price of products. This is called **price mechanism**.
- What to produce is solved by **producing the most-demanded goods** for which people spend a lot, as their only motive is to generate a high profit.
- How to produce is solved by **using the cheapest yet efficient combination of resources** – capital or labour- in order to maximise profits.
- For whom to produce is solved by **producing for people who are willing and able to pay for goods** at a high price.

Advantages:

- A **wide variety of quality goods and services** will be produced as different firms will compete to satisfy consumer wants and make profits. Quality is ensured to make sure that consumers buy from them. There is consumer sovereignty.
- Firms will **respond quickly to consumer changes in demand**. When there is a change in demand, they will quickly allocate resources to satisfying that demand, so as to maintain profits.
- **High efficiency** will exist. Since producers want to maximise profits, they will use resources very efficiently (producing more with less resources).
- Since there is hardly any government intervention (in the form of regulations, extra fees and fines etc. for example), firms will find it easy and inexpensive to start and run businesses.

Disadvantages:

- Only profitable goods and services are produced. **Public goods*** and some **merit goods*** for which there is no demand **may not be produced**, which is a drawback and affects the economic development.
- Firms will **only produce for consumers who can pay for them**. Poor people who cannot spend much won't be produced for, as it would be non-profitable.
- **Only profitable resources will be employed**. Some resources will be left unused. In a market economy, capital-intensive production is favoured over labour- intensive production (because it's more cost-efficient). This can lead to unemployment.
- Harmful (**demerit**) **goods may be produced** if it is profitable to do so.
- Negative impacts on society (**externalities**) may be ignored by producers, as their sole motive is to keep consumers satisfied and generate a high profit.
- A firm that is able to dominate or control the market supply of a product is called a **monopoly**. They may use their power to restrict supply from other producers, and even charge consumers a high price since they are the only producer of the product and consumers have no choice but to buy from them.
- Due to high competition between firms, **duplication of products** may take place, which is a waste of resources.

***Public goods:** goods that can be used by the general public, from which they will benefit. Their consumption can't be measured, and thus cannot be charged a price for (this is why a market economy doesn't produce them). Examples are street lights and roads.

***Merit goods:** goods which create a positive effect on the community and ought to be consumed more. Examples are schools, hospitals, food. The opposite is called demerit goods which includes alcohol and cigarettes

***Subsidies:** financial grants made to firms to lower their cost of production in order to lower prices for their products.

Before we dive into what market failure is, let's get familiar with some terms related to market failure:

Public goods: goods that can be used by the general public, from which they will benefit. Their consumption can't be measured, and thus cannot be charged a price for (this is why a market economy doesn't produce them). Examples include street lights and roads.

Merit goods: goods which create a positive effect on the society and ought to be consumed more. Examples include schools and hospitals. The opposite is called demerit goods which include alcohol and cigarettes.

External costs (negative externalities) are the negative impacts on society (third-parties) due to production or consumption of goods and services. Example: the pollution from a factory.

External benefits (positive externalities) are the positive impacts on society due to production or consumption of goods and services. Example: better roads in a neighbourhood due to the opening of a new business.

Private costs are the costs to the producer and consumer due to production and consumption respectively. Example: the cost of production.

Private benefits are the benefits to the producer or consumer due to production and consumption respectively. Example: the better immunity received by a consumer when he receives a vaccine.

Social Costs = External costs + Private Costs

Social Benefits = External benefits + Private benefits

Market Failure

Market failure occurs when the price mechanism fails to allocate resources effectively. This is the most disadvantageous aspect to the market economy. Causes of market failure are:

- **When social costs exceed social benefits** (especially where negative externalities (external costs) are high).
- **Over-provision of demerit goods** like alcohol and tobacco: the external costs arising from demerit goods are not reflected in the market and so they are overproduced.
- **Under-provision of merit goods** such as schools, hospitals and public transport, since the external benefits of these goods are not reflected in the market, they are underproduced.
- **Lack of public goods** such as roads, bus terminals and street lights: since their consumption cannot be measured and charged a price for, they are not produced by the private sector.
- **Immobility of resources:** when resources cannot move between their optimal uses and thus are not used to the maximum. For example, when workers (labour) don't have occupational or geographic mobility.
- **Information failure:** when information between consumers, producers and the government are not efficiently and correctly communicated. Example: a cosmetics firm advertises its products as healthy when it is in fact not. The consumers who believe the firm and use its products might suffer skin damage.

- **Abuse of monopoly* powers:** monopolistic businesses may use their powers to charge consumers a high price and only produce products they wish to, since they know consumers have no choice but to buy from them.
*Monopoly: a single supplier who supplies the entire market with a particular product, without any competition. Example: public utilities like water, gas and electricity in many countries are provided by their respective governments with no other producer allowed in the market.

2.11 – Mixed Economic System

In a mixed economic system, **both the market and government intervention co-exist**. Examples include almost all countries in the world (India, UK, Brazil etc.). This is because it overrides all the disadvantages of both the market and planned (govt. only) economies. It identifies the importance of the price mechanism in operating an efficient resource allocation and also the role of the government in correcting (any) market failures.

Features:

- both the public and the private sector exists
- planning and final decisions are made by the govt. while the market system can determine allocation of resources owned by it, along with the public organizations.

Advantages:

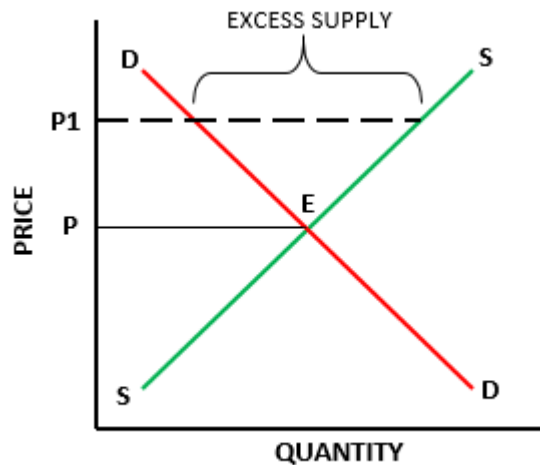
- The govt. can provide public goods, necessities and merit goods. The private businesses can provide profitable and most-demanded goods (luxury goods, superior goods). Thus, everyone is provided for.
- The govt. will keep externalities, monopolies, harmful goods etc. in control.
- The govt. can provide jobs in the public sector (so there is better job security).
- The govt. can also provide financial help to collapsing private organizations, so jobs are kept secure.

Disadvantages:

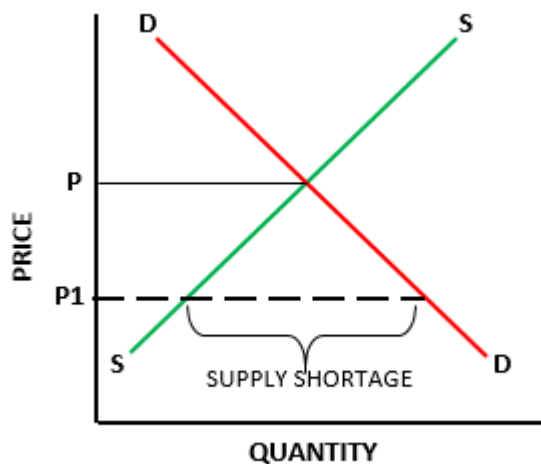
- Taxes will be imposed, which will raise prices and also reduce work incentive.
- Laws and regulations can increase production costs and reduce production in the economy.
- Public sector organizations will still be inefficient and will produce low quality goods and services.

The specific ways in which the government, in a mixed economic system, can **correct market failures** of the market:

- **Legislation and regulation** – the government can make laws that regulate market activity, for example, prohibit smoking in public (which would cause a negative externality). One important kind of legislation the govt. can undertake is **price controls** – setting a minimum price or maximum price on goods.



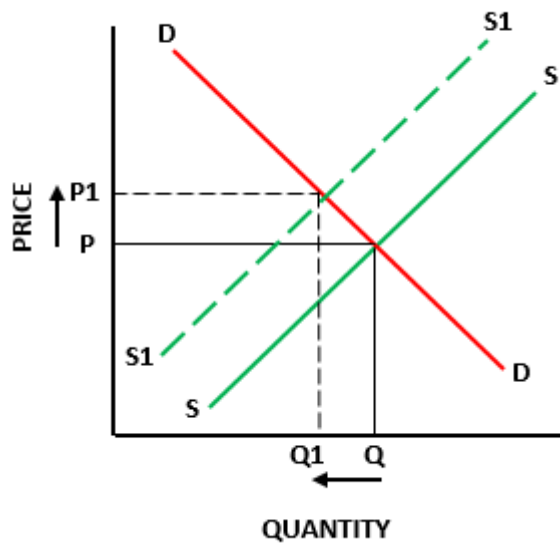
Minimum price or price floor is set to control a decreasing tendency of price. The minimum wage laws in many countries are an example of minimum price. The government sets the minimum wage above the existing market equilibrium wage, to ensure that all workers get a basic minimum wage to sustain them. But even as low-income workers now get better wages, the higher wage will cause the demand for labour to contract, as shown in the diagram to the left. There will also be higher supply of labour (workers who want work) because of higher wages. A reduced demand and increased supply will cause excess supply of labour i.e., unemployment.



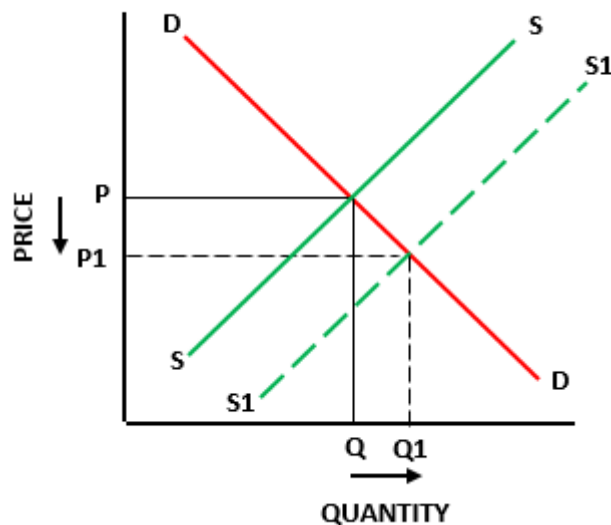
Maximum price or price ceiling is set to control an increasing tendency of price. It is usually set on rent (this is called rent control), to ensure that low-income tenants can afford to rent homes. But as a result of the lower rent, landlord will stop renting more homes, causing supply to contract, as shown in the diagram to the left. At the same time, lower rent will increase the demand for homes. A reduced supply of homes and higher demand for them will cause a shortage of supply in relation to demand.

- **Direct provision of merit and public goods** – since there is little incentive for the price mechanism to supply these goods, governments usually provide them. For example, free education, free healthcare, public parks. One way the govt. can do this is by **nationalising** certain products it considers essential to be provided by a governing authority, rather than the market. For example, in India, the government operates the only railway network because only it can provide cheap services to its millions of poor, daily passengers.
- **Taxation** on products – imposing a tax on products (indirect taxes) with negative externalities can discourage its production and consumption. For example, a tax on tobacco will make it expensive to produce and consume. In the diagram below, a tax has been imposed on a product, causing its supply to shift from S to S1. The price rises from P to P1 because of the additional tax

amount, and the quantity traded in the market falls from Q to Q_1 .



- **Subsidies** – a subsidy is a grant (financial aid) on products that have a positive externality. Subsidising, for example, cooking gas for the poor, will increase the living standard of the population. In the diagram below, a subsidy has been imposed on a good, causing its supply to shift from S to S_1 . It results in a fall in price from P to P_1 and subsequently, an increase in the quantity traded in the market from Q to Q_1 .



**Note: movements along a demand or supply curve of a good only happen as a result of a direct change in price of the good; changes caused by any other factor, tax and subsidy included, is represented by a shift in the curves.*

- **Tradable permits** – firms will have to buy permits from the government to do something, for example, pollute at a certain level, and these can be traded among firms. Since permits require money, firms will be encouraged to pollute less.
- **Extension of property rights** – one of the main reasons for pollution in public spaces is that it is public – it does not harm a specific private individual – the resource is the government's who cannot charge compensations easily. So the government can extend property rights (right to own property) of public places to private individuals. This will effectively **privatise** resources, create a market for these spaces and then individuals can be fined for polluting

- **International cooperation** among governments – governments work together on issues that affect the future of the environment.
As you can see, market failure can be corrected by governments in a variety of ways and the presence of a government is quite indispensable in any modern economy. Planned (government-only) economies are too inefficient and free market (no government) economies result in market failures. So a mixed economic system tries to balance both sides. That being said, there are certain drawbacks to government intervention in an economy.
- **Political incentives:** this occurs when there is a clash between political and economics (because a government is a political entity with political incentives). For example, even though mining companies cause a lot of environmental damage, the government may encourage and promote their activities to garner political and financial support from them.
- **Lack of incentives:** in the free market, individuals have a profit incentive to innovate and cut costs, but in the public sector, such an incentive is absent since the government will pay them salaries regardless of their performance. So, even as the government provides certain public and merit goods directly to the people at low costs, they tend to be very inefficient.
- **Time lags, information failure:** these are some of the government failure arising because of a lack of incentive. Government offices and employees don't have an incentive to provide timely services or give accurate information and this leads to very inefficient systems.
- **Welfare effects of policies:** government policies such as taxation and welfare payments distort the market. This means that such policies will influence demand and supply in the economy and cause markets to move away from the efficient points produced by a market system. For example, high corporate taxes will deter companies from expanding their operations and making more profits or deter new enterprises from entering the market. Unemployment benefits given out by the government may cause people to stay unemployed and receive free benefits instead of working.

Money

What is money?

A medium of exchange of goods and services.

Why do we need money?

We need money in order to exchange goods and services with one another. This is because we aren't self-sufficient – we can't produce all our wants by ourselves. Thus, there is a need for exchange.

In the past, **barter system** (exchanging a good or service for another good or service) prevailed. This had a lot of problems such as the need for the double coincidence of wants (if the person wants a table and he has a chair to exchange, he must find a person who has a table to exchange and is also willing to buy a chair), the goods being perishable and non-durable, the indivisibility of goods, lack of portability etc.

Thus the money we use today is in the form of currency notes and coins, which are **durable, uniform, divisible** (can be divided into 10's, 50's, 100's etc), **portable and is generally accepted**. These are the characteristics of what is considered 'good money'.

The functions of money:

- Money is a **medium of exchange**, as explained above.
- Money is a **measure of value**. Money acts as a unit of account, allowing us to compare and state the worth of different goods and services.
- Money is a **store of value**. It holds its value for a long time, allowing us to save it for future purposes.

- Money is a means of **deferred payment**. Deferred payments are purchases on credit – where the consumer can pay later for the goods or service they buy.

Banking

Banks are financial institutions that act as an intermediary between borrowers and savers. It is the money we save at banks that is lent out as loans to other individuals and businesses.

Commercial banks are those banks that have many retail branches located in most cities and towns. Example: HSBC. There is also a **central bank** that governs all other commercial banks in a country. Example: The Reserve Bank Of India (RBI).

Functions of a commercial bank:

- Accept deposits in the form of savings.
- Aid customers in making and receiving payments via their bank accounts.
- Give loans to businesses and private individuals.
- Buying and selling shares on customers' behalf.
- Provide insurance (protection in the form of money against damage/theft of personal property).
- Exchange foreign currencies.
- Provide financial planning advice.

Functions of a central bank:

- It issues notes and coins of the national currency.
- It manages all payments relating to the government.
- It manages national debt. Central banks can issue and repay public debts on the government's behalf.
- It supervises and controls all the other banks in the whole economy, even holding their deposits and transferring funds between them.
- It is the lender of 'last resort' to commercial banks. When other banks are having financial difficulties, the central bank can lend them money to prevent them from going bankrupt.
- It manages the country's gold and foreign currency reserves. These reserves are used to make international payments and adjust their currency value (adjust the exchange rate).
- It operates the monetary policy in an economy. (This will be explained in a later chapter)

Disposable income is the income of a person after all income-related taxes and charges have been deducted.

Spending (Consumption)

The buying of goods and services is called **consumption**. The money spent on consumption is called **consumer expenditure**.

People consume in order to satisfy their needs and wants and give them satisfaction.

Factors affecting consumption:

- **Disposable income:** the more the disposable income, the more people consume.
- **Wealth:** the more wealthy (having assets such as property, jewels, company shares) a person is, the more he spends.
- **Consumer confidence:** if consumers are confident of keeping their jobs and their future incomes, then they might be encouraged to spend more now, without worries.
- **Interest rates:** if interest rates provided by banks on saving are high, consumers might save more so they can earn interest and thus consumer expenditure will fall.

Saving

Saving is income not spent (or delaying consumption until some later date). People can save money by depositing in banks, and withdraw it a later date with the interest.

Factors affecting saving:

- **Saving for consumption:** people save so that they can consume later. They save money so that they can make bigger purchases in the future (a house, a car etc). Thus, saving can depend on the consumers' future plans.
- **Disposable income:** if the amount of disposable income people have is high, the more likely that they will save. Thus, rich people save a higher proportion of their incomes than poor people.
- **Interest rates:** people also save so that their savings may increase overtime with the interest added. **Interest** is the return on saving; the longer you save an amount and the higher the amount, the higher the interest received.
- **Consumer confidence:** if the consumer is not confident about his job security and incomes in the future, he may save more now.
- **Availability of saving schemes:** banks now offer a variety of saving schemes. When there are more attractive schemes that can benefit consumers, they might resort to saving rather than spending.

Borrowing

Borrowing, as the word suggests, is simply the borrowing of money from a person/institution. The lender gives the borrower money. The lender is usually the bank which gives out loans to customers.

Factors affecting borrowing:

- **Interest rates:** interest is also the cost of borrowing. When a person takes a loan, he must repay the entire amount at the end of a fixed period while also paying an amount of interest periodically. When the interest rates rise, people will be reluctant to borrow and vice versa.
- **Wealth/Income:** banks will be more willing to lend to wealthy and high-income earning people, because they are more likely to be able to repay the loan, rather than the poor. So even if they would like to borrow, the poor end up being able to borrow much lesser than the rich.
- **Consumer confidence:** how confident people feel about their financial situation in the future may affect borrowing too. For example, if they think that prices will rise (inflation) in the future, they might borrow now, to make big purchases now.
- **Ways of borrowing:** the no. of ways to borrow can influence borrowing. Nowadays there are many borrowing facilities such as overdrafts, bank loans etc. and there are more credit (future payment) options such as hire purchases (payment is done in installments overtime), credit cards etc.

Expenditure patterns between income groups

The richer people spend, save and borrow more amounts than the poor.

The poor spend higher proportions of their disposable income, especially on necessities, than the rich.

The poor save lesser proportions of their disposable income in comparison with the rich.

Labour Market

Labourers need wages to satisfy their wants and needs.

Payments for labour:

- **Time-rate wage:** wage given based on the no. of hours the employee has worked. **Overtime wages** are given to workers who have worked extra no. of hours, which will usually be 1.5 times or even twice the normal time rate.
- **Piece-rate wage:** wage given based on the amount of output produced. The more output an employee produced, the more wage he/she earns. This is used in industries where output can be easily measured and gives employees an incentive to increase their productivity.
- **Salary:** monthly payments made to workers, usually managers, office staff etc. usually in non-manual jobs.
- **Performance-related payments:** payments given to individual workers or teams of workers who have performed very well. **Commissions** given to salespersons for selling to a targeted no. of customers is a form of performance-related pay.

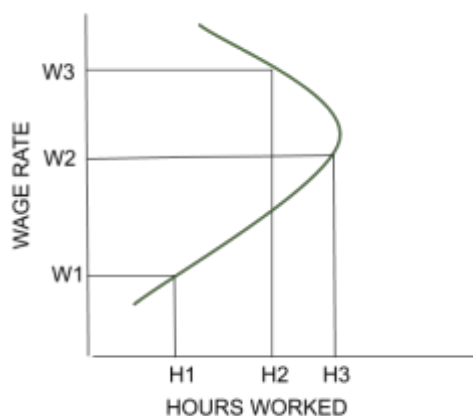
What affects an individual's choice of occupation?

- **Wage factors:** the wage conditions of a job/firm such as the pay rate, the prospect for performance-related payments and bonuses etc. will be considered by the individual before he chooses a job.
- **Non-wage factors:** This will include:
 - hours of work
 - holiday entitlements
 - promotion prospects
 - quality of working environment
 - job security
 - fringe benefits (free medical insurance, company car, price discounts on company products etc.)
 - training opportunities
 - distance from home to workplace
 - pension entitlement

Labour demand is the **number of workers demanded by firms at a given wage rate**. Labour demand is called '**derived demand**', since the level of demand of a product determines that industry's demand for labour. That is, **the higher the demand for a product, the more labour producers will demand** to increase supply of the product.

When the wage increases, the demand for labour contracts (and vice versa).

Labour supply is the **number of workers available and ready to work in an industry at a given wage rate**. When the wage rate increases, the supply of labour extends, and vice versa.



We also know that as the number of hours worked increases, the wage rate also increases. However, when a person gets to a very high position and his wages/salary increases highly, the number of hours he/she works may decrease. This can be shown in this diagram, called a **backward-**

bending labour supply curve. CEOs and executive managers at the top of the management tend to have backward-bending labour supply curves.

Just like in a demand and supply curve analysis, **labour demand and supply will extend and contract due to changes in the wage rate. Other factors that cause changes in demand and supply of labour will result in a shift in the demand and supply curve of labour.**

Factors that cause a shift in the labour demand curve:

- **Consumer demand for goods and services:** the higher the demand for products, the higher the demand for labour.
- **Productivity of labour:** the more productive labour is, the more the demand for labour.
- **Price and productivity of capital:** capital is a substitute resource for labour. If the price of capital were to lower and its productivity to rise, firms will demand more of capital and labour demand will fall (labour demand curve shifts to the left).
- **Non-wage employment costs:** wages are not the only cost to a firm of employing workers. Sometimes, employment tax, welfare insurance for each employee etc. will have to be paid by the firm. If these costs increase, firms will demand less labour.

Factors that cause a shift in the labour supply curve:

- **Advantages of an occupation:** the different advantages a job can offer to employees will affect the supply of labour- the people willing to do that job. Example: if the number of working hours in the airline industry increases, the labour supply there will shift to the left.
- **Availability and quality of education and training:** if quality training and education for a particular job, say pilots, is lacking, then the labour supply for it will be low. When new education and training institutes open, the labour supply will rise (labour supply curve shifts to the right).
- **Demographic changes:** the size and age structure of the population in an economy can affect the labour supply. The labour supply curve will shift to the right when more people come into a country from outside (immigration) and when the birth rate increases (more young people will be available for work).

Why would a person's wage rate change overtime?

As a beginner, the individual would have a low wage rate since he/she is new to the job and has no experience. Overtime, as his/her experience increases and skills develop, he/she will earn a higher wage rate. If he/she gets promoted and has more responsibilities, his/her wage rate will further increase. When he/she nears retirement age, the wage rate is likely to decrease as their productivity and skills are likely to weaken.

Wage Differentials

Why do different jobs have different wages?

- **Different abilities and qualifications:** when the job requires more skills and qualifications, it will have a higher wage rate.
- **Risk involved in the job:** risky jobs such as rescue operation teams will gain a higher wage rate for the risks they undertake.
- **Unsociable hours:** jobs that require night shifts and work at other unsociable hours are paid more.
- **Lack of information about other jobs and wages:** Sometimes people work for less wage rates simply because they do not know about other jobs with higher wage rates.
- **Labour immobility:** the ease with which workers can move between different occupations and areas of an economy is called **labour mobility**. If labour mobility is high, workers can move to jobs with a higher pay. Labour immobility causes people to work at a low wage rate because they don't have the skills or opportunities to move to jobs with a higher wage.

- **Fringe benefits:** jobs which offer a lot of fringe benefits have low wages. But sometimes the highest-paid jobs are also given a lot of fringe benefits, to attract skilled labour.

Why do wages differ between people doing the same job?

- **Regional differences in labour demand and supply:** for example, if the demand in an area for accountants is very high, the wage rate for accountants will be high; whereas, in an area of low demand for accountants, the wage rate for accountants will be low. Similarly, a high supply of accountants will cause their wages to be low, while a low supply (scarcity) of accountants will cause their wages to be high. It's the law of demand and supply!
- **Fringe benefits:** some firms which pay a lot of fringe benefits, will pay less wages, while firms (in the same industry) which pay lesser fringe benefits will have higher wages.
- **Discrimination:** workers doing the same work may be discriminated by gender, race, religion or age.
- **Length of service:** some firms provide extra pay for workers who have worked in the firm for a long time, while other firms may not.
- **Local pay agreements:** some trade unions may agree a national wage rate for all their members – therefore all their members (labourers) will get a higher wage rate than those who do the same job but are not in the trade union. This depends on the relative bargaining power of the trade union.
- **Government labour policies:** wages will be fairer in an economy where the government has set a minimum wage policy. The government's corporate tax policies can also influence the amount of wage firms will be willing to pay out.

Other wage differentials:

- **Public-private sector pay gap:** public sector jobs usually have a high wage rate. But sometimes public sector wages are lower than that of the private sector's because low wages can be compensated by the public sector's high job security and pension prospects.
- **Economic sector:** workers in primary activities such as agriculture receive very low wages in comparison to those in the other sectors because the value of output they produce is lower. Further still, workers in the manufacturing sector may earn lesser than those in the services sector. But it comes down to the nature of the job itself. A computer engineer in the manufacturing sector does earn more than a waiter at a restaurant after all.
- **Skilled and unskilled workers:** Skilled workers have a higher pay than unskilled workers, because they are more productive and efficient and make lesser mistakes.
- **Gender pay gap:** Men are usually given a higher pay than women. This is because women tend to go for jobs that don't require as much skill as that is required by men's jobs (teaching, nursing, retailing); they take career breaks to raise children, which will cause less experience and career progress (making way for low wages); more women work part-time than full-time. Sometimes, even if both men and women are working equally hard and effectively, discrimination can occur against women.
- **International wage differentials:** developed countries usually have high wage rates due to high incomes, large supply of skilled workers, high demand for goods and services etc; while in a less-developed economy, wage rates will be low due to a large supply unskilled labour.

Division of Labour/Specialisation

Division of labour is the concept of dividing the production process into different stages enabling workers to specialise in specific tasks. This will help increase efficiency and productivity. Division of labour is widely used in modern economies. From the making of iPhones (the designs, processors, screens, batteries, camera lenses, software etc. are made by different people in different parts of the world) to this very website (where notes, mindmaps, illustrations, design etc. are all managed by different people).

Advantages to workers:

- **Become skilled:** workers can get skilled and experienced in a specific task which will help their future job prospects
- **Better future job prospects:** because of the skill and training they acquire, workers will, in the future, be able to get better jobs in the same field.
- Saves time and expenses in training

Disadvantages to workers:

- **Monotony:** doing the same task repetitively might make it boring and lower worker's morale.
- **Margin for errors increases:** as the job gets repetitive, there also arises a chance for mistakes.
- **Alienation:** since they're confined to just the task they're doing, workers will feel socially alienated from each other.
- **Lower mobility of labour:** division of labour can also cause a reduced mobility of labour. Since a worker is only specialised in doing one specific task(s), it will be difficult for him/her to do a different job.
- **Increased chance of unemployment:** when division of labour is introduced, many excess workers will have to be laid off. Additionally, if one loses the job, it will be harder for him/her to find other jobs that require the same specialisation.

Advantages to firms:

- **Increased productivity:** when people specialise in particular tasks, the total output will increase.
- **Increased quality** of products: because workers work on tasks they are best suited for, the quality of the final output will be high.
- **Low costs:** workers only need to be trained in the tasks they specialise in and not the entire process; and tools and equipment required for a task will only be needed for a few workers who specialise in the task, and not for everybody else.
- **Faster:** when everyone focuses on a particular task and there is no need for workers to shift from one task to another, the production will speed up
- **Efficient movement of goods:** raw materials and half-finished goods will easily move around the firm from one task to the next.
- Better selection of workers: since workers are selected to do tasks best suited for them, division of labour will help firms to choose the best set of workers for their operations.
- Aids a **streamlined production process:** the production process will be smooth and clearly defined, and so the firm can easily adapt to a mass production scale.
- **Increased profits:** lower costs and increased productivity will help boost profits.

Disadvantages o firms:

- **Increased dependency:** The production may come to a halt if one or more workers doing a specific task is absent. The production is dependent on all workers being present to do their jobs.
- **Danger of overproduction:** as division of labour facilitates mass production, the supply of the product may exceed its demand, and cause a problem of excess stocks of finished goods. Firms need to ensure that they're not producing too much if there is not enough demand for the product in the first place.

Advantages to the economy:

- **Better utilisation of human resources** in the economy as workers do the job they're best at, helping the economy achieve its maximum output.
- **Establishment of efficient firms** and industries, as the higher profits from division of labour will attract entrepreneurs to invest and produce.
- Inventions arise: as workers become skilled in particular areas, they can innovate and invent new methods and products in that field.

Disadvantages to the economy:

- **Labour immobility:** occupational immobility may arise because workers can only specialise in a specific field.

- **Reduces the creative instinct** of the labour force in the long-run as they are only able to do a single task repetitively and the previous skills they acquired die out.
- **Creates a factory culture**, which brings with it the evils of exploitation, poor working conditions, and forced monotony.

Trade Unions are organizations of workers that aim at promoting and protecting the interest of their members (workers). They aim on improving wage rates, working conditions and other job-related aspects.

The functions of a trade union:

- Negotiating **improvements in non-wage benefits** with employers.
- **Defending employees' rights.**
- **Improving working conditions**, such as better working hours and better safety measures.
- **Improving pay and other benefits.**
- **Supporting workers who have been unfairly dismissed or discriminated against.**
- Developing the skills of members, by providing training and education.
- Providing recreational activities for the members.
- **Taking industrial actions** (strikes, overtime ban etc.) when employers don't satisfy their needs. These are explained later in this topic.

Collective bargaining: the process of negotiating over pay and working conditions between trade unions and employers.

When can trade unions argue for higher wages and better working conditions?

- **Prices are rising** (inflation): the cost of living increases when prices increase and workers will want higher wages to consume products and raise their families.
- **The sales and demand of the firm has increased.**
- **Workers in other firms are getting a higher pay.**
- **The productivity of the members has increased.**

Industrial disputes

When firms don't satisfy trade union wants or refuse to agree to their terms, the members of a trade union can organize industrial disputes. Here are some:

- **Overtime ban:** workers refuse to work more than their normal hours.
- **Go-slow:** workers deliberately slow down production, so the firm's sales and profits go down.
- **Strike:** workers refuse to work and may also protest or picket outside their workplace to stop deliveries and prevent other non-union members from entering. They don't receive any wages during this time. This will halt all production of the firm.

Trade union activity has several impacts:

Advantages to workers:

- Workers benefit from collective bargaining power by being able to establish **better terms of labour.**
- Workers feel a **sense of unity and feel represented**, increasing morale.
- **Lesser chance of being discriminated and exploited.**

Disadvantages to workers:

- **Workers might get lesser wages or none if they go on strike** – as the output and profits of the firm falls and they refuse to pay.

Advantages to firms:

- **Time is saved in negotiating** with a union when compared to negotiating with individuals workers.
- When making changes in work schedules and practices, a trade union's cooperation can help **organise workers efficiently**.
- Mutual respect and good relationships between unions and firms are **good for business morale and increases productivity**.

Disadvantages to firms:

- **Decision making may be long** as there will be need of lengthy discussions with trade unions in major business decisions.
- Trade unions may **make demands that the firm may not be able to meet** – they will have to choose between profitability and workers' interests.
- Higher wages bargained by trade unions will reduce the firm's profitability.
- **Businesses will have high costs and low output if unions organise agitations**. Their revenue and profits will go down and they will enter a loss. They may also lose a lot of customers to competing firms.

Advantages to the economy:

- Ensures that the labour force in the economy is **not exploited** and that **their interests are being represented**

Disadvantages to the economy:

- Can negatively **impact total output** of the economy.
- Firms may decide to substitute labour for capital if they can't meet trade unions' expensive demands, and so **unemployment may rise**.
- Higher wages resulting from trade union activity can make the nation's exports expensive and thus **less competitive in the international market**

In modern times, the powers of trade unions have drastically weakened.

Globalisation, liberalisation and privatisation of economies are making markets more competitive. Firms have more incentive to reduce costs of production to a minimum in order to remain competitive and profitable. Therefore, it is much harder for unions to force employers to increase wages. Most unions operating nowadays are more focused on bettering working conditions and non-monetary benefits.

Classification of Firms

Firms can be classified in terms of the sectors they operate in and their relative sizes.

Firms are classified into the following three categories based on the type of operations undertaken by them:

- **Primary:** all economic activity involving extraction of raw natural materials. This includes agriculture, mining, fishing etc. In pre-modern times, most economic activity and employment was in this sector, mostly in the form of subsistence farming (farming for self-consumption).
- **Secondary:** all economic activity dealing with producing finished goods. This includes construction, manufacturing, utilities etc. This sector gained importance during the industrial revolution of the 19th and 20th centuries and still makes up a huge part of the modern economy.
- **Tertiary:** all economic activity offering intangible goods and services to consumers. This includes retail, leisure, transport, IT services, banking, communications etc. This sector is now the fastest-growing sector as consumer demand for services have increased in developed and developing nations.

Firms can also be classified on the basis of whether they are publicly owned or privately owned:

- **Public:** this includes all firms owned and run by the government. Usually, the defence, arms and nuclear industries of an economy are completely public. Public firms don't have a profit motive, but aim to provide essential services to the economy it governs. Governments do also run their own schools, hospitals, postal services, electricity firms etc.
- **Private:** this includes all firms owned and run by private individuals. Private firms aim at making profits and so their products are those that are highly demanded in the economy. Firms can also be classified on their relative size as small, medium or large depending on the output, market share, organisation (no. of departments and subsidiaries etc).

Small Firms

A small firm is an independently owned and operated enterprise that is **limited in size and in revenue** depending on the industry. They require relatively less capital, less workforce and less or no machinery. These businesses are ideally suited to operate on a small scale to serve a local community and to provide profits to the owners.

Advantages of small businesses:

- **Independence:** owner(s) are free to run the business as he/she pleases.
- **Control:** the owner(s) has full control over the business, unlike in a large business where multiple managers, departments and branches will exist.
- **Flexibility:** small businesses can adapt to quick changes as the owner is more involved in the decision-making.
- **Better communication:** since there are fewer employees, information can be intimated easily and quickly.
- **Innovation:** small businesses can tend to be innovative because they have less to lose and are willing to take risks.

Disadvantages of small businesses:

- **Higher costs:** small firms cannot exploit economies of scale – their average costs will be higher than larger rivals.
- **Lack of finance:** struggles to raise finance as choice of sources of acquiring finance is limited.
- **Difficult to attract experienced employees:** a small business may be unable to afford the wage and training required for skilled workers.
- **Vulnerability:** when economic conditions change, it is harder for small businesses to survive as they lack resources.

Small firms still exist in the economy for several reasons:

- **Size of the market:** when there is only a small market for a product, a firm will see no point in growing to a larger size. The market maybe small because:
 - the market is local – for example, the local hairdresser.
 - the final product maybe an expensive luxury item which only require small-scale production (e.g. custom-made paintings)
 - personalised/custom services can only be given by small firms, unlike large firms that mostly give standardised services (e.g. wedding cake makers).
- **Access to capital is limited,** so owners can't grow the firm.

- **Owner(s) prefer to stay small:** a lot of entrepreneurs don't want to take risks by growing the firm and they are quite satisfied with running a small business.
- **Small firms can co-operate:** co-operation between small firms can lead them to set up jointly owned enterprises which allow them to enjoy many of the benefits that large firms have.
- **Governments help small firms:** governments usually provide help to small scale firms because small firms are an important provider of employment and generate innovation in the production process. In most countries, it is the medium and small industries that contribute much of the employment.

Growth of Firms

When a firm grows, its scale of production increases. Firms can grow in to ways: internally or externally.

Internal Growth/Organic Growth

This involves **expanding the scale of production of the firm's existing operations**. This can be done by purchasing more machinery/equipment, opening more branches, selling new products, expanding business premises, employing more workers etc.

External Growth

This involves **two or more firms joining together to form a larger business**. This is called **integration**. This can be done it two ways: mergers or takeovers.

A takeover or acquisition happens when a company buys enough shares of another firm that they can take full control. The firm taken over loses its identity and becomes a part of what is known as the holding company. A well-known example would be Facebook's acquisition of Whatsapp in 2014.

A merger occurs when the owners of two or more companies agree to join together to form a firm.

Mergers can happen in three ways:

- **Horizontal Integration:** integration of firms engaged in the production of the same type of good at the same level of production. Example: a cloth manufacturing company merges with another cloth manufacturing company.

Advantages:

-
- **Exploit internal economies of scale:** including bulk-buying, technical economies, financial economies.
- **Save costs:** when merging, a lot of the duplicate assets including employees can be laid off.
- Potential to secure '**revenue synergies**' by creating and selling a wider range of products.
- **Reduces competition:** by merging with key rivals, the two firms together can increase market share.

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Disadvantages:

-
- **Risk of diseconomies of scale:** a larger business will bring with a lot of managerial and operational issues leading to higher costs.
- **Reduced flexibility:** the addition of more employees and processes means the need for more transparency and therefore more accountability and red tape, which can slow down the rate of innovating and producing new products and processes.

- **Vertical Integration:** integration of firms engaged in the production of the same type of good but at different levels of production (primary/secondary/tertiary). Example: a cloth manufacturing company (secondary sector) merges with a cotton growing firm (primary sector).
- **Forward vertical integration:** when a firm integrates with a firm that is at a later stage of production than theirs. Example: a dairy farm integrates with a cheese manufacturing company.
- **Backward vertical integration:** when a firm integrates with a firm that is at an earlier stage of production than theirs. Example: a chocolate retailer integrates with a chocolate manufacturing company.

Advantages:

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- It can give a firm **assured supplies or outlets** for their products. If a coffee brand merged with coffee plantation, the manufacturers would get assured supplies of coffee beans from the plantation. If the coffee brand merged with a coffee shop chain, they would have a permanent outlet to sell their coffee from.
- Similarly, one firm **can prevent the other firm from supplying materials or selling products to competitors**. The coffee brand can have the coffee plantation to only supply them their coffee beans. The coffee brand can also have the coffee shop chain only selling coffee with their coffee powder.
- The profit margins of the merged firm can now be absorbed into the merging firm.
- The firms can increase their market share and become more competitive in the market.

Disadvantages:

-
- **Risk of diseconomies of scale:** a larger business will bring with a lot of managerial and operational issues leading to higher costs
- **Reduced flexibility:** the addition of more employees and processes means the need for more transparency and therefore more accountability and red tape, which can slow down the rate of innovating and producing new products and processes
- **It's a difficult process:** The firms, when vertically integrated, are entering into a stage of production/sector they're not familiar with, and this will require staff of either firm to be educated and trained. Some might even lose their jobs. It can be expensive as well.
- **Lateral/Conglomerate integration:** this occurs when firms producing different type of products integrate. They could be at the same or different stages of production. Example: a housing company integrates with a dairy farm. Thus, the firm can produce a wide range of products. This helps **diversify** a firm's operations.

Advantages:

-
- **Diversify risks:** conglomerate integration allows businesses to have activities in more than one market. This allows the firms to spread their risks. In case one market is in decline, it still has another source of profit.
- **Creates new markets:** merging with a firm in a different industry will open up the firm to a new customer base, helping it to market its core products to this new market.
- **Transfer of ideas:** there could be a transfer of ideas and resources between the two businesses even though they are in different industries. This transfer of ideas could help improve the quality and demand for the two products.

Disadvantages:

- **Inexperience can lead to mismanagement:** if the firms are in entirely different industries and have no experience in the other's industry, cooperating and managing the two industries may be difficult and could turn disastrous.
- **Lose focus:** merging with and focusing on an entirely new industry could cause the firm to lose focus of its core product.
- **Culture clash:** as with all kinds of mergers, there could be a culture clash between the two firms' employees on practices, standards and 'how things are done'.

Scale of Production

As a firm's scale of production increases its average costs decrease. **Cost saving from a large-scale production is called economies of scale.**

Internal economies of scale are decisions taken within the firm that can bring about economies (advantages). Some internal economies of scale are:

- **Purchasing economies:** large firms can buy raw materials and components in bulk because of their large scale of production. Supplier will usually offer price discounts for bulk purchases, which will cut purchasing costs for the firm.
- **Marketing economies:** large firms can afford their own vehicles to distribute their products, which is much cheaper than hiring other firms to distribute them. Also, the costs of advertising is spread over a much large output in large firms when compared to small firms.
- **Financial economies:** banks are more willing to lend money to large firms since they are more financially secure (than small firms) to repay loans. They are also likely to get lower rates of interest. Large firms also have the ability to sell shares to raise capital (which do not have to be repaid). Thus, they get more capital at lower costs.
- **Technical economies:** large firms are more financially able to invest in good technology, skilled workers, machinery etc. which are very efficient and cut costs for the firm.
- **Risk-bearing economies:** large firms with a high output can sell into different markets (even overseas). They are able to produce a variety of products (**diversification** in production). This means that their risks are spread over a wider range of products or markets; even if a market or product is not successful, they have other products and markets to continue business in. Thus, costs are less.

External economies of scale occur when firms benefit from the entire industry being large. This may include:

- **Access to skilled workers:** large firms can recruit workers trained by other firms. For example: when a new training institution for pilots and airline staff opens, all airline firms can enjoy economies of scale of having access to skilled workers, who are more efficient and productive, and cuts costs.
- **Ancillary firms:** they are firms that supply and provide materials/services to larger firms. When ancillary firms such as a marketing firm locates close to a company, the company can cut costs by using their services more cheaply than other firms.
- **Joint marketing benefits:** when firms in the same industry locate close to each other, they may share an enhanced reputation and customer base.
- **Shared infrastructure:** development in the infrastructure of an industry or the economy can benefit large firms. Examples: more roads and bridges by the govt. can cut transport costs for firms, a new power station can provide cheaper electricity for firms.

Diseconomies of scale occur when a firms grows too large and average costs start to rise. Some common diseconomies are:

- **Management diseconomies:** large firms have a wide internal organisation with lots of managers and employees. This makes communication difficult and decision-making very slow. Gradually, it leads to inefficient running of the firms and increases costs.
- **Too much output** may require a large supply of raw materials, power etc. which can lead to shortage and halt production, increasing costs.

- Large firms may use automated production with lots of capital equipment. Workers operating these machines may feel bored in doing the repetitive tasks and thus become **demotivated and less cooperative**. Many workers may leave or go on strikes, stopping production and increasing costs.
- **Agglomeration diseconomies**: this occurs when firms merge/acquire too many different firms producing different products, and the managers and owners can't coordinate and organise all activities, leading to higher costs.
- More shares sold into the market and bought means more owners coming into the business. Having a lot of owners can lead to a lot of **disputes and conflicts** among themselves.
- A lot of large firms can face diseconomies when their **products become too standardised and less of a variety in the market**. This will reduce sales and profits and increase average costs. A firm that doubles all its inputs (resources) and is able to more than double its output as a result, experiences **increasing returns to scale**. A firm that doubles all its inputs and fails to double its output as a result, experiences a decreasing or **diminishing returns to scale**.

Demand for Factors of Production

Some factors that determine the demand of factors of production:

- **The demand for the product**: if more goods and services are demanded by consumers, more factors of production will be demanded by firms to produce and satisfy the demand. That is, the demand for factors of production is derived demand, as it is determined by the demand for the goods and services (just like labour demand).
- **The availability of factors**: firms will also demand factors that are easily available and accessible to them. If the firm is located in a region where there is a large pool of skilled labour, it will demand more labour as opposed to capital.
- **The price of factors**: If labour is more expensive than capital, firms will demand more capital (and vice versa), as they want to reduce costs and maximize profits.
- **The productivity of factors**: If labour is more productive than capital, then more labour is demanded, and vice versa.

Labour-intensive and Capital-intensive production

Labour-intensive production is where more labourers are employed than other factors, say capital. Production is mainly dependent on labour. It is usually adopted in small-scale industries, especially those that produce personalised, handmade products. Examples: hotels and restaurants.

Advantages:

- **Flexibility**: labour, unlike most machinery can be used flexibly to meet changing levels of consumer demand, e.g., part-time workers.
- **Personal services**: labour can provide a personal touch to customer needs and wants.
- **Personalised services**: labourers can provide custom products for different customers. Machinery is not flexible enough to provide tailored products for individual customers.
- **Gives feedback**: labour can give feedback that provides ideas for continuous improvements in the firm.
- **Essential**: labour is essential in case of machine breakdowns. After all, machines are only as good as the labour that builds, maintains and operates them..

Disadvantages:

- **Relatively expensive**: in the long-term, when compared to machinery, labour has higher per unit costs due to lower levels of productivity.

- **Inefficient and inconsistent:** compared to machinery, labour is relatively less efficient and tends to be inconsistent with their productivity, with various personal, psychological and physical matters influencing their quantity and quality of work.
 - **Labour relation problems:** firms will have to put up with labour demands and grievances. They could stage an overtime ban or strike if their demands are not met.
- Capital refers to the machinery, equipment, tools, buildings and vehicles used in production. It also means the investment required to do production. **Capital-intensive production** is where more capital is employed than other factors. It is a production which requires a relatively high level of capital investment compared to the labour cost. Most capital-intensive production is automated (example: car-manufacturing).

Advantages:

- **Less likely to make errors:** Machines, since they're mechanically or digitally programmed to do tasks, won't make the mistakes that labourers will.
- **More efficient:** machinery doesn't need breaks or holidays, has no demands and makes no mistakes.
- **Consistent:** since they won't have human problems and are programmed to repeat tasks, they are very consistent in the output produced.
- **Technical economies of scale:** increased efficiency can reduce average costs

Disadvantages:

- **Expensive:** the initial costs of investment is high, as well as possible training costs.
- **Lack of flexibility:** machines need not be as flexible as labourers are to meet changes in demand.
- **Machinery lacks initiative:** machines don't have the intuitive or creative power that human labour can provide the business, and improve production.

Production and Productivity

A firm combines scarce resources of land, labour and capital (inputs) to make (produce) goods and services (output). **Production is thus, the transformation of raw materials (input) to finished or semi-finished goods and services (output).** In other words, production is the adding of value to inputs to create outputs. It is the production that gives the inputs value.

Some factors that influence production:

- **Demand for product:** the more the demand from consumers, the more the production.
- **Price and availability of factors of production:** if factors of production are cheap and readily available, there will be more production.
- **Capital:** the more capital that is available to producers, the more the investment in production.
- **Profitability:** the more profitable producing and selling a product is, the more the production of the product will be.
- **Government support:** if governments give money in grants, subsidies, tax breaks and so on, more production will take place in the economy.

Productivity measures the amount of output that can be produced from a given amount of input over a period of time.

Productivity = Total output produced per period / Total input used per period

Productivity increases when:

- more output or revenue is produced from the same amount of resources

- the same output or revenue is produced using fewer resources.
(Labour productivity is the measure of the amount of output that can be produced by each worker in a business).

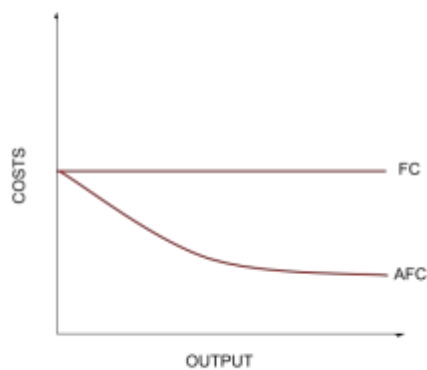
Factors that influence productivity:

- Division of labour:** division of labour is when tasks are divided among labourers. Each labourer specializes in a particular task, and thus this will increase productivity.
- Skills and experience of labour force:** a skilled and experienced workforce will be more productive.
- Workers' motivation:** the more motivated the workforce is, the more productive they will be. Better pay, working conditions, reasonable working hours etc. can improve productivity.
- Technology:** more technology introduced into the production process will increase productivity.
- Quality of factors of production:** replacing old machinery with new ones, preferably with latest technologies, can increase efficiency and productivity. In the case of labour, training the workforce will increase productivity.
- Investment:** introducing new production processes which will reduce wastage, increase speed, improve quality and raise output will raise productivity. This is known as lean production.

Costs of Production

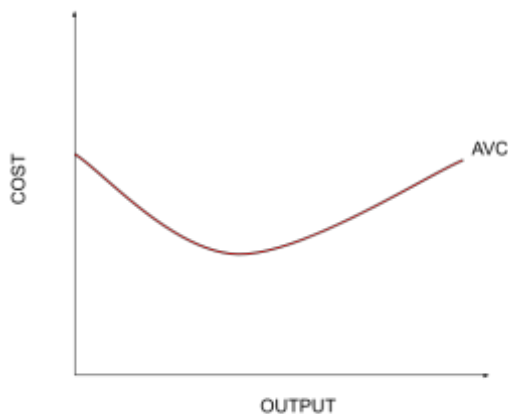
Fixed costs (FC) are costs that are fixed in the short-term running of a business and have to be paid even when no production is taking place. Examples: rent, interest on bank loans, telephone bills. These costs do not depend on the amount of output produced.

Average Fixed Cost (AFC) = Total Fixed Cost (TFC) / Total Output

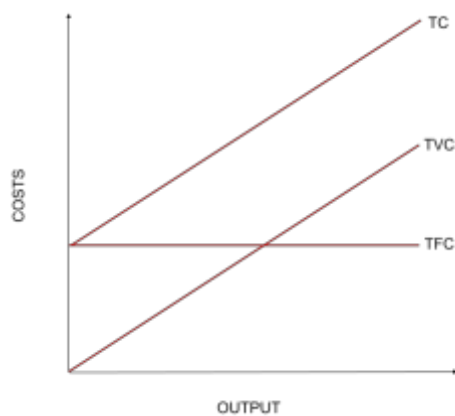


Variable costs (VC) are costs that are variable in the short-term running of a business and are paid according to the output produced. The more the production, the more the variable costs are. Examples: wages, electricity bill, cost of raw materials.

Average Variable Cost (AVC) = Total Variable Costs (TVC) / Total Output



Total Costs (TC) = Total Fixed Costs (TFC) + Total Variable Costs (TVC)

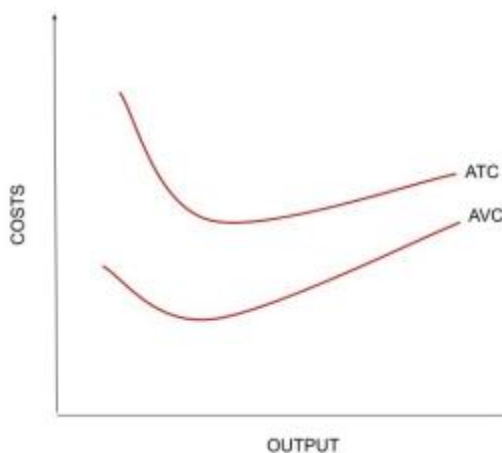


This is a simple graph showing the relation between TC, FC and VC. The gap between the TC and TVC indicates the TFC

Average cost or Average total Cost (ATC) is the cost per unit of output.

Average Total Cost (ATC) = Total Cost (TC) / Total Output or

Average Cost (AC) = Average Variable Cost (AVC) + Average Fixed Cost (AFC)



(Remember 'average' means 'per unit' and so will involve dividing the particular cost by the total output produced. In the graphs above you will notice that the average variable costs and average total costs first fall and then start rising. This is because of economies of scale and diseconomies of scale respectively. As the firm increases its output, the average costs decline but as it starts growing beyond a limit, the average costs rise).

Let's calculate some costs in an example:

Suppose, a TV manufacturer produces 1000 TVs a month. The firm's fixed costs in rent is \$900, and variable cost per unit is \$500. What would its TFC, TVC, AVC, AFC, AC and TC be, in a month?

No. of units of TVs produced = 1000

Total Fixed Costs for one month = \$900

Average Fixed Cost = $\$900 / 1000 = \0.9 per unit

Variable Cost of producing one unit of TV = \$500

Total Variable Costs for producing 1000 TVs in a month = $\$500 * 1000 = \$500,000$

Average Variable Cost = $\$500,000 / 1000 = \500 (AVC is the same as VC per unit!)

Total Costs = Total Fixed Costs + Total Variable Costs ==> $\$900 + \$500,000 = \$500,900$

Average Costs = Total Costs / Total Output ==> $\$500,900 / 1000 = \500.9

or Average Costs = AFC + AVC ==> $\$0.9 + \$500 ==> \$500.9$

Revenue

Revenue is the total income a firm earns from the sale of its goods and services. The more the sales, the more the revenue.

Total Revenue (TR) = No. of units sold (Sales) * Price per unit (P)

Average Revenue = Total Revenue (TR) / No. of units sold (Sales) (= Price per unit (P)!))

Suppose, from the example above, a TV is sold at \$800 and the firm sells all the units it produces, what is the firm's Total Revenue and Average Revenue, for a month?

No. of units sold (Sales) in a month = No. of units produced in a month = 1000

Total Revenue = Sales * Price ==> $1000 * \$800 = \$800,000$

Average Revenue = Total Revenue / Sales = $\$800,000 / 1000 = \800

Total Revenue – Total Cost = Profit

Objectives of Firms

Objectives vary with different businesses due to size, sector and many other factors. However, many business in the private sector aim to achieve the following objectives.

- **Survival:** new or small firms usually have survival as a primary objective. Firms in a highly competitive market will also be more concerned with survival rather than any other objective.

To achieve this, firms could decide to lower prices, which would mean forsaking other objectives such as profit maximization.

- **Profit:** profit is the income of a business from its activities after deducting total costs from total revenue. Private sector firms usually have profit making as a primary objective. This is because profits are required for **further investment** into the business as well as for the **payment of return** to the shareholders/owners of the business. Usually, firms aim to **maximise their profits** by either minimising costs, or maximising revenue, or both.
- **Growth:** once a business has passed its survival stage it will aim for growth and expansion. This is usually measured by value of sales or output. Aiming for business growth can be very beneficial. A larger business can ensure **greater job security** and **salaries** for employees. The business can also benefit from higher **market share** and **economies of scale**.
- **Market share:** market share can be defined as the sales in proportion to total market sales achieved by a business. Increased market share can bring about many benefits to the business such as increased customer loyalty, setting up of brand image, etc.
- **Service to the society:** Some operations in the private sectors such as social enterprises do not aim for profits and prefer to set more social objectives. They aim to better the society by aiding society financially or otherwise.

A business' objectives do not remain the same forever. As market situations change and as the business itself develops, its objectives will change to reflect its current market and economic position. For example, a firm facing serious economic recession could change its objective from profit maximization to short term survival.

Competitive Markets

Firms compete in the market to increase their customer base, sales, market share and profits.

Price competition involves competing to offer consumers the lowest or best possible prices of a product. **Non-price competition** is competing on all other features of the product (quality, after-sales care, warranty etc.) other than price.

Informative advertising means providing information about the product to consumers. Examples include advertising of phones, computers, home appliances etc. which include specific information about their technical features.

Persuasive advertising is designed to create a consumer want and persuade them to buy the product in order to boost sales. Examples include advertisements of perfumes, clothes, chocolates etc.

Pricing Strategies

What can influence the price that producers fix on a product?

- Level and strength of consumer demand.
- The amount of competition from rival producers in the market.
- The cost of production and the level of profit targeted.

Price skimming: When a new and unique product enters the market, its producers charge a very high price for it initially as consumers will be willing to pay more for the new product. As more competitors begin to launch similar products, producers may lower prices. Apple's iPhones are good examples – they are very expensive at launch and get cheaper overtime.

Penetration pricing: when producers set a very low price which encourages consumers to try the product, helping expand sales and increase loyalty. This way, the product is able to penetrate a market, especially useful when there are a lot of

existing rival products. Netflix, when it first started out as a DVD rental service, used penetration pricing (\$1 monthly subscription!) to encourage customers to try their service which helped it create a large customer base.

Destruction pricing (predatory pricing): prices are kept very low (lower than the cost of production per unit) in order to 'destroy' the sales of existing products, as consumers will turn to the lowest priced products. Once the product is successful, it can raise prices and cover costs. India's Reliance Jio, a telecom company, was accused of predatory pricing during its initial launch years. Predatory pricing is illegal in many countries as it creates a non-competitive business environment and encourages monopoly practices.

Price wars happen when competing firms continually trying to undercut each other's prices.

Cost-plus-pricing: this involves calculating the average cost of producing each unit of output and then adding a mark-up value for profit.

Price = (Total Cost/Total Output) + Mark-up

This ensures that the cost of production is covered and that each unit produces a profit.

Perfect Competition

In a perfectly competitive market, there will be **many sellers and many buyers** – a lot of different firms compete to supply an identical product.

As there is fierce competition, **neither producers nor consumers can influence market price – they are all price takers**. If any firm did try to sell at a high price, it would lose customers to competitors. If the price is too low, they may incur a loss. There will also be a huge amount of output in the market.

Advantages:

- **High consumer sovereignty**: consumers will have a wide variety of goods and services to choose from, as many producers sell similar products. Products are also likely to be of high quality, in order to attract consumers.
- **Low prices**: as competition is fierce, producers will try and keep prices low to attract customers and increase sales.
- **Efficiency**: to keep profits high and lower costs, firms will be very efficient. If they aren't efficient, they would become less profitable. This will cause them to raise prices which would discourage consumers from buying their product. Inefficiency could also lead to poor quality products.

Disadvantages:

- **Wasteful competition**: in order to keep up with other firms, producers will duplicate items; this is considered a waste of resources.
- **Mislead customers**: to gain more customers and sales, firms might give false and exaggerated claims about their product, which would disadvantage both customers and competitors.

Monopoly

Dominant firms who have market power to restrict competition in the market are called monopolies. In a pure monopoly, there is only a **single seller** who supplies a good or service. Example: Indian Railways. Since customers have no other firms to buy from, monopolies can raise prices – that is they are **able to influence prices** as it will not affect their profitability. These high prices result in monopolies generating excessive or abnormal profits.

Monopolies don't face competition because the market faces high **barriers to entry** – obstacles preventing new firms from entering the market. That is, there might

be high start-up costs (sunk costs), expensive paperwork, regulations etc. If the monopoly has a very high brand loyalty or pricing structures that other firm couldn't possibly compete with, those also act as barriers to entry.

Disadvantages:

- There is **less consumer sovereignty**: as there are no (or very little) other firms selling the product, output is low and thus there is little consumer choice.
- Monopolies may **not respond quickly to customer demands**.
- **Higher prices**.
- **Lower quality**: as there is little or no competition, monopolies have no incentive to raise quality, as consumers will have to buy from them anyway. (But since they make a lot of profit, they may invest a lot in research and development and increase quality).
- **Inefficiency**: With high prices, they may create high enough profits that, costs due to inefficiency won't create a significant problem in their profitability and so they can continue being inefficient.

Why monopolies are not always bad?

- As only a single producer exists, it will produce more output than what individual firms in a competition do, and thus benefit from economies of scale.
- They can still face competition from overseas firms.
- They could sell products at lower price and high quality if they fear new firms may enter the market in the future.

The Role of Government

The public sector in every economy plays a major role, as a producer and employer. Governments work locally, nationally and internationally. Here are the roles they play in the economy:

- As a producer, it provides, at all levels of government:
 - merit goods (educational institutions, health services etc.)
 - public goods (streetlights, parks etc.)
 - welfare services (unemployment benefits, pensions, child benefits etc.)
 - public services (police stations, fire stations, waste management etc.)
 - infrastructure (roads, telecommunications, electricity etc.).
- As an employer, it provides at all levels of government, employment to a large population, who work to provide the above mentioned goods and services. It also creates employment by contracting projects, such as building roads, to private firms.
- Support agriculture and other prime industries that need public support.
- Help vulnerable groups of people in society through redistributing income and welfare schemes.
- Manage the macroeconomy in terms of prices, employment, growth, income redistribution etc.
- Governments also manage its trade in goods and services with other countries by negotiating international trade deals.

Government Macroeconomic Aims

The government's major macroeconomic objectives are:

- **Economic Growth**: economic growth refers to an increase in the gross domestic product (**GDP**), the amount of goods and services produced in the economy, over a period of time. More output means economic growth. But if output falls over time (economic recession), it can cause:
 - fall in employment, incomes and living standards of the people
 - fall in the tax revenue the govt. collects from goods and services and incomes, which will, in turn, lead to a cut in govt. spending

- fall in the revenues and profits of firms
- low investments, that is, people won't invest in production as economic conditions are poor and they will yield low profits.
- **Price Stability:** inflation is the continuous rise in the average price levels in an economy during a time period. Governments usually target an inflation rate it should maintain in a year, say 3%. If prices rise too quickly it can negatively affect the economy because it:
 - reduces people's purchasing powers as people will be able to buy less with the money they have now than before
 - causes hardship for the poor
 - increases business costs especially as workers will demand higher wages to support their livelihood
 - makes products more expensive than products of other countries with low inflation. This will make exports less competitive in the international market.
- **Full Employment:** if there is a high level of unemployment in a country, the following may happen:
 - the total national output (goods produced) will fall
 - government will have to give out welfare payments (unemployment benefits) to the unemployed, increasing public expenditure while income taxes fall – causing a budget deficit
 - large unemployment causes public unrest and anger towards the government.
- **Balance of Payments Stability:** economies **export** (sell) many of their products to overseas residents, and receive income and investment from abroad; they also **import** (buy) goods and services from other economies, and make investments in other countries. These are recorded in a country's Balance of Payments (BoP).
 Exports > Imports = Surplus in BoP
 Exports < Imports = Deficit in BoP
 All economies try to balance this inflow and outflow of international trade and payments and try to avoid any deficits because:
 - if it exports too little and imports too much, the economy may run out of foreign currency to buy further imports
 - a BoP deficit causes the value of its currency to fall against other foreign currencies and make imports more expensive to buy, while a BoP surplus causes its currency to rise against other foreign currencies and make its exports more expensive in the international market.
- **Income Redistribution:** to reduce the inequality of income among its citizens, the government will redistribute incomes from the rich to the poor by imposing taxes on the rich and using it to finance welfare schemes for the poor. All governments struggle with income inequality and try to solve it because:
 - widening inequality means higher levels of poverty
 - poverty and hardship restricts the economy from reaching its maximum productive capacity.

Conflict of Macroeconomic Aims

When a policy is introduced to achieve one macroeconomic aim, it tends to conflict with one or more other aims. In other words, as one aim is achieved, another aim is undone. Let's look at some conflicts of government macroeconomic aims.

Full Employment v/s Price Stability

Low rates of unemployment will boost incomes of businesses and workers. This rise in incomes, mean higher demand and consumption in the economy, which causes firms to raise their prices – resulting in inflation. This is probably the most prominent policy conflict in the study of Economics.

Economic Growth & Full Employment v/s BoP Stability

Once again, as incomes rise due to economic growth and low unemployment,

people will import more foreign products and consume relatively less domestic products. This will cause a rise in imports relative to exports and a deficit may arise in the balance of payments.

Economic Growth v/s Full Employment

In the long run, when economic growth is continuous, firms may start investing in more capital (machinery/equipment). More capital-intensive production will make a lot of people unemployed.

Budget: a financial statement showing the forecasted government revenue and expenditure in the coming fiscal year. It lays out the amount the government expects to receive as revenue in taxes and other incomes and how and where it will use this revenue to finance its various spending endeavours. Governments aim for its budgets to be balanced.

Government spending

Governments spend on all kinds of public goods and services, not just out of political and social responsibility, but also out of economic responsibility. Government spending is a part of the aggregate demand in the economy and influences its well-being. The main areas of government spending includes defence and arms, road and transport, electricity, water, education, health, food stocks, government salaries, pensions, subsidies, grants etc.

Reasons governments spend:

- To supply goods and services that the private sector would fail to do, such as **public goods**, including defence, roads and streetlights; **merit goods**, such as hospitals and schools; and **welfare payments and benefits**, including unemployment and child benefits.
- To **achieve supply-side improvements** in the economy, such as spending on education and training to improve labour productivity.
- To spend on policies to **reduce negative externalities**, such as pollution controls.
- To **subsidise industries** which may need financial support, and which is not available from the private sector, usually agriculture and related industries.
- To help **redistribute income** and improve income inequality.
- To inject spending into the economy to **aid economic growth**.

Effects of government spending

- Increased government spending will lead to higher demand in the economy and thus **aid economic growth**, but it can also lead to inflation if the increasing demand causes prices to rise faster than output.
- Increased government spending on public goods and merit goods, especially in infrastructure, can lead to **increased productivity and growth** in the long run.
- Increased government spending on welfare schemes and benefits will increase living standards, and help **reduce inequality**.
- However too much government spending can also cause '**crowding out**' of private sector **investments** – private investments will reduce if the increase in government spending is financed by increased taxes and borrowing (large government borrowing will drive up interest rates and discourage private investment).

Tax

Governments earn revenue through interests on government bonds and loans, incomes from fines, penalties, escheats, grants in aid, income from public property, dividends and profits on government establishments, printing of currency etc; but its major source of revenue comes from taxation. Taxes are a **compulsory payment made to the government** by all people in an economy. There are many reasons for levying taxes from the economy:

- **It is a source of government revenue:** if the government has to spend on public goods and services it needs money that is funded from the economy itself. People pay taxes knowing that it is required to fund their collective welfare.
- **To redistribute income:** governments levy taxes from those who earn higher incomes and have a lot of wealth. This is then used to fund welfare schemes for the poor.
- **To reduce consumption and production of demerit goods:** a much higher tax is levied on demerit goods like alcohol and tobacco than other goods to drive up its prices and costs in order to discourage its consumption and production. Such a tax on a specific good is called **excise duty**.
- **To protect home industries:** taxes are also levied on foreign goods entering the domestic market. This makes foreign goods relatively more expensive in the domestic market, enabling domestic products to compete with them. Such a tax on foreign goods and services is called **customs duty**.
- **To manage the economy:** as we will discuss shortly, taxation is also a tool for demand and supply side management. Lowering taxes increase aggregate demand and supply in the economy, thereby facilitating growth. Similarly, during high inflation, the government will increase taxes to reduce demand and thus bring down prices. More on this below.

Classification of Taxes

Taxes can be classified into direct or indirect and progressive, regressive or proportional.

Direct Taxes are taxes on incomes. The burden of tax payment falls directly on the person or individual responsible for paying it.

- **Income tax:** paid from an individual's income. Disposable income is the income left after deducting income tax from it. When income tax rises, there is little disposable income to spend on goods and services, so firms will face lower demand and sales, and will cut production, increasing unemployment. Lower income taxes will encourage more spending and thus higher production.
- **Corporate Tax:** tax paid on a company's profits. When the corporate tax rate is increased, businesses will have lower profits left over to put back into the business and will thus find it hard to expand and produce more. It will also cause shareholders/owners to receive lower dividends/returns for their investments. This will discourage people from investing in businesses and economic growth could slow down. Reducing corporate tax will encourage more production and investment.
- **Capital gains tax:** taxes on any profits or gains that arise from the sale of assets held for more than a year.
- **Inheritance tax:** tax levied on inherited wealth.
- **Property tax:** tax levied on property/land.

Advantages:

- **High revenue:** as all people above a certain income level have to pay income taxes, the revenue from this tax is very high.
- **Can reduce inequalities in income and wealth:** as they are progressive in nature – heavier taxes on the rich than the poor- they help in reducing income inequality.

Disadvantages:

- **Reduces work incentives:** people may rather stay unemployed (and receive govt. unemployment benefits) rather than be employed if it means they would have to pay a high amount of tax. Those already employed may not work productively, since for any extra income they make, the more tax they will have to pay.
- **Reduces enterprise incentives:** corporate taxes may demotivate entrepreneurs to set up new firms, as a good part of the profits they make will have to be given as tax.
- **Tax evasion:** a lot of people find legal loopholes and escape having to pay any tax. Thus tax revenue falls and the govt. has to use more resources to catch those who evade taxes.

Indirect Taxes are taxes on goods and services sold. It is added to the prices of goods and services and it is paid while purchasing the good or service. It is called indirect because it indirectly takes money as tax from consumer expenditure. Some examples are:

- **GST/VAT:** these are included in the price of goods and services. Increasing these indirect taxes will increase the prices of goods and services and reduce demand and in turn profits. Reducing these taxes will increase demand.
- **Customs duty:** includes import and export tariffs on goods and services flowing between countries. Increasing tariffs will reduce demand for the products.
- **Excise Duty:** tax on demerit goods like alcohol and tobacco, to reduce its demand.

Advantages:

- **Cost-effective:** the cost of collecting indirect taxes is low compared to collecting direct taxes.
- **Expanded tax-base:** direct taxes are paid by those who make a good income, but indirect taxes are paid by all people (young, old, unemployed etc.) who consume goods and services, so there is a larger tax base.
- **Can achieve specific aims:** for example, excise duty (tax on demerit goods) can discourage the consumption of harmful goods; similarly, higher and lower taxes on particular products can influence their consumption.
- **Flexible:** indirect tax rates are easier and quicker to alter/change than direct tax rates. Thus their effects are immediate in an economy.

Disadvantages:

- **Inflationary:** The prices of products will increase when indirect taxes are added to it, causing inflation.
- **Regressive:** since all people pay the same amount of money, irrespective of their income levels, the tax will fall heavily on the poor than the rich as it takes more proportion of their income.
- **Tax evasion:** high tariffs on imported goods or excise duty on demerit goods can encourage illegal smuggling of the good.

Progressive Taxes are those taxes which burdens the rich more than the poor, in that **the rate of taxation increases as incomes increase**. An income tax is the perfect example of progressive taxation. The more income you earn, the more proportion of the income you have to pay in taxes, as defined by income tax brackets.

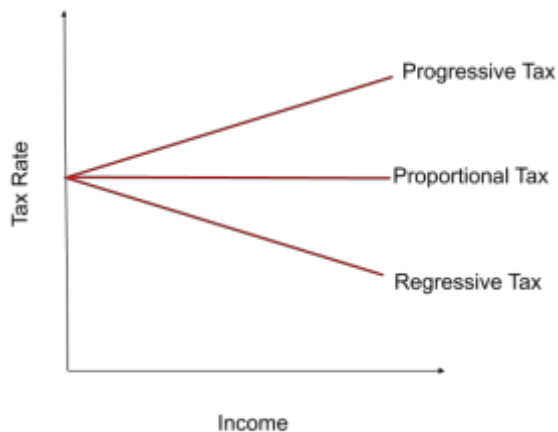
For example, a person earning above \$100,000 a month will have to pay a tax rate of 20%, while a person earning above \$200,000 a month will have to pay a tax rate of 25%.

Regressive Taxes are those taxes which burden the poor more than the rich, in that **the rate of taxation falls as incomes increase**. An indirect tax like GST is an example of a regressive tax because everyone has to pay the same tax when they are paying for the product, rich or poor.

For example, suppose the GST on a kilo of rice is \$1; for a person who earns \$500 dollars a month, this tax will amount to 0.2% of his income, while for a richer person who earns \$50,000 a month, this tax will amount of just 0.002% of his income.

The **burden on the poor is higher than on the rich**, making its regressive.

Proportional Taxes are those taxes which burden the poor and rich equally, in that **the rate of taxation remains equal as incomes rise or fall**. An example is corporate tax. All companies have to pay the same proportion of their profits in tax. For example, if the corporate tax is 30%, then whatever the profits of two companies, they both will have to pay 30% of their profits in corporate tax.



Qualities of a good tax system (the canons of taxation):

- **Equity:** the tax rate should be justifiable rate based on the ability of the taxpayer.
- **Certainty:** information about the amount of tax to be paid, when to pay it, and how to pay it should all be informed to the taxpayer.
- **Economy:** the cost of collecting taxes must be kept to a minimum and shouldn't exceed the tax revenue itself.
- **Convenience:** the tax must be levied at a convenient time, for example, after a person receives his salary.
- **Elasticity:** the tax imposition and collection system must be flexible so that tax rates can be easily changed as the person's income changes.
- **Simplicity:** the tax system must be simple so that both the collectors and payers understand it well.

Impacts of taxation

Taxes can have various direct impacts on consumers, producers, government and thus, the entire economy.

- The main purpose of tax is to raise income for the government which can lead to higher spending on health care and education. The impact depends on **what the government spends the money on**. For example, whether it is used to fund infrastructure projects or to fund the government's debt repayment.
- Consumers will have **less disposable income to spend** after income tax has been deducted. This is likely to lead to lower levels of spending and saving. However, if the government spends the tax revenue in effective ways to boost demand, it shouldn't affect the economy.
- Higher income tax reduces disposable income and can **reduce the incentive to work**. Workers may be less willing to work overtime or might leave the labour market altogether. However, there are two conflicting effects of higher tax:
 - **Substitution effect:** higher tax leads to lower disposable income, and work becomes relatively less attractive than leisure – workers will prefer to work less.

- **Income effect:** if higher tax leads to lower disposable income, then a worker may feel the need to work longer hours to maintain his desired level of income – workers feel the need to work longer to earn more.
- The impact of tax then depends on which effect is greater. If the substitution effect is greater, then people will work less, but if income effect is greater, people will work more
- Producers will have **less incentive to produce** if the corporate taxes are too high. Private firm aim on making profits, and if a major chunk of their profits are eaten away by taxes, they might not bother producing more and might decide to close shop.

Fiscal Policy

Fiscal policy is a government policy which adjusts government spending and taxation to influence the economy. It is the budgetary policy, because it manages the government expenditure and revenue. Government aims for a balance budget and tries to achieve it using fiscal policy.

A budget is in surplus, when government revenue exceed government spending.

While this is good it also means that the economy hasn't reached its full potential.

The government is keeping more than it is spending, and if this surplus is very large, it can trigger a slowdown of the economy.

When there is a budget surplus, the government employs an expansionary fiscal policy where govt. spending is increased and tax rates are cut.

A budget is in deficit, when government expenditure exceeds government revenue.

This is undesirable because if there is not enough revenue to finance the expenditure, the government will have to borrow and then be in debt.

When there is a budget deficit, the government employs contractionary fiscal policy, where govt. spending is cut and tax rates are increased.

Fiscal policy helps the government achieve its aim of economic growth, by being able to influence the demand and spending in the economy. It also indirectly helps maintain price stability, via the effects of tax and spending.

Expansionary fiscal policy will stimulate growth, employment and help increase prices. Contractionary fiscal policy will help control inflation resulting from too much growth. But as we will see later on, controlling inflation by reducing growth can lead to increased unemployment as output and production falls.

The money supply is the total value of money available in an economy at a point of time. The government can control money supply through a variety of tools including open market operations (buying and selling of government bonds) and changing reserve requirements of banks. (The syllabus doesn't require you to study these in depth)

The **interest rate is the cost of borrowing money.** When a person borrows money from a bank, he/she has to pay an interest (monthly or annually) calculated on the amount he/she borrowed. Interest is also be earned on the money deposited by individuals in a bank.

(The interest on borrowing is higher than the interest on deposits, helping the banks make a profit).

Higher interest rates will discourage borrowing and therefore, investments; it will also encourage people to save rather than consume (fall in consumption also discourage firms from investing and producing more).

Lower interest rates will encourage borrowing and investments, and encourage people to consume rather than save (rise in consumption also encourage firms to invest and produce more).

The monetary authority of the country cannot directly change the general interest rate in the economy. Instead, it changes the interest rates of borrowing between the central bank and commercial banks, as well as the interest on its bonds and securities. These will then influence the interest rate provided by commercial banks on loans and deposits to individuals and businesses.

Monetary Policy

Monetary policy is a government policy controls money supply (availability and cost of money) in an economy in order to attain growth and stability. It is usually conducted by the country's central bank and usually used to maintain price stability, low unemployment and economic growth.

Expansionary monetary policy is where the government **increases money supply by cutting interest rates**. Low interest rates will mean more people will resort to spending rather than saving, and businesses will invest more as they will have to pay lower interest on their borrowings. Thus, the higher money supply will mean more money being circulated among the government, producers and consumers, increasing economic activity. **Economic growth** and an **improvement in the balance of payments** will be experienced and **employment will rise**.

Contractionary monetary policy is where the government **decreases money supply by increasing interest rates**. Higher interest rates will mean more people will resort to saving rather than spending, and businesses will be reluctant to invest as they will have to pay high interest on their borrowings. Thus, the lower money supply will mean less money being circulated among the government, producers and consumers, reducing economic activity. This helps slow down economic growth and **reduce inflation**, but at the cost of possible **unemployment** resulting from the fall in output.

Supply side policies are microeconomic policies aimed at increasing supply and productivity in the economy, to enable long-term economic growth. Some of these policies include:

- **Public sector investments:** investments in infrastructure such as transport and communication can greatly help the economy by making the flow of resources quick and easy, and facilitate faster growth.
- **Improving education and vocational training:** the government can invest in education and skills training to improve the quality and quantity of labour to increase productivity.
- **Spending on health:** accessible, affordable and good quality health services will improve the health of the population, helping reduce the hours lost to illnesses and increasing productivity.
- **Investment on housing:** as more housing spaces are built, the geographical mobility of the population will increase, helping increase output.
- **Privatization:** transferring some public corporations to private ownership will increase efficiency and increase output, as the private sector has a profit-motive absent in public sector.
- **Income tax cuts:** reducing income tax will increase people's willingness to work more and earn more, helping increase the supply in the economy.
- **Subsidies** are financial grants made to industries that need it. More subsidies mean more money for producers to produce more, thereby increasing supply.

- **Deregulation:** removing or easing the laws and regulations required to start and run businesses so they can operate and produce more output with reduced costs and hassle, encouraging investments.
- **Removing trade barriers:** the govt. can reduce or withdraw import duties, quotas etc. on imports so that more resources, goods and services may be imported to increase productivity and efficiency in the domestic economy. It can also reduce export duties to increase export of resources, goods and services to other nations, thereby encouraging domestic firms to increase production.
- **Labour market reforms:** making laws that would reduce trade union powers would reduce business costs and increase output. Minimum wages could be reduced or done away with to allow more jobs to be created. Welfare payments like unemployment benefits could be reduced so that more people would be motivated to look for jobs rather than rely on the benefits alone to live. These will not only increase the incentive to work but also increase the incentive to invest.

For example, India, in the early 1990's undertook massive privatisation, liberalisation and deregulation measures; abolishing its heavy licensing and red tape policies, allowing private firms to easily enter the market and operate, and opening up its economy to foreign trade by reducing the excessive trade tariffs and regulations. This led to a period of high economic growth and helped India become the emerging economy it is today.

Supply-side policies have the direct effect of increasing **economic growth** as the productive capacity of the economy is realised. In doing so, it can also create more job opportunities and help **reduce unemployment**. Trade reforms will also enable it to **improve its balance of payments**.

However, the reliance on public expenditure and tax cuts mean that the government may run large budget deficits. Deregulation and privatisation will also reduce government intervention in the economy, which may prompt market failure.

Economic growth is an increase in the amount of goods and services produced per head of the population over a period of time.

The total value of output of goods and services produced is known as the **national output**. This can be calculated in three ways: using output, income or expenditure.

GDP (Gross Domestic Product): the total market value of all final goods and services provided within an economy by its factors of production in a given period of time.

Nominal GDP: the value of output produced in an economy in a period of time, measured at their current market values or prices is the nominal GDP.

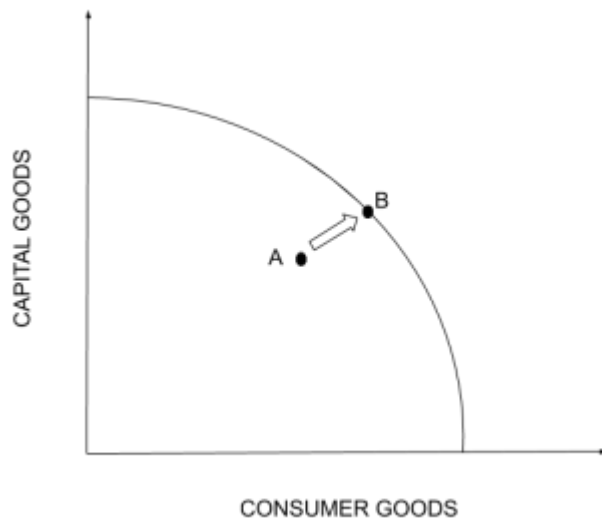
Real GDP: the value of output produced in an economy in a period of time, measured assuming the prices are unchanged over time. This GDP, in constant prices, provides a measure of the real output of a country.

GDP per head/capita: this measures the average output/ income per person in an economy. Since this takes into account the population, it provides a good measure of the living standards of an economy.

$$\text{GDP per capita} = \text{GDP} / \text{Population}$$

An increase in real GDP over time indicates economic growth as goods and services produced have increased. It indicates that the economy is utilizing its resources better or its productive capacity has increased. On a PPC, economic growth will be shown by an outward shift of the PPC, which is also called 'potential

growth'. 'Actual growth' occurs when the economy moves from a point inside the PPC to a point closer to the PPC.



This diagram shows 'actual growth' as the economy realizes its potential growth. In order to experience potential economic growth, the PPC would have to shift outwards.

Causes of economic growth

- **Discovery of more natural resources:** more resources mean more the production capacity. The discovery of oil and gas reserves have enabled a lot of economies (Norway, Saudi Arabia, Venezuela etc.) to grow rapidly.
- **Investment in new capital and infrastructure:** investment in new machinery, buildings, technology etc. has enabled firms and economies to expand their production capacities. Investment in modern infrastructure such as airports, roads, harbours etc. have improved access and communication in economies, helping in achieving quicker and more efficient production.
- **Technical progress:** New inventions, production processes etc. can increase the productivity of existing resources in industries and help boost economic growth.
- **Increasing the quantity and quality of factors of production:** A larger and more productive workforce will increase GDP. More skilled, knowledgeable and productive human resources thus help increase economic growth. Similarly, good quality capital, use of better natural resources, innovative entrepreneurs all aid economic growth in the long run.
- **Reallocating resources:** Moving resources from less-productive uses to more-productive uses will improve economic growth.

The benefits of economic growth:

- **Greater availability of goods and services** to satisfy consumer wants and needs.
- **Increased employment** opportunities and incomes.
- In underdeveloped or developing economies, economic growth can drastically **improve living standards** and bring people out of poverty.
- **Increased sales, profits** and business opportunities.
- Rising output and demand will encourage **investment in capital goods** for further production, which will help achieve long run economic growth.
- **Low and stable inflation**, if growth in output matches growth in demand.
- **Increased tax revenue** for government (as incomes and spending rise) that can be invested in public goods and services.

The drawbacks of economic growth:

- Technical progress may cause capital to replace labour, causing a rise in **unemployment**. This will be disastrous for highly populated underdeveloped and developing economies, pulling more people into poverty
- Scarce resources are used up rapidly when production rises. **Natural resources may get depleted** over time.
- Increasing production can **increase negative externalities** such as pollution, deforestation, health problems etc. Climate change is a consequence of rapid global economic growth.
- If the economy produces over its productive capacity and if the growth in demand outstrips the growth in output, economic growth **may cause inflation**
- Economic growth has also been accused of **widening income inequalities** in developing economies, because rich investors and businessmen gain more than the working class and poor during growth – the benefits of growth are not evenly distributed. This will cause **relative poverty to rise**.

Governments aim for **sustainable economic growth** which refers to a rate of growth which can be maintained without creating significant economic problems for future generations, such as depletion of resources and a degraded natural environment.

Recession

Recession is the phase where there is **negative economic growth**, that is real GDP is falling. This usually happens after there is rapid economic growth. High inflation during the boom period will cause consumer spending to fall and cause this downturn. Workers will demand more wages as the cost of living increase, and the price of raw materials will also rise, leading to firms cutting down production and laying off workers. Unemployment starts to rise and incomes fall.

Causes of recession:

- **Financial crises:** if banks have a shortage of liquidity, they reduce lending and this reduces investment.
- **Rise in interest rates:** increases the cost of borrowing and reduces demand.
- **Fall in real wages:** usually caused when wages do not increase in line with inflation leading to falling incomes and demand.
- **Fall in consumer/business confidence:** reduces both supply and demand.
- **Cut in govt. spending:** when government cuts spending, demand falls.
- **Trade wars:** uncertainty in markets, and thus businesses will be reluctant to invest during a trade war, causing supply to fall.
- **Supply-side shocks:** e.g. rise in oil prices cause inflation and lower purchasing power.
- **Black swan events:** black swan events are unexpected events that are very hard to predict. For example, COVID-19 pandemic in 2020 which disrupted travel, supply chains and normal business activity, as well consumer demand, has caused recessions in many countries.

Consequences of recession:

- **Firms go out of business:** as demand falls, firms will be forced to either reduce production to a level that is sustainable or close shop.
- **Unemployment:** cuts in production will cause a lot of people to lose work.
- **Fall in income:** cuts in production also causes fall in incomes.
- **Rise in poverty and inequality:** unemployment and lack of incomes will pull a lot of people into poverty, and increase inequality (as the rich will still find ways to earn).
- **Fall in asset prices** (e.g. fall in house prices/stock market): recessions trigger a crash in the stock markets and other asset markets as investors' and consumers' confidence in the well-being fall of the economy during a recession. The shares owned by investors will be worth less.
- **Higher budget deficit:** due to falling consumption and incomes, the government will see a fall in tax revenue, causing a budget deficit to grow.
- **Permanently lost output:** as firms go out of business and employment falls, it results in a permanent loss of output, as **the economy moves inwards from its PPC**.



If the economy was producing at A on its PPC, a recession will cause production to fall to B.

Policies to promote economic growth

Expansionary fiscal and monetary policies (demand-side policies) and supply-side policies described in the previous sections can be employed to promote economic growth, depending on the nature of the problems that are holding back the economy from growing. For example, if it is the poor quality of human capital (labour) that is preventing the economy from achieving its maximum productive capacity, the government should invest in education and vocational training centres to improve the quality of the labour force and increase productivity. If it is a lack of effective demand causing slow growth, the government should focus on cutting income taxes, indirect taxes and interest rates to boost spending.

Effectiveness of such policies:

- Demand-side policies that increase the rate of growth above the long-run trend rate will **cause inflation** and quickly lead to a recession if not controlled.
- Supply-side policies can **take considerable time to take effect**. For example, if the government invested in better education and training, it could take several years for this to lead to higher labour productivity.
- In a recession, supply-side policies won't solve the fundamental problem of deficiency of aggregate demand. Increasing the flexibility of labour markets and encouraging investment may help to some extent, but unless there is sufficient demand, firms will be reluctant to increase production and make new investments.

Some terms to be familiar with while we're discussing employment:

Labour force – the working population of an economy, i.e. all people of working age who are willing and able to work.

Dependent population – people not in the labour force and thus depend on the labour force to supply them with goods and services to fulfill their needs and wants. This includes students in education, retired people, stay at home parents, prisoners or similar institutions as well as those choosing not to work.

Employment is defined as an engagement of a person in the labour force in some occupation, business, trade, or profession.

Unemployment is a situation where people in the labour force are actively looking for jobs but are currently unemployed.

All governments have a macroeconomic objective of maintaining a low unemployment rate.

Full Employment is the situation where the entire labour force is employed. That is, all the people who are able and willing to work are employed – unemployment rate is 0%.

Changing patterns and level of employment

Over time, patterns and levels of employment change. It could be due to the effects of the business cycle that every economy goes through from time to time (growth and recession). It could be due to the changes to the demographics (population- age and gender) of the country. It could also be due to structural changes (dramatic shifts in how an economy operates). Let's look at some ways in which this happens:

As an economy develops, it undergoes a structural change as output and **employment shifts from primary sector to manufacturing and then to the tertiary (services) sectors**. This can be seen in rapidly developing countries like India where there employment in agriculture and allied industries are rapidly falling and people are moving towards the fast-growing service sector, especially IT and retail.

In the same way, **employment moves from the informal sector** (formally unrecognised trades such as street vending- output is not included in GDP and incomes are not taxed) **to the formal sector** (recognised – included in GDP and taxes) as economies develops. People who previously worked as street vendors may work in registered firms, as the economy develops.

Overtime, as an economy develops, the labour force also sees an **increase in the proportion of female labour**. As social attitudes become progressive and women are encouraged to work, more women will enter the labour force and contribute to growth.

Similarly, as the country develops, the **proportion of old people may increase** in proportion to young and working people (because death and birth rates fall). This will cause the labour force to shrink and cause a huge burden on the economy. Japan is now facing this problem as their birth rates are falling and it is up to a shrinking labour force to support a growing dependent senior population.

As economies become more market-oriented (government enterprises and interventions decline), the economy will naturally see a large proportion of the **labour force shift to the private sector**.

Measuring unemployment

Economies periodically calculate the number of people unemployed in their economies, to check the unemployment rate and see what policies they should implement to reduce it if it is too high. They can do this in two ways:

- **Claimant count:** unemployed people can file for unemployment claims (benefits/allowances provided to the unemployed job seekers) by the government. The government can count the total number of unemployment claims made in the economy to measure unemployment.
- **Labour surveys:** economies conduct surveys among the entire labour force to collect data about it. This will include data on the number of people unemployed.

Unemployment rate = number of people unemployed / total no. of people in the labour force

There are some problems with measuring employment.

Under-employment: people may be officially classed as employed but they may be working fewer hours than they would like. For example, they may have a part-time job, but want a full-time job. This is considered as unemployment because they may not fulfil the working hours needed to be considered employed.

Inactivity rates: the long-term unemployed may become discouraged and leave the labour market completely. In effect they are not working, but they are classed as economically inactive rather than unemployed. So, the unemployment rate can be understated.

The causes/types of unemployment

- **Frictional unemployment:** this occurs as a result of workers leaving one job and spending time looking for a new one. This type of unemployment is short-lived.
- **Cyclical unemployment:** this occurs as a result of fall in aggregate demand due to an economic recession. When demand falls, firms will cut their production and workers will lose their jobs. There will be a nation-wide rise in unemployment.
- **Structural unemployment:** this occurs due to the long-term change in the structure of an economy. Workers end up having the wrong skills in the wrong place – causing them to be unfit for employment. This can be explained by dividing it further:
 - **Technological unemployment:** this has rose in recent times as industrial robots, machinery and other technology are being substituted for labour, leaving people jobless.
 - **Sectoral unemployment:** unemployed caused as a sector/industry declines and leave its workers unemployed.
- **Seasonal unemployment:** this occurs as a result of the demand for a product being seasonal. For example, the demand for umbrellas will fall in non-monsoon seasons, and so workers in umbrella manufacturing firms will become unemployed over those seasons.
- **Voluntary unemployment:** when people choose not to work for various reasons – they want to pursue higher education, would like to take a break etc. Because they're not actively looking for work, voluntarily unemployed people do not belong to the labour force!

The consequences of unemployment

- People will **lose their working skills** if they remain unemployed for a long time and may find it even harder to find suitable jobs. As people remain unemployed, their incomes will be low, and **living standards will fall**.
- Unemployment will also **lead to poverty**, homelessness and ill health and encourage people to steal and commit other crimes to make money- **crime rates** will rise.
- People losing skills is not just detrimental to the unemployed individuals but also to firms who may employ these people in the future. They will have to **retrain these workers**.
- Firms will have to **pay redundancy payments** to workers they lay off.
- **Workers will be demotivated** as they fear they could lose their jobs, especially in a recession.
- As firms lay off workers, they will be left with spare capacity- unutilised machinery, tools and factory spaces, leading to **higher average costs**.
- Due to low incomes, people's purchasing power/consumption will fall, causing **demand to fall**.
- People will need to rely on charity or **government unemployment benefits** to support themselves. These benefits are provided from tax revenue. But now, as incomes have fallen tax revenue will also fall. This might mean that people remaining in work will have to pay more of their income as tax, so that it can be distributed as unemployment benefits to the unemployed. The **burden on tax-payers will rise**.

- Public expenditure on other projects such as schools, roads etc. will have to be cut down to make way for benefits. There is **opportunity cost** involved here.
- The economy doesn't reach its maximum productive capacity, i.e., they are **economically inefficient** on the PPC. The economy **loses output**.

Policies to reduce unemployment

Both demand-side policies and supply-side policies help reduce unemployment.

Demand-side policies will address the unemployment caused by demand deficiency (cyclical) while the supply-side measures will curtail structural and frictional unemployment.

- **Expansionary policies to increase demand:** expansionary fiscal policies like cutting down taxes and increasing government spending (which also creates jobs) and expansionary monetary policies like cutting interest rates will help boost demand in the economy, to keep production and employment high. However these will take effect only with a time lag. Cutting tax rates won't help if people don't have confidence in the economy and prefer to save. Similarly, cutting interest rates will also be ineffective if banks are unwilling to lend to businesses, due to low confidence in the economy.
- **Depreciate the exchange rate:** as the currency depreciates, the country's exports will become cheaper and so export demand from abroad will increase, helping boost production and employment in the export industries.
- **Control inflation:** higher inflation causes firms to lay off workers to reduce costs. So if the government tries to control inflation via monetary tools, it will help reduce firm costs and increase employment. But there is also the argument that as unemployment rises, incomes will also rise, driving up prices again.
- **Cutting unemployment benefits** to provide incentive to work: many people don't work because they are comfortable living off the unemployment benefits provided by the government. Cutting down on these benefits, will persuade them to look for work and earn. But this would of course, go against the welfare principle of the government.
- **Restricting imports and encourage exports:** a lot of unemployment occurs when good quality and cheaper foreign products put domestic industries out of business. Controlling imports using import tariffs and quotas will encourage domestic firms to emerge and increase production and thus increase employment. Similarly, easing controls on labour-intensive export industries will open up new job opportunities. However such protectionist measures can harm the country in the long-run as efficient competition from abroad reduces.
- **Cutting down minimum wages:** minimum wages increase firms' labour costs and so they will lay off workers. Lowering the minimum wages will encourage firms to employ more labour.
- **Remove labour market regulations:** letting the market have a free hand in the labour market – abolishing maximum working weeks, minimum wages, making it easier to hire and fire workers – will improve the labour market flexibility, can improve imperfections in the labour market. However, this can cause temporary unemployment and cause greater job insecurity.
- **Training/Retraining:** structural employment issues can be eliminated by retraining the unemployed in skills required in modern industries. This will also improve occupational mobility. This is very expensive when done on a large scale across the economy, requiring training centres to be built, and trainers to be employed. The benefits of providing skills and training will only be reaped in the long-term.
- **Promote industries in unemployed areas:** a lot of employment is created when government provide subsidies and tax breaks for new industries which set up shop in certain backward regions.
- **Increase geographical mobility of labour:** frictional unemployment is caused because people can't move around to find good jobs. The government can improve labour mobility by investing in transport and housing services.

- **Provide information:** frictional unemployment can be eliminated to an extent by making information available about job vacancies to the unemployed through job centres, newspapers, government websites etc.

nflation

Inflation is the general and sustained rise in the level of prices of goods and services in an economy over a period of time.

For example, the inflation rate in UK in 2010 was 4.7%. This means that the average price of goods and services sold in the UK rose by 4.7% during that year.

Inflation is **measured using a consumer price index (CPI)** or retail price index (RPI).

The consumer price index is calculated in this way:

- A selection of goods and services normally purchased by a typical family or household is identified.
- The prices of these '**basket of goods and services**' will then be monitored at a number of different retail outlets across the country.
- The average price of the basket in the first year or '**base year**' is **given a value of 100**.
- The **average change in price of these goods and services over the year is calculated**.
- If it rises by an average of 25%, the new index is $125\% \times 100 = 125\%$. If in the next year there is a further average increase of 10%, the price index is $110\% \times 125 = 137.5\%$. The average inflation rate over the two years is thus $137.5 - 100 = 37.5\%$

Causes of Inflation

- **Demand-pull inflation:** inflation caused by an **increase in aggregate demand** is called demand-pull inflation. This is also defined as the increase in price **due to aggregate demand exceeding aggregate supply**. Demand could rise due to higher incomes, lower taxes etc. The demand curve will shift right, causing an extension in supply and a rise in price.
- **Cost-push inflation:** inflation caused by an **increase in cost of production** in the economy. The cost of production could rise due to higher wage rate, higher indirect taxes, higher cost of raw materials, higher interest on capital etc. The supply curve will shift left causing a contraction in demand and a rise in price.
- A lot of economists agree that **a rise in money supply in contrast with output is the key reason for inflation**. If the GDP isn't accelerating as much as the money supply, then there will be a higher demand which could exceed supply, leading to inflation.

The consequences of inflation

- **Lower purchasing power:** when the price level rises, the lesser number of goods and services you can buy with the same amount of money. This is called a fall in the purchasing power. When purchasing power falls, consumers will have to make choices on spending.
- **Exports are less internationally competitive:** if the prices of exports are high, its competitiveness in international markets will fall as lower priced foreign goods will rival it. This could lead to a current account deficit if exports lower (especially if they are price elastic).
- **Inflation causing inflation:** during inflation, the cost of living in the economy rises as you have to pay more for goods and services. This might cause workers to demand higher wages increasing the cost of production. If the price of raw materials also increase, the cost of production again increases, causing cost-push inflation.
- **Fixed income groups, lenders, and savers lose:** a person who has a fixed income will lose as he cannot press for higher wages during inflation (his/her real wages fall as purchasing power of his/her wages fall). Lenders who lent money before inflation and receive the money back during inflation will lose the value on their money. The same amount of money is now worth less (here, the people who borrowed gain purchasing power). Savers also lose because the interest they're earning on savings in banks does not increase as much as the inflation, and savers will lose the value on their money.

Policies to control inflation

- **Contractionary monetary policy** that will reduce demand: contractionary monetary policy is the most popular policy employed to curtail inflation. Raising interest rates will discourage spending and investing (as cost of borrowing rises) and reduce the money supply in the economy, helping cut down on demand. But this depends a lot on the consumer and business confidence in the economy. Spending and investing may still continue to rise as confidence remains high. There is also a considerable time lag for monetary policy to take effect.
- **Contractionary fiscal policy** that will reduce demand: raising taxes will discourage spending and investing and cutting down on government spending will reduce aggregate demand in the economy, helping bring down the price level. However, this is an unpopular policy only employed when inflation is severe.
- **Supply side policies:** supply-side policies such as privatisation and deregulation hope to make firms competitive and efficient, and thus avoid inflationary pressures. But this is a long-term policy only helping to keep the long-term inflation rate stable. Sudden surges in inflation cannot be addressed using supply side measures
- **Exchange rate policy:** Appreciating the domestic currency can lower import prices helping reduce cost-push inflation arising from expensive imported raw materials. It also makes export more expensive, helping lower the export demand in the economy as well as creating incentives for exporting firms to cut costs to remain competitive.

Deflation

Deflation is the **general fall in the price level**.

Deflation is also **measured using CPI**, but instead of showing figures above 100, it will show an index below 100 denoting a deflation. For example, a drop in the average prices of the basket of goods in a year is 10%, the deflation will be $100 - (90\% * 100 = 90\%) = 10\%$.

Causes of deflation

- **Aggregate supply exceeding aggregate demand:** when supply exceeds demand, there is an excess of output in the economy not consumed, causing prices to fall.
- **Demand has fallen** in the economy: during a recession, a fall in demand in the economy causes general prices to fall and cause a deflation.
- **Labour productivity has risen:** higher output will lead to lower average costs, which could reflect as lower prices for products.
- **Technological advance** has reduced cost of production, pulling down cost-push inflation.

Consequences of deflation

- Lower prices will discourage production, resulting in **unemployment**.
- As demand and prices fall, **investors will be discouraged to invest**, lowering the output/GDP.
- Deflation can **cause recession** as demand and prices continue to fall and firms are forced to close down as enough profits are not being made.
- **Tax revenue of the government will fall** as economic activity and incomes falls. They might be forced to borrow money to finance public expenditure.
- **Borrowers will lose** during a deflation because now the value of the debt they owe is higher than when they borrowed the money.
- Deflation will **increase the real debt burden of the government** as the value of debt money increases.

Policies to control deflation

- **Expansionary monetary policy** to revive demand: cutting interest rates will encourage more spending and investment in the economy which will stimulate prices to rise. However, if interest rate is already at a very low point where decreasing it any further won't increase spending, because people still prefer to save some money and pay off debts, and banks are not willing to lend at a very low interest rate, (this situation is called a liquidity trap), then cutting interest rates will have no effect on spending.
- **Expansionary fiscal policy** to revive demand: increasing government spending in the economy, especially in infrastructure will help raise demand, along with cuts in direct taxes. The money

for this expenditure can be created via quantitative easing (selling government bonds to the public).

- **Devaluation:** devaluing the currency through selling domestic currency and/or increasing the money supply will cause export prices to fall, encouraging production of exports, resulting in higher demand; and also increase prices of imported products which will raise costs and prices for products in the economy.
- **Change inflation expectations:** when a deflation is expected, businesses won't increase wages and consumers won't pay higher prices (because they expect prices to fall in the future). This will cause the deflation they expected. But if the monetary authorities indicate that they expect higher inflation, firms will pay their workers more and consumer will spend more now, avoiding a deflation.

Living standards or standards of living refer to all the factors that contribute to a person's well-being and happiness

Measuring Living Standards

- **GDP per head/capita:** this measures the average income per person in an economy.

Real GDP per capita = Real GDP / Population

Merits of using GDP per capita to measure living standards:

- GDP is a useful measure of the total production taking place in the country, and so indicates the material well-being of the economy
- it also takes population into consideration, adding emphasis on the goods and services available to individuals
- since it is calculated on output, is a good indicator of the jobs being created
- GDP data is readily available so is population data

Limitations of using GDP per capita to measure living standards:

- **it takes no account of what people can buy using their incomes.** A country with a high GDP per head may be no better off than a country with a low GDP per head, if there are far fewer products to choose from
- similarly, **GDP doesn't consider changes in technology that can have a large impact on living standards.** People might have had more income in the last decade but they couldn't benefit from all the technology available today
- **distribution of income is very unequal in reality**, so the GDP per head isn't accurate. Some people might be very rich while others very poor, but the GDP per head will only give the average incomes
- real GDP **excludes the unpaid work people do** for charities and voluntary organizations etc. Thus, it understates the total output
- GDP **also doesn't differentiate between the positive and negative values economies place on different output/expenditure.** For example, if the output rises because the sales of tobacco, alcohol or pornographic materials, it might show in the records as a rise in GDP per head but might not actually make people better off. Similarly, GDP might rise if the government has to rebuild after a natural disaster, which doesn't mean living standards have risen
- the official GDP figures can be overstated due to technical errors or by political manipulation to look good, and give a wrong picture of living standards
- this measure doesn't consider leisure activities, health and education levels, environmental quality- **all that determines people's happiness** and well-being

- in order to effectively compare GDP per head across countries, they need to be converted to a common currency and adjusted for differing purchasing power in different countries
 - comparing GDP per head can also be unreliable as GDP accounting methods can be different for different countries.
 -
 - **Human Development Index (HDI):** used by the United Nations to compare living standards across the globe, the HDI combines different measures into one to give a HDI value from 0 (lowest) to 1 (highest). These are:
 - **Income index**, measured using the average national income – **GNI per head** adjusted for differences in exchange rate and prices in different countries (purchasing power parity)
 - **Education index**, measured by how many years on average, a person aged 25 will have spent on education (**mean years of schooling**) and how many years a young child entering school can now be expected to spend in education in his entire life (**expected years of schooling**)
 - **Healthcare index**: measured by **average life expectancy at birth**
- The benefits of using HDI to measure living standards:

- it takes into account some **major indicators** of living standards
- recognises that it is not just output or income that determines living standards, but also **social factors**
- it is a useful method to **compare global living standards**– it shows clear patterns of living standards
- it is very **useful and reliable** measure since its produced by the UN and is thus also widely used and recognised

The limitations of HDI to compare living standards:

-
- **it combines a set of separate indicators into one**, so a country with good literacy rates and living standards but poor life expectancy can have a low HDI value
- **there are wide divergences in HDI within countries**
- **GNI per head doesn't say anything about inequalities in income and wealth within countries**
- **it doesn't consider other factors** such as environmental quality, access to safe drinking water, political freedom, crime rates etc. which are also important indicators of living standards
- **the HDI information for all countries may not be available** such as war-struck countries where civilisation has been disrupted

In the 2019 HDI index published by the UN, Norway comes first with an HDI index of 0.954 while Niger comes last with an index of just 0.377 owing to very low levels of education and GNI per head. See the full list

at <http://hdr.undp.org/en/content/2019-human-development-index-ranking>

Reasons for differences in living standards and income distribution within and between countries

These have been discussed above in the merits and limitations of using GDP per capita and HDI. More will be discussed in the coming chapters. Some other reasons are discussed below

Difference in living Standards within a country: there can be variations in living standards within a country. An excellent example of this is the high living standards of the Indian state of Kerala (where IGCSE AID is based!) which has a HDI index of 0.779 while the poorest state of Bihar stands at 0.567 (2018).

- **Regional variances in income and consumption**
- **Major type of sectors/jobs:** manufacturing and services heavy regions will have higher incomes, education and health services compared to agricultural regions
- **Local government provisions of education and health**
- **Difference in living Standards between countries:**
- **Productivity of industries:** more productive industries yield more output and incomes
- **Major industries:** what makes countries like Qatar and Norway achieve some of the world's highest per capita incomes is that their income comes mostly from petroleum industries that are scarce and highly demanded internationally
- **Population:** dense population lower per capita income and put pressure on scarce resource
- **Ability of citizens pay taxes:** higher tax-base and taxable incomes allow governments to invest in infrastructure and welfare programmes
- **Provision of health and educational facilities**
- **Variety of goods/services produced:** if citizens can choose from a wide variety of products, living standards rise. Western countries like US enjoy this
- **War, crime and natural disasters:** war-struck countries of Asia, the high crime rates of Latin America and frequent natural disasters in island countries, drive down their living standards as they damage infrastructure and put people into hardship

Absolute poverty: the inability to afford basic necessities needed to live (food, water, education, health care and shelter). This is measured by the number of people living below a certain income threshold (called a poverty line).

Relative poverty: the condition of having fewer resources than others in the same society. It is measured by the extent to which a person's or household's financial resources fall below the average income level in the economy. Relative poverty is basically a measurement of income inequality since a high relative poverty should indicate a higher income inequality.

Causes of poverty

- **Unemployment:** when people are unemployed and have to go without income for a long time, they may end up having to sell their possessions, consume less and go into poverty.
- **Low education levels:** this means that people are uneducated, unskilled and unable to find better jobs, keeping them in poverty.
- **The size of family:** more family members with only a few people earning, means more costs of living, pulling the family into poverty if they're not earning much.
- **Age:** older people are likely to have more health problems and be less suitable for further employment, causing poverty. Young people are still employable and may find ways to earn an income.

- **Poor government support** for basic services.
- **Poor health:** ill mental and physical health is both a cause and result of poverty.
- **Overpopulation:** high population density will put pressure on scarce resource and the economy may not be able to produce and provide for everyone, causing poverty.
- **Minority group/ethnicity/migrants:** will face discrimination from bureaucrats, employers and the society at large and so won't be able to access and enjoy all services. E.g.: African-Americans in the US tend to be poorer than their white counterparts.
- **Gender:** women usually face discrimination, especially in employment and end up being poorer than men.

Policies to alleviate poverty

- Introduce measures to **reduce unemployment:** an expansionary fiscal/monetary policy will increase aggregate demand and increase employment opportunities. Income and standards of living will rise.
- **Impose progressive taxes:** income taxes are progressive, that is, they increase as income increases. Imposing these will mean that people on higher incomes will pay a large percentage of their incomes as tax and help reduce relative poverty.
- Introduce **welfare services:** money from taxes can be provided as income support to people with very low incomes. It can also be used to provide free or low-cost homes, healthcare and education.
- Introduce **minimum wage legislation** to raise the wage of low-paid employees.
- Increase the quantity and **quality of education.**
- Attract and invite **inward investments** from firms abroad to provide jobs and incomes for people.
- **Overseas aid** could be gained from foreign governments and aid agencies. This will include food aid, financial aid, technological aid, loans and debt relief.

Population is the total number of people inhabiting a specific area. Two-hundred years ago, the world population was just over a billion, now it is about 7.7 billion, with China and India having populations above 1 billion each! It is projected to hit 10 billion by 2056.

Factors that affect population

- **Birth rates:** the average number of children born in a country each year compared to the total population of an economy is known as the birth rate. This is usually expressed as the number of births for every 1000 people in the population.

Why do different countries have different birth rates?

- **Living standards:** improved quality and availability of food, housing, clean water and medical care result in fewer babies dying. Countries where children often die due to poor living standards have higher birth rates (people have more children fearing that some of their children might die. These children can then work to produce food and earn incomes).
- **Contraception:** increased use of contraception and legalisation of abortion have reduced birth rates in developed countries.
- **Customs and religion:** many religious beliefs don't allow the use of contraceptive pills, so birth rates in those communities rise. In developed economies it is now less fashionable to have large families, so birth rates have fallen.
- **Changes in female employment:** more females in developed countries entering the labour force has resulted in falling birth rates since they do not want motherhood to affect their careers.
- **Marriage:** in developed countries, people are tending to marry later in life, so birth rates have reduced.

- **Death rates:** the number of people who die each year compared to every 1000 people of the population is the death rate of an economy.

Reasons for differing death rates in different economies:

- **Living standards:** just as birth rates, death rates also tend to be very high in less-developed economies due to lack of good-quality food, shelter and medical care. Malnutrition remains the major cause of high death rates in these countries. In developed countries, the major causes of death include lifestyle diseases, mostly caused by unhealthy diets.
- **Medical advances and health care:** lack of medical care and infrastructure in less-developed countries continue to be a cause for high death rates.
- **Natural disasters and wars:** hurricanes, floods, earthquakes and famine due to lack of rain and poor harvests, and wars and civil conflicts increase death rates.
- **Net Migration:** migration refers to the number of people entering (immigration) and leaving (emigration) the country. **Net migration** measures the difference between the immigration and emigration to and from an economy. A **net inward migration** will increase the working population of the economy, but can put pressure on governments finances as demand for housing, education and welfare increase. A **net outward migration** may increase the income per capita (if the emigrants send money to families back home) and thus the HDI, but can result in loss of skilled workers.

Reasons for differing net migration in different economies:

- **Living standards:** people move to countries where living standards are high which they can benefit from.
- **Employment/wages:** people migrate mainly to seek better job opportunities. Widespread unemployment and low wages in the home country will cause people to move to countries with better employment opportunities and higher wages.
- **Climate:** very cold or very warm countries/regions will face more emigration than other countries.

Population structure

The structure of a population can be analyzed using:

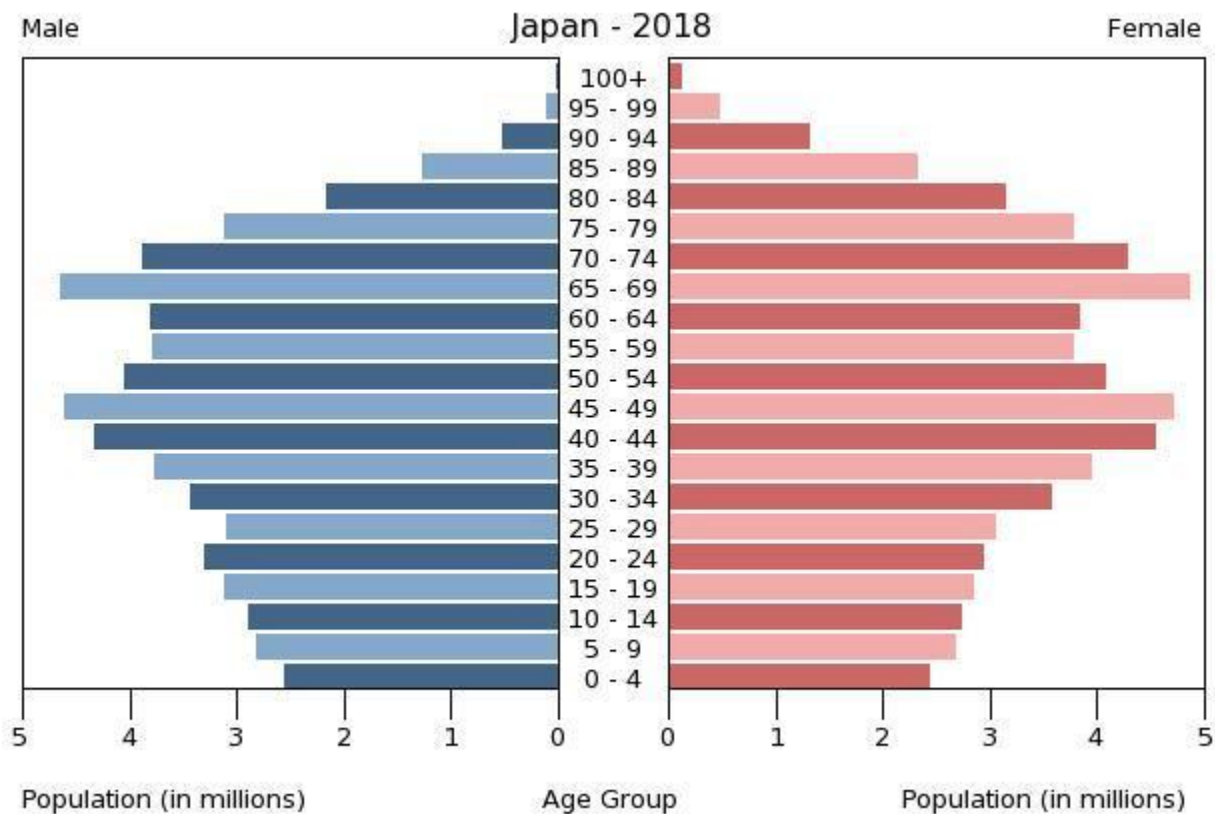
- **Age distribution:** the number of people in each age-group.
Falling birth and death rates mean that the average age in developed countries are rising whereas in developing and less-developed economies, high death and birth rates result in low average ages. The median age in developed Monaco is highest at 53.1 while in under-developed Niger it is just 15.3. Dependency ratio is the ratio of the dependent population (those outside the labour force – children and senior citizens – who depend on the labour force to supply them with goods and services by paying taxes) to the total population in an economy. A high dependency population, such as in Japan, put pressure on the government to increase taxes rates in order to raise more revenue to support the dependents, putting pressure on the labour force.

Consequences of an ageing population:

- The workforce will decline and there will be much dependence on the tax-paying population to fund the welfare of old people.
- Increase in demand for products for old people including healthcare.
- The government will have to spend more on housing, old age welfare schemes etc.
- Old people are less mobile and so the economy will be slow to adapt to new technologies.
- **Gender distribution:** the balance of males and females. The sex ratio measures the no. of males to the no. of females (the global sex ratio is 101:100; while Arab countries have sex ratios as high as 2.87, island countries register low sex ratios). Since the average female lives longer than the average male, there are more females in the older age-groups than males. **Gender imbalance** is an excess of males or females and is caused by
 - Wars killing many young males

- Violence towards females (honour killings, rapes)
 - Sex-specific immigration – more males immigrate to a country looking for work
- Consequences of changes in the gender distribution:
- having more females will encourage birth rates to rise and increase population growth
 - more females in employment will increase productivity
 - more females in education and employment will increase living standards
 - a more balanced gender distribution can aid better social equality as social attitudes towards women in education and employment become progressive

Population pyramids display the age and gender distribution of an economy. The vertical axes show the age groups and the horizontal axes show the gender groups- males on the left and females on the right.



- **Geographic distribution:** where people live. 90% of the world population live in developing countries. This puts a lot of pressure on scarce resources in these countries. About half of the world population live in urban areas, and this continues to rise, which has helped increase production and living standards but resulted in rapid consumption of natural resources and high levels of pollution and congestion.
- **Occupational distribution:** what jobs people work in. In developed economies, more people work in the service sector while in less-developed economies, most people work in agriculture. In developing economies, there is a huge migration of workers from primary production to manufacturing and service sectors. Female employment and self-employment are also rising, which will add to production and higher living standards.

An optimal population is one where the output of goods and services per head of the population is maximised. An economy is underpopulated when it does not have enough labour to make the best use of its resources; and it is overpopulated when the population is too large given the resources it has.

Effects of increasing population size

- **Increases size of the home market** and thus potential for increase in aggregate demand in the long-run.
- Higher demand and incomes will lead to **more economic growth** and expansion.
- **Increased supply of labour.**
- Puts more **pressure on already scarce resources**, especially land.
- **More capital goods** will have to be produced to sustain and satisfy the needs and wants of the enlarged population.
- **Fall in rate of productivity** in line with the law of diminishing returns – too many people working on limited resources means low productivity.
- **Shift of employment and output from the primary sector towards the services sector** because land for primary activities is fixed, but want for services is practically infinite as population grows, and the emergence of mechanisation and technologies will force people out of the primary sector.
- **Congestion of urban centres:** as population and incomes rise, people will move to cities and towns which will become crowded. There will be need for heavy transport, communications, housing, waste management infrastructure spending.

Economic development refers to the increase in the economic welfare of people through growth in productive scale and wealth of an economy. Governments aim for their countries to expand from developing economies to developed economies.

Developed countries are characterised by high GDP per capita, high life expectancy, high literacy rate, a stable or dwindling population growth, excellent infrastructure, high levels of foreign investments, excellent healthcare, high productivity, and a relatively large tertiary sector. Example: Japan

Under-developed economies or less-developed economies are characterised by very low GDP per capita, high population growth, poor infrastructure, healthcare and education, low literacy rates, low levels of foreign investments, poor productivity, and a relatively large primary sector.

Example: Somalia

Developing economies are countries that are becoming more developed through expansion of the industrial sector and fewer people suffering the extremes of poverty. They may attract high levels of foreign investments and will be undergoing major economic shifts towards the tertiary sector. However they may still have a low standard of living, owing to high population growth. Example: India

The reasons for low economic development

- **Over-dependence on agriculture:** farming is the most common work in less-developed economies. Most people work to feed themselves and their families and sell off any surplus. This means that there is little or no trade happening, which results in poor incomes, no economic growth or development.
- **Domination of international trade by developed economies:** the more wealthier developed economies have exploited poorer countries by buying up their natural resources at low prices and selling products made from them in international markets at higher prices. Rich countries also protect their industries by paying subsidies to domestic producers, increasing global supply, and in turn, lowering prices. Poor economies cannot compete with these very low prices, and they lose their jobs and incomes.
- **Low levels of savings** because of low incomes and widespread poverty.
- **Lack of capital:** low incomes in under-developed economies lead to a lack of savings that could be invested in industries.
- **Poor investment in infrastructure:** good infrastructure in transport, health and education is essential for growth and development.

- **High population growth:** rapidly expanding populations (due to high birth rates) in less-developed countries will reduce the real GDP/income per head.
 - **Wars and conflicts deplete resources:** there is little scope for development when the country is a war zone.
 - **Corrupt and/or unstable governments:** causes neglect of economy and citizens' welfare
- The opposites are true for developed economies.

Some **development indicators** that are used to measure how developed an economy are: GDP per capita, population living on less than \$1 a day, life expectancy at birth, adult literacy rate, access to safe water supplies and sanitation, proportion of workers in different sectors of production etc.

Specialisation is when a nation **concentrates its productive efforts on producing a limited variety of goods and services in which they're really efficient and productive at and have an advantage over other economies in.**

For example, due to the existence of vast oil and gas reserves in the region, Middle-Eastern countries concentrate their production on petroleum and have made a fortune off of it.

Specialisation is determined on the basis of either resource allocation or of cost of production.

Absolute advantage: when **one country can produce more efficiently than another** either by producing more of a good or service with same amount of resources or producing the same amount of a good or service with fewer resources. For example, India has an absolute advantage in operating call centres because of its abundant and cheap labour force, compared to western countries.

Comparative advantage: **when one country can produce a good at a lower opportunity cost** (in terms of other goods and services being forgone) than another country. It takes into account the opportunity cost incurred in producing each good. For example, India may have an absolute advantage in operating call centres (against Philippines), but it has lower opportunity costs in other IT industries, than Philippines. Thus, Philippines has, in recent years, seen a growing call centre industry while India has seen theirs decline.

Note: you are not required by the syllabus to know the terms 'absolute advantage' and 'comparative advantage', but only the principles.

Advantages of international specialisation:

- **Economies of scale and efficiency:** just like specialisation by individuals, countries can specialise in what they do best, and this leads to efficiency and economies of scale. It can therefore increase output while reducing costs. When more countries specialise, world output increases.
- **Job creation:** specialisation leads to increased output and therefore it could lead to more investment and thus jobs are created. Moreover, it requires skilled labour and thus earnings are higher.

- **Allows more international trade** to take place. Therefore goods and services produced under the most efficient conditions can be traded and all countries can benefit from them.
- **Revenue to the government:** as income increases and more trade takes place, it can increase government revenue from taxes.
- **Wider markets:** specialisation and trade allow firms to sell their products to international markets, helping them build international brands and increase market shares and profits.
- **Consumer sovereignty:** consumer across the globe will now be able to buy cheap and high quality products from around the world. Because of specialisation and trade, we now can get the best chocolate from Switzerland, the best coffee from Ghana and Colombia, cheap IT services from India, oil from the Middle East, and budget cars from Japan.

Disadvantages of international specialization:

- **Structural unemployment:** even though national level specialisation usually creates more jobs, there is a risk that certain types of structural unemployment might occur. As the country moves towards specialisation, the workers in the declining industries will be put out of work.
- **Over-exploitation of resources:** output maybe increased by over-exploiting Today, international specialization and trade is causing rapid depletion of non-renewable resources like oil and coal. Middle Eastern countries are depleting their oil resources so quickly, they are now building new industries to sustain them in the future.
- **Threat of foreign competition:** non-specialised industries of a country will face fierce competition from the foreign countries that specialise in them.
- **Risk of over-specialisation:** because of more international dependence on other countries for trade (they will have to sell their specialised products to other countries and buy other products they need from abroad), any global economic change will greatly affect highly specialised countries. For example, petroleum-exporting countries have seen their revenues dip when oil prices fall. They are now trying to diversify into other products like tourism to sustain them.
- **Strategic vulnerability:** relying on other countries for vital goods and services makes a country dependent on those countries. Political or economic changes abroad may impact the supply of goods or services available to the country.

Globalisation is a process of interaction and integration among the people, companies, and governments of different nations, a process driven by international trade and investment and aided by information technology [definition by: www.globalization101.org]

Multinational Corporations (MNCs)

Businesses which **have their operations, factories and assembly plants in more than one country** are known as multinational businesses. Examples include Starbucks, IKEA, Toyota, Adidas etc. The country they are based in is called the home country, and the countries they operate in are called host countries.

Advantages to home country:

- MNCs create **opportunities for marketing** the products produced in the home country throughout the world.

- They create **employment opportunities** to the people of home country, both at home and abroad.
- It aids and encourages the **economic growth** and development of the home country.
- MNCs help to maintain **favourable balance of payments** of the home country in the long run as they export their products abroad.

Advantages to host country:

- Provides significant **employment and training** to the labour force in the host country.
- **Transfers of skills and expertise**, helping to develop the quality of the host labour force.
- MNCs add to the host country's **GDP** through their spending, for example with local suppliers and through capital investment.
- **Competition** from MNCs acts as an incentive for domestic firms in the host country to improve their competitiveness and efficiency.
- MNCs **extend consumer and business choice** in the host country.
- MNCs bring with them **efficient business practices, technologies and standards** from across the world, which can influence the industries in the home country.
- Profitable MNCs are a source of **significant tax revenues** for the host economy (for example on profits earned as well as payroll and sales-related taxes).

Disadvantages to home country:

- MNCs **transfer capital** from the home country to various host countries causing unfavourable balance of payments.
- MNCs **may not create employment** opportunities to the people of home country if it employs labour from other countries, perhaps due to lower costs or better skills.
- As investments in foreign countries is more profitable, MNCs **may neglect the home country's industrial and economic development**.

Disadvantages to host country

- Domestic **businesses may not be able to compete** with MNC's efficiency, low costs, low prices and brand image, and may be forced to close shop.
- MNCs may **not act ethically or in a socially responsible way**, especially by taking advantage of weak countries who gain a lot from the MNCs presence in their country. For example, exploiting workers with low wages and poor working conditions in a country where labour laws are weak.
- MNCs may be **accused of imposing their culture** on the host country, perhaps at the expense of the richness of local culture.
- **Profits earned by MNCs may be remitted back** to the MNC's home country rather than being reinvested in the host economy.
- MNCs may make use of transfer pricing and other tax avoidance measures to **reduce the profits on which they pay tax** to the government in the host country.

Free Trade and Protection

Free trade is when there are no restrictions for trade between economies.

The advantages of free trade:

- **Allows countries to benefit from specialisation:** if there was no international trade, then countries wouldn't be able to specialise – that is, they would have to become self-sufficient by producing all the goods and services they require themselves. Total output would lower and costs would rise. With specialization and free trade, output, incomes and living standards will improve.
- **Increases consumer choice:** consumers can now enjoy a variety of products from around the globe.

- **Increases competition and efficiency:** international trade means that there will be more competition among firms in different countries. This would help increase efficiency.
- **Creates new business opportunities:** free trade will allow businesses to produce and sell goods for overseas consumers and expand and grow their operations by doing so. Profits and revenue would rise.
- **Enables firms and economies to benefit from the best workforces, resources and technologies** from around the world.
- **Increases economic inter-dependency** and thus fosters cooperation and reduces potential for international conflicts.

The disadvantages of free trade:

- **Free trade may reduce opportunities for growth in less-developed economies and threaten jobs in developed economies.** Small businesses in developing countries may not be able to compete with larger foreign firms. Established businesses in developed countries may lose market share as new firms keep entering the market. The US has seen considerable unemployment in manufacturing sectors since China joined the WTO and flooded international markets with their cheap products.
- **Causes rapid resource depletion and climate change** as more resources are used up by firms.
- **Exploitation of workers and the environment:** free trade has allowed firms to relocate to countries with lower costs (usually lower wages), where workers and the environment can be exploited (as health, safety and environmental laws in such countries are likely to be relaxed).
- **Income inequality worsens:** multinational firms and consumers have dominated the international supply and demand. This means that the rich keep getting richer (by buying and selling more products) while the poor lose out on products and resources.

Protection involves the use of trade barriers by governments to restrict international market access and competition. Trade barriers include:

- **Tariffs:** these are indirect taxes on imported (or exported) goods that make them more expensive, imposed in order to discourage domestic consumers from buying them.
- **Subsidies:** government allows subsidies to domestic producers so that they can increase their output and reduce costs and in turn reduce prices, in the hope that consumers will be encouraged to buy inexpensive domestic goods rather than imports.
- **Quotas:** this is a limit on the number of imports allowed into a country in a given period. Restricting supply will push up their market prices and discourage consumption of those imports.
- **Embargo:** this is a complete ban on imports of a good to a country.
- **Excessive quality standards:** imports may only enter a country after extensive quality checks which will be costly and so foreign producers will be discouraged to sell their products in the country, reducing imports.

Reasons for protection:

- **To protect infant industries:** trade barriers will help protect infant/sunrise industries (industries that are new and are hoping to grow). Lesser competition from foreign firms will increase their chances of survival and growth.
- **To protect sunset industries:** sunset industries are those that are on their declining stage. They would still employ many people and closure of firms in that industry will result in high unemployment. Lesser competition from foreign firms will decrease their rate of decline.
- **To protect strategic industries:** strategic industries will include transport, energy, defence etc. and governments will want to protect these so they are not dependent on supplies from overseas. If foreign firms supplied these, they would restrict output and raise prices.
- **To limit over-specialization:** if a country specializes in the production of a narrow range of products and there is a global fall in demand for one of them, then the economy is at risk. Protectionism will ensure diversification into producing more products and reduce this risk.

- **To protect domestic firms from dumping:** dumping is a kind of predatory pricing, that occurs when imports are sold at a price either below the price charged in the home market or below its cost of production. As a result, domestic firms will be unable to compete and be forced to go out of business. Once this happens, the foreign firms will raise their prices and enjoy monopolistic power. Trade barriers will eliminate the risk of dumping.
- **To correct a trade imbalance:** protectionism can reduce the imports coming into a country and thus reduce expenditure on imports by domestic consumers. If a country is experiencing a deficit (imports exceeding exports), then protectionism will correct this imbalance.
- Because other countries use trade barriers.

Consequences of protection:

- **They restrict consumer choice.**
- **They restrict new revenue and employment opportunities.**
- High levels of import tariffs and quotas will increase the costs of production at home and drive up prices, causing **cost-push inflation**.
- **They protect inefficient domestic firms:** when trade barriers are used to protect domestic industries, it might include inefficient industries. Protectionism means that even very inefficient industries are protected (when they are better off being exposed to foreign competition and being forced to become more efficient).
- **Other countries may retaliate:** if a country introduces trade barriers to restrict imports from other countries, the countries that are affected by this will also impose similar trade barriers. A trade war may develop. Relations between countries will worsen.
- In today's globalised society, being heavily protected and not engaging in free trade will result in the country being out of pace with the rest of the world. They will be **unable to grow and develop, and will lose out on the benefits of free trade**.

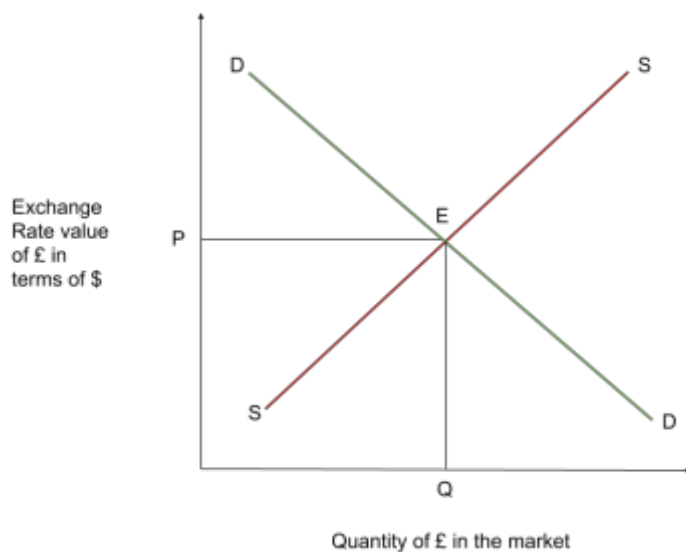
Tip: If you have trouble remembering all the pros and cons listed above, just remember this: basically, the advantages of free trade are the disadvantages of protectionism and the disadvantages of free trade are the advantages of protectionism.

The foreign exchange rate is the value or price of a currency expressed in terms of another currency. For example, £1 = \$1.2

This exchange rate will be used when these countries trade and need to convert money. So if a person were to convert £100 into dollars, he would get \$120 (100×1.2).

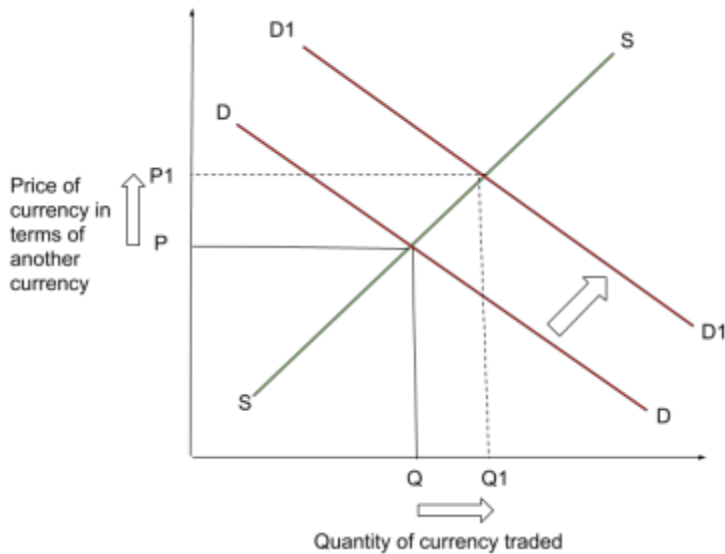
The foreign exchange rate of each currency is **determined by the market demand and supply of the currency**.

- Demand for the a currency, say the pound sterling, exists when foreign consumers want to buy and import goods and services from the UK, when overseas companies buy pounds to invest in the UK etc. Here, the UK's currency is being demanded abroad.
- Supply of a currency, say the pound sterling, exists as UK consumers want to buy and import goods and services from other countries, when UK companies buy foreign currencies to invest abroad. Here the UK's currency is being supplied abroad.
- The price at which demand and supply of the currency equals is the **equilibrium market foreign exchange rate** value of a currency against another currency. An increased supply and decreased demand causes the exchange rate to fall, while a decreased supply and increased demand causes the exchange rate to rise.

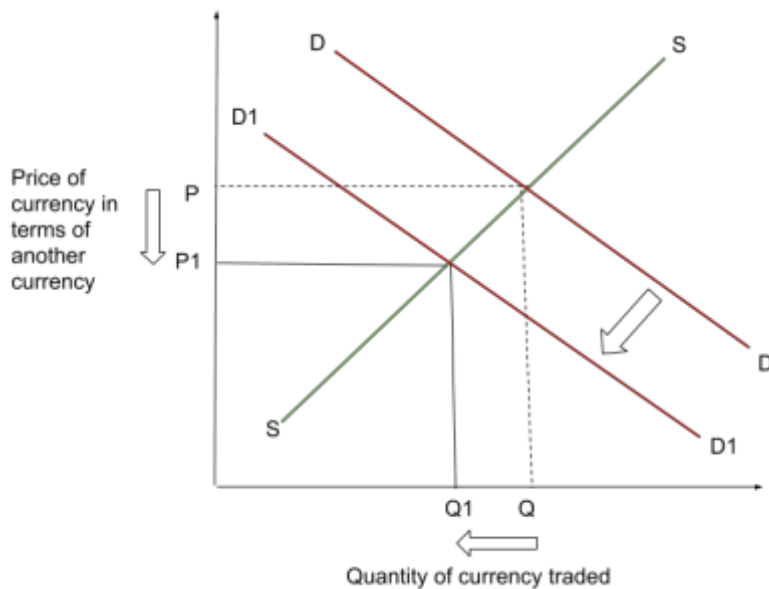


Causes of foreign exchange rates fluctuations:

- **Changes in the demand for exports and imports:** when a country's import value is greater than its export value (which is a deficit), it means that more of their currency is being supplied (going out) than being demanded. The exchange rate for the country's currency will fall. If there is a surplus in the current account, the exchange rate will rise.
- **Inflation:** if the inflation in a country is higher than that of other countries it trades with, the price of that country's goods in the international market will be higher compared to goods from other countries. The demand for the country's goods will fall and the so the currency demand will also fall, causing a fall in exchange rate.
- **Changes in interest rate:** if a country's interest rate rises, overseas residents may be keen to save or invest money in that country. The demand for the currency will rise, and the exchange rate will rise. If interest rates fall below that of other countries, the currency will fall in value as overseas demand falls.
- **FDI/MNCs:** globalisation and economic activities of multinational companies mean that investment in overseas production plants requires the use of foreign currencies. For example, a US company with factories in UK needs to pay its labour with pound sterling, not with the US dollars, increasing the demand for the pound. Thus, inward FDI will boost the demand for a currency and increase its foreign exchange value. In contrast, outward FDI will increase the supply of a currency and cause its foreign exchange rate to fall.
- **Speculation:** foreign exchange traders and investment companies move money around the world to take advantage of higher interest rates and variations in exchange rates to earn a profit. As huge sums of money are involved (known as 'hot money'), this can cause exchange rate fluctuations, at least in the short run. If speculators lack confidence in the economy they will withdraw their investments in that country, thereby causing a fall in the value of the currency. In contrast, high confidence in the economy will invite investments and raise the foreign exchange value of the currency.
- **Government intervention:** government intervention in the foreign exchange market can affect the exchange rate. For example, if greater demand for British goods causes a rise in the value of the pound, the Bank of England (UK's central bank) can sell their reserves of pound sterling in the foreign exchange market to increase its supply and cause a fall in its value.



A rise in demand for the domestic currency (or a fall in supply – shift of supply curve to the left) causes its exchange rate to rise



A fall in demand for the domestic currency (or a rise in supply – shift of supply curve to the right) causes its exchange rate to fall

Consequences of foreign exchange rate fluctuations:

- **A fall in the foreign exchange rate causes import prices to rise and export prices to fall.**

A fall in the foreign exchange rate, of say the pound, means that now you have to pay more pounds when you're importing an American good to the UK, for example. If the initial exchange rate is $\$1 = \text{£}0.8$, and the original price of the good was $\$2$, you'd have to pay $\text{£}1.6$ ($\$2 * \text{£}0.8$) to buy the good. Now, suppose the exchange rate falls to $\$1 = \text{£}1$ (a pound is now worth lesser dollars), you'd have to pay $\text{£}2$ ($\$2 * \text{£}1$) to buy the good.

Similarly, a fall in the foreign exchange rate of the pound means that now you'll get fewer dollars when you're exporting a British good to America. Using the initial exchange rate as described above, an export initially costing $\text{£}2$ means American consumers will have to pay $\$2.5$ ($\text{£}2 / \text{£}0.8$) to buy it. After the exchange rate falls, they will have to pay only $\$2$ ($\text{£}2 / \text{£}1$).

- **A rise in the foreign exchange rate causes import prices to fall and export prices to rise.**

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If the initial exchange rate is $\$1 = \text{£}0.8$, and the original price of the good was $\$2$, you'd have to pay $\text{£}1.6$ ($\$2 * \text{£}0.8$) to buy the good. Now, suppose the exchange rate rises to $\$1 = \text{£}0.5$ (a pound is now worth more dollars), you'd have to pay only $\text{£}1$ ($\$2 * \text{£}0.5$) to buy the good.

Similarly, a rise in the foreign exchange rate of the pound means that now you'll get more dollars when you're exporting a British good to America. Using the initial exchange rate as described above, an export initially costing $\text{£}2$ means American consumers will have to pay $\$2.5$ ($\text{£}2 / \text{£}0.8$) to buy it. After the exchange rate rises, they will have to pay $\$4$ ($\text{£}2 / \text{£}0.5$).

- Now, if the country's export and import demands are price elastic (relatively more sensitive to price changes), a fluctuation in the exchange rate and the subsequent changes in the prices of exports and imports will change the demand for them.

A fall in exchange rate will make imports expensive and exports cheap, so import demand and spending will fall and export demand and spending will rise.

A rise in the exchange rate will make imports cheaper and exports expensive, so import demand and spending will rise and export demand and spending will fall.

- Hence, we can conclude that when $PED > 1$ (elastic), **a fall in foreign exchange rate will improve the trade balance** (reduce deficits) of the country as exports will rise relative to imports.
- On the other hand, when $PED < 1$ (inelastic), **a rise in foreign exchange rate will worsen the trade balance** (but reduce surplus) of the country as imports will rise relative to exports.

Floating Foreign Exchange Rate

This is an **exchange rate that is determined freely by market demand and supply** conditions, and so will fluctuate regularly.

The rise in the value of one currency against others, on a floating exchange rate is known as **appreciation** of the currency.

The fall in the value of one currency against others, on a floating exchange rate is known as **depreciation** of the currency.

Advantages:

- **Automatic Stabilisation:** any disequilibrium in the balance of payments would be automatically corrected by a change in the exchange rate. For example, if a country suffers from a deficit in the balance of payments, then the country's currency should depreciate. This would cause the country's import demand to fall (as imports become expensive) and export demand to rise (as export prices fall). The balance of payments equilibrium would therefore be restored. Similarly, a surplus would be eliminated as the currency appreciates.
- **Frees up internal policy:** a floating exchange rate allows a government to pursue internal policy objectives such as full employment and growth, not having to worry about balance of payments imbalances as they will be automatically adjusted.
- **Insulated from external changes:** a floating exchange rate helps to insulate a country from inflation elsewhere. If a country were on a fixed exchange rate then it would 'import' inflation through higher import prices. A floating exchange rate would automatically adjust demand and supply in the economy and avoid such external disturbances.
- **Don't need too much foreign reserves:** under a floating exchange rate system, there is no need to maintain reserves to deliberately change the exchange rate. These reserves can therefore be used to import capital goods in order to promote faster economic growth.

Disadvantages:

- **Uncertainty:** since currency values fluctuate constantly, businesses, investors and consumers will be uncertain about the economy and its future. They may lose confidence in the economy if it fluctuates too rapidly.
- **Lack of Investment:** the uncertainty introduced by floating exchange rates may discourage direct foreign investment. They will prefer to invest in countries with fixed exchange rate systems where they can effectively predict economic conditions and act accordingly.

- **Speculation:** the day-to-day fluctuations in exchange rates may encourage speculative movements of 'hot money' from country to another, thereby causing more exchange rate fluctuations.
- **Lack of Discipline:** the need to maintain an exchange rate imposes a discipline upon the national economy, which is absent in a floating exchange rate regime. With a floating exchange rate, short-run problems such as domestic inflation may be ignored until they lead to a crisis.

Fixed Foreign Exchange Rate

A fixed exchange rate is one that is fixed and controlled by the central bank, acting on behalf of the government of the country. The central bank will intervene in the market by **buying and selling its currency in the foreign exchange market** to maintain a fixed exchange rate.

A deliberate fall in the value of a fixed exchange rate is called a **devaluation**.

A deliberate rise in the value of a fixed exchange rate is called a **revaluation**.

Advantages:

- **Certainty:** since the currency value is kept in check, there will be more certainty in the economy and businesses, consumers, investors and governments won't have to worry about the effects of automatic changes in exchange rate.
- **Stability encourages investment:** a fixed exchange rate provides greater certainty and encourages firms to invest. One of the reasons Japanese firms are reluctant to invest in UK is because the pound works on a flexible exchange rate (unlike the Euro which is on a fixed system), causing uncertainty about the economy.
- **Keep inflation low:** depreciation of a currency can cause inflation as demand, prices and costs for firms rise. On a fixed exchange rate, firms have an incentive to keep cutting costs to remain competitive.
- **Balance of Payments stability:** since the exchange rate is not determined by market forces, sudden changes in the balance of payments will be eliminated, keeping it stable instead.

Disadvantages:

- **Conflict with other macroeconomic objectives:** the goal to maintain a fixed exchange rate may conflict with other macroeconomic objectives when the government intervenes with its policies. For example, if it raises interest rates to remove the pressure of the currency to fall, economic growth might be adversely affected.
- **Less flexibility:** in a fixed exchange rate, it is difficult to respond to temporary shocks. For example, if the price of oil increases, a country which is a net oil importer will see a deterioration in the current account balance of payments. But since the country operates a fixed exchange rate, it cannot devalue the currency too much and thus cannot make an effective intervention to improve the current account.
- **Risk of overvaluation or undervaluation:** it is difficult to know the right rate to fix the exchange rate at. If the rate is too high, it will make exports uncompetitive. If it is too low, it could cause inflation. It is difficult to ascertain the optimum foreign exchange rate.

The Balance of payments is a record of all the monetary transactions between residents of a country and the rest of the world over a given period of time. It is divided into three main accounts: the current account, the capital account and the financial account.

(In the explanation below, we'll look at the balance of payments from the point of view of the UK)

The Current Account

The Current account records the following:

- **The visible trade** (in goods)
 - **The invisible trade** (in services)
 - **Net income received or made in payment for the use of factors of production (also called net primary income):**
 - income debits (outflows) include wages paid to overseas residents working in the UK, interests, profits and dividends paid out to overseas residents and firms who have invested in the UK
 - income credits (inflows) include wages paid to UK residents working overseas, interests, profits and dividends earned by UK residents and firms on investments they have in other countries
 - **income received – income paid = net income received**
 - **Net current transfers**, which include payments between governments for international co-operation and other transactions that **involve payments for non-productive activities (also called net secondary income):**
 - debits (outflows) will include financial aid, donations, pension payments etc. paid to overseas residents and foreign governments, and tax and excise duties paid by UK residents on foreign purchases
 - credits (inflows) will include financial aid, donations, grants, pension payments etc. received from overseas residents and foreign governments, and tax and excise duties paid by overseas residents on UK purchases
 - **transfers received – transfers paid = net transfers**
- A current account example:

Item	\$ (billion)
Visible exports (X_v)	784.2
Visible imports (M_v)	1230.4
Balance of trade ($X_v - M_v = A$)	-446.2
Invisible exports (X_i)	286.4
Invisible imports (M_i)	219.9
Balance on services ($X_i - M_i = B$)	66.5
Net primary income (C)	21.0

Net secondary income (D)

-58.6

Current account balance (A + B + C + D)

-417.3

Current Account Deficit

When the financial outflows in the current account exceed the financial inflows, the current account is in deficit.

Causes:

- **Higher exchange rate:** if the currency is overvalued, imports will be cheaper and therefore there will be a higher quantity of imports. Exports will become uncompetitive and therefore there will be a fall in the quantity of exports.
- **Economic growth:** if there is an increase in aggregate demand and national income increases, people will have more disposable income to consume goods. If producers cannot meet the domestic demand, consumers will have to import goods from abroad. Thus faster economic growth enables the possibility of a current account deficit developing.
- **Decline in competitiveness:** if export industries are in decline and cannot compete with foreign countries, the exports fall, ushering in a deficit. This is a major reason for many countries today experiencing current account deficits.
- **Inflation:** this makes exports less competitive and imports more competitive (cheaper).
- **Recession in other countries:** if the country's main trading partners experience negative economic growth then they will buy less of the country's exports, worsening the current account.
- **Borrowing money:** if countries are borrowing money from other countries to finance their expenditure and growth, current account deficits will develop.

Consequences:

- **Low growth:** a deficit leads to lower aggregate demand and therefore slower growth.
- **Unemployment:** deficit can lead to loss of jobs in domestic industries as demand for exports is low and demand for imports is high.
- **Lowers standard of living:** in the long run, persistent trade deficits undermine the standard of living as demand and income fall, especially if the net incomes and transfers show a negative balance.
- **Capital outflow:** currency weakness can lead to investors losing confidence in the economy and taking capital away.
- **Loss of foreign currency reserves:** countries may run short of vital foreign currency reserves as more foreign currency is being spent on imports and foreign currency revenues from exports is falling.
- **Increased Borrowing:** countries need to borrow money or attract foreign investment in order to rectify their current account deficits. In addition, there is an opportunity cost of debt repayment, as the government cannot use this money to stimulate economic growth.
- **Lower exchange rate:** a fall in demand for exports and/or a rise in the demand for imports reduces the exchange rate. While a lower exchange rate can mean exports becoming more price-competitive, it also means that essential imports (such as oil and foodstuffs) will become more expensive. This can lead to imported inflation.

The severity of these consequences depends on the size and duration of the deficit. Persistent deficits can harm the economy in the long-run as low export growth causes unemployment.

Correcting a current account deficit

- **Do nothing because a floating exchange rate should correct it:** if there is a trade deficit, a depreciation will occur as more currency is being spent than received. Depreciation will make imports more expensive and exports cheaper. As a result, domestic demand for imports will fall and foreign demand for exports will rise, reducing the deficit.
- **Use contractionary fiscal policy:** a government can cut public expenditure and increase taxes to reduce total demand in the economy, which will reduce demand for imports and improve the trade balance. However, a fall in demand may affect firms in the economy who may cut output and employment in response.
- **Use contractionary monetary policy:** a higher interest rate will attract more direct inward investments and nullify the trade deficit. Higher interest rates will also make borrowing from banks more expensive and increase the incentive to save, thus discouraging consumers from spending. The govt. can also devalue the exchange rate to improve export competitiveness and demand.
- **Protectionist measures:** these measures reduce the competitiveness of imports, thereby making domestic consumption more attractive. For example, tariffs raise the price of imports while quotas limit the amount of imports in the economy.

Current Account Surplus

When the financial inflows in the current account exceed the financial outflows, the current account is in surplus.

Causes:

- **Improved competitiveness:** exports may have become more price-competitive in the international market, due to perhaps, better labour productivity or low prices.
- **Growth in foreign countries:** export demand may have risen due to trading partners experiencing growth and higher incomes.
- **High foreign direct investment:** strong export growth can be the result of a high level of foreign direct investment.
- **Depreciation:** a trade surplus might result from a currency depreciation .
- **High domestic savings rates:** high levels of domestic savings and low domestic consumption of goods and services cause more products to be exported and imports to fall.
- **Closed economy:** some countries have a low share of national income taken up by imports, perhaps because of a range of tariff and non-tariff barriers.

Consequences:

- **Economic growth:** net exports is a component of GDP, so a rise in exports and incomes will cause economic growth.
- **Appreciation:** as exports increase, the demand for the currency increases and therefore the value of the currency increases, which will make exports more expensive and cause its demand to fall.
- **Employment:** since exports have increased, jobs in the export industries will have increased too.
- **Better standards of living:** higher net incomes, transfers and export revenue make the country's citizens better off.
- **Inflation:** higher demand for exports can lead to demand-pull inflation. This can diminish the international competitiveness of the country over time as the price of exports rises due to inflation.

Correcting a current account surplus:

- **Do nothing because a floating exchange rate should correct it:** if there is a trade surplus, an appreciation will occur as more currency is being demanded. An appreciation will make imports cheaper and exports expensive. As a result, foreign demand for exports will fall and domestic demand for imports will rise, reducing a trade surplus.

- **Use expansionary fiscal policy:** increasing public expenditure and cutting taxes can boost total demand in an economy for imported goods and services.
- **Use expansionary monetary policy:** lower interest rates will make borrowing from banks cheaper and increase the incentive to spend, thus encouraging consumers to spend on imports and correct a trade surplus.
- **Remove protectionist measures:** reducing tariffs and quotas cause imports to rise and close a surplus in the current account.